

UNDERGROUND CBRN-HARDENED BLAST-RESISTANT PROTECTIVE STRUCTURE, WITH SENTRY POST

Pune, Maharashtra · Master project report

Item	State
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Architectural revision	Rev F
Structural revision	Phase 2 Rev A + M1
Design report revision (historical)	Rev D
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Governing authority	master/MASTER_PROJECT_STATE.md — Parts A, B, F and L
Status	FOR REVIEW — NOT FOR CONSTRUCTION

This report states the project once, in the order an engineer would need it to reproduce the design: the science first, then the site, then the loads, then the calculations, then what is drawn, then what is still open.

It is a statement of the design, not a history of it. Every value carries the evidence class the master gives it. Where the project does not hold a number, this report says so in the same sentence rather than filling it in.

Where this report and the master disagree, the master governs.

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PART 1 — THE PROJECT AND THIS DOCUMENT

1.1 What this report is

This is the **master project report** for a buried reinforced-concrete protective structure at Pune, Maharashtra, and for the separate sentry post that watches its approach. It states the project once, in the order an engineer would need it to reproduce the design: the physics first, then the ground, then the loads, then the calculations, then what is drawn, then what is still open.

It is written to stand on its own. Everything needed to follow an element from a first-principles load to a bar mark on a drawing is in here, with the clause of the Indian Standard that permits each step named beside it.

It is a statement of the design, not a history of it. The revision history is summarised in Appendix D and held in full in the master; this report gives the current position and says how it was reached only where the reasoning is part of the design.

1.2 Authority, and what governs what

Document	Role
master/MASTER_PROJECT_STATE.md	The single source of truth. Parts A, B, F and L are the authoritative current design. Where this report and the master disagree, the master governs
master/QUICK_STATE.md	Orientation digest only. Not an authority
This report	The project stated once, in full, for reading — design basis, science, calculations, services, works management, open items
Discipline packages	Structural CAD/·Drainage/·HVAC/·Electrical/·EMP Protection/·Fire and Life Safety/·Schedule of Finishes/·Site Selection and Geotechnical/·Site and Concealment/·WORKS MANAGEMENT/·Owner Rulings RC4/·DRAWING QAQC/
Drawings	80 DXF. The ten Rev F architectural files are source , not build artefacts; the discipline sheets are generated and must be regenerated, never hand-edited

This report changes no design value. No dimension, level, load, thickness, bar, quantity, rate, date or float is altered by it, no .std file is touched, and **no evidence tag is converted, downgraded or deleted.** Where the project does not hold a number, this report says so in the same sentence rather than filling it in.

1.3 Evidence classes — and why they are the point of the document

Every value in this project carries a class. They are the master's own and they are used here unchanged. A reader who ignores them will read an assumption as a measurement, which is the one mistake this document set exists to prevent.

Tag	Meaning	How to treat it
[C] CONFIRMED	Read directly from a project file, or computed in the project from confirmed inputs	Use as fact
[R] RECONSTRUCTED	Derived by back-calculation from partial evidence	Use, but state the derivation when relying on it
[A] ASSUMED	An engineering assumption. No test, no client confirmation	Must be re-confirmed before construction
[U] UNRESOLVED	Two or more conflicting values and insufficient evidence to choose	Do not guess
[N] NOT AVAILABLE	The information does not exist in any project material	Do not invent

The register is not decoration. Seven assumptions remain open (Part 18.2) and **every one of them is a physical test on this plot** — a borehole, a piezometer, a plate load test, a percolation test, a packer test. Eighteen open items remain (Part 18.3), and **four of them are the ones a reviewer should look at first**: an investigation that reached about 1.5 m against a formation at (–)6.800; two 1 400 mm holes in the protective boundary whose bonding is still undesigned; a CBRN shelter that cannot state which way a release drifts; and a sealed box with a 4 kW heat surplus and an unmodelled rejection path.

1.4 Project identity

Field	Value	Class
Project title	Underground CBRN-hardened, blast-resistant protective structure with associated sentry post	[C]
Location	Pune, Maharashtra, India — CTW / College of Military Engineering campus	[C]

Field	Value	Class
Project type	Buried reinforced-concrete protective structure, military-operational, for actual construction	[C]
Academic context	B.E. Civil Engineering, final semester, three-presentation capstone	[C]
Objective	Protect 9 occupants for 96 h against a nuclear air-blast design basis threat, with CBRN, EMP and fallout hardening	[C]
Architectural revision	Rev F — the ten uploaded DXF files	[C]
Design report revision	Rev D — older than the drawings; the largest historical reconciliation issue, now closed (Part 18.4)	[C]
Structural revision	Phase 2 Rev A , incorporating Modification M1	[C]
Presentation status	P1 delivered. P2 (AutoCAD + STAAD + manual calculations) is the active deliverable. P3 not yet scoped	[C]
Software	STAAD.Pro (Bentley), AutoCAD/DXF, plus a purpose-built dependency-free Python DXF toolchain	[C]

1.5 The three structures, and they are deliberately not equal

#	Structure	Status
1	Main underground box — 8 bays, 22.0 × 6.2 m external, roof 2.0 m below grade	Blast designed. This is the protected volume
2	Entry headhouse and covered approach stairwell — above-ground bermed RC, sitting on the shelter roof and on adjacent fill	Headhouse blast designed ; the stairwell is NOT , and is declared expendable
3	Sentry post — two-storey RC framed building, ≥ 10 m clear of the shelter excavation, on its own footings on in-situ rock	NOT blast designed. A recorded decision, not an omission — Part 11.1

1.5.1 The protective boundary — the single most important idea in the design

The protective boundary is **Blast Doors 1 and 2 at level (–)6.100, together with walls W6 and W7, the perimeter walls, the mat and the pressure slab.**

Everything above the blast doors — the stair shaft, the headhouse, the entry stairwell — is OUTSIDE it. That is deliberate. The stair shaft is open to atmosphere through the roof void, and the design accepts that it fills with blast pressure rather than pretending it does not. That single admission is what produced **Finding F1** and **Modification M1** (Part 9.2), and it is the reason W6 and W7 are 400 mm thick instead of 200.

Gas-tight envelope = Bays 1 to 6 only — 67.80 m² floor, 216.96 m³ (stated on the issued sheet as 67.8 / 217.0) [C]. Bay 7 (stair shaft) and Bay 8 (generator) are **outside** it; Bay 8 is the "grey zone".

The project holds **TWO protective boundaries and they do not coincide.** The gas-tight boundary is bays 1–6. The **EMP boundary is bays 1–8** — ruled by the project owner at RC4, because otherwise the 15 kVA generator, its control panel and every cable in bay 8 would be unprotected against the pulse. **Only one of the two has ever been drawn.** Part 14.4.

1.5.2 Access and egress

Route	Path
Primary	Grade → covered entry stairwell, 12R @ 166.667 / 300 → platform (–)2.000 → inner security door → headhouse → stair void → main dog-leg stair, 24R @ 170.8333 / 280, 3 flights × 8 → arrival landing (–)6.100 → Blast Door 1 , 1200 × 2100, ≥ 7 bar → Bay 6
Secondary	Blast Door 2 , 1200 × 2100, from the stair shaft into Bay 8 (generator / grey zone)
Emergency	Two 1 400 mm dia vertical escape shafts, ESC 1 (Bay 1) and ESC 2 (Bay 8), 250 mm RC collars, heads at +0.150 and +0.700

R1, the main stair, is the only route that does not require climbing a shaft, and the only one usable by an injured or unconscious person. ESC 1 and ESC 2 are **6.250 m** and **6.800 m** vertical climbs (Part 14.5).

1.6 The mission, in numbers

	Value	Class
Occupancy	9 persons	[C]
Endurance	96 hours	[C]
Gas-tight floor area / volume	67.80 m² / 216.96 m³	[C]
Gross internal volume, bays 1–8	20.800 × 5.000 × 3.200 = 332.8 m³	[C]
Design ventilation flow	300 m³/h , 2 × 300 trains, true N+1	[C]
Air changes on the gross volume	≈ 0.9 per hour	[R]

	Value	Class
Closed-mode CO ₂ limit without scrubbing	9.9 h to 1.0 %	[C]
Closed-mode limit on the soda-lime store	48 h	[C]
Oxygen store	2 × 50 L at 150 bar = 15 m ³ = 80 h at 4.5 m ³ /day	[C]
Standby power	15 kVA generator, bay 8; connected load 7.360 kVA = 49 %	[C] / [R]
Battery autonomy, essential loads	4 h , 149 Ah at 48 V	[C] by ruling
Potable water	1 000 L tank, bay 1 = 27.8 L/person/day. Not firefighting water	[C]

1.7 Working rules this project is held to

These are not general good practice — they are the specific disciplines this project adopted, and several of them exist because something once went wrong.

1. **The master governs.** A digest is never an authority.
2. **Inspect native files before modifying them.** Parse the DXF; read the .std. Never edit from memory or from a summary.
3. **Never guess or silently resolve** an [UNRESOLVED], [ASSUMED] or [NOT AVAILABLE] item. Do not convert one into a confirmed fact, and do not delete or downgrade a tag.
4. **Do not fabricate missing generator scripts or unavailable output drawings.**
5. **Distinguish file reconciliation from analysis.** STAAD.Pro is not available in this environment. **Editing and verifying a .std file is not running an analysis** and must never be reported as one.
6. **No drawing shows an arrangement that differs from a calculation.** If the steel changes, the drawing changes in the same turn.
7. **Every significant change gets a revision identifier** and is recorded; **no revision is overwritten without preserving it.**
8. **A new drawing package registers itself with the QA tool.** Twelve sheets were invisible to the drawing index for four revisions because this was not done (Part 13.3).
9. **A change that stops at "reinforcement" and never reaches "drawings" is not finished.**

1.7.1 Frozen geometry

The main staircase is frozen: 24 risers, 170.8333 mm riser, 280 mm tread, 3 flights × 8, total rise 4 100 mm. The landing levels, the 1 200 flight widths, the 200 mm well and the 2 533 mm headroom go with it. **Nothing in this report changes any of it,** and no service, duct, pipe, drain or diffuser is placed in the flights, the well or the landings.

1.8 How this report is arranged

Part	Contents
1	The project, the authority, the evidence classes, the rules
2	The engineering science behind the design — blast physics, structural dynamics, radiation attenuation, buoyancy, earth pressure, filtration, EMP coupling, seismic response
3	Codes and standards, clause by clause
4	Site selection and geotechnics
5	Geometry and configuration — the complete dimensional register
6	Materials, durability, cover, development and lap lengths
7	Loading — every load derived and every combination stated
8	Analysis models — what they are, what they prove, what they cannot
9–11	Structural design calculations , element by element: the box; the stairs, headhouse and appurtenances; the sentry post
12	Reinforcement register and quantities
13	Drawings and drawing quality
14	Services — HVAC and CBRN, drainage, electrical, EMP, fire
15	Site layout, external works, finishes and concealment
16	Works management — bill, cost, programme, procurement, QA/QC, risk
17	What this design does not demonstrate
18	Assumptions, open items and owner rulings

Part	Contents
19	The verification performed for this report
A–D	Notation · IS clause index · reproducing the project · revision history

PART 2 — THE ENGINEERING SCIENCE BEHIND THE DESIGN

What this part is. The design decisions in Parts 6 to 11 are only defensible if the physics underneath them is understood. This part sets out that physics: where the loads come from, why the code rules take the form they do, and which physical effect each dimension was bought for.

It introduces no project value. Every number quoted here is carried from the project's own register and is cross-referenced to the Part where it is used. Where the physics would allow a quantity the project does not hold — a weapon yield, an ambient design temperature, a soil modulus — **the gap is named, not filled.**

2.1 The blast wave

2.1.1 What a shock front is

An explosion releases energy in a time far shorter than the time the surrounding air needs to move out of the way. The air is compressed faster than an acoustic wave can carry the information away, so the disturbance steepens into a **shock front**: a near-discontinuity across which pressure, density, temperature and particle velocity all jump.

Across that front, conservation of mass, momentum and energy give the **Rankine–Hugoniot** relations. For a strong shock in air the two consequences that matter to a structural engineer are:

STATIC OVERPRESSURE	p_{so}	the rise in ambient pressure at the front. It acts EQUALLY IN ALL DIRECTIONS, like any pressure. A buried roof, a buried wall and a free-standing wall all feel it.
DYNAMIC PRESSURE	q	the kinetic pressure of the air actually moving behind the front, $q = 0.5 \rho u^2$. It acts ONLY in the direction of flow, and it is what produces DRAG on an obstacle standing in it.

They are different loads with different physics, and the design uses both. The roof and walls of the buried box are loaded by p_{so} ; the drag term q is what the older Rev D calculation used for the bermed headhouse walls before this project raised them to the full envelope (Part 7.5).

2.1.2 Reflection, and why the reflected pressure is not simply doubled

When a shock meets a rigid surface head-on, the air behind the front is brought to rest and its momentum is converted into further compression. For a weak (acoustic) wave the reflected pressure would be exactly twice the incident. For a strong shock in a gas that can be heated and partially dissociated, the reflection coefficient rises well above 2 — for the design basis threat of this project it is close to 4:

$p_{so} = 344.7 \text{ kPa (50 psi)}$	incident static overpressure	[C]
$p_r = 1366 \text{ kPa}$	reflected pressure	[C]
$q = 282 \text{ kPa}$	dynamic pressure	[C]
reflection coefficient $p_r / p_{so} = 1366 / 344.7 = 3.96$		

That reflected value is not used on the buried box — a buried surface is not a free-standing reflecting face — but it **is** the number that decides the sentry post. Part 11.1 computes the reflected force on the sentry post's $4.0 \times 6.7 \text{ m}$ face as **36 600 kN, about 3 730 tonnes**, and that single figure is why hardening it was rejected rather than attempted.

2.1.3 Positive phase duration and impulse

Behind the front, pressure decays back to ambient over the **positive phase duration** t_d , then falls below ambient into a **negative phase** in which the flow reverses. The area under the pressure–time curve is the **impulse**, and for short loading it is impulse rather than peak pressure that damages a structure.

$t_d = 0.13 \text{ to } 1.33 \text{ s}$	the design basis positive phase duration	[C]
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The yield behind those numbers is deliberately not stated. $p_{so} = 344.7$ kPa and $t_d = 0.13$ – 1.33 s **are** the design inputs, and every load case in this project uses them. The weapon yield that would produce them is [N] and was **ruled to stay unstated rather than invented** — owner ruling 6, RC4 (Part 18.4). Nothing in the design requires it. The one thing it would change is the prompt-radiation cover depth, and that is said openly in 2.4.

The negative phase is also not designed for as a separate load case, and that is recorded in Part 17. Where it matters it has been allowed for physically rather than numerically: the escape-shaft head hatches take **quarter-turn dogs so the leaf resists uplift as well as downward pressure**, because the negative phase and the rebound both lift it (Part 10.5).

2.1.4 Why a buried roof takes the full overpressure

A metre of soil is not a shield against overpressure. The wave in the air couples into the ground and propagates as a stress wave; the soil is a medium that **transmits** pressure, it does not absorb it in the way intuition suggests. Codified practice handles this with a **soil transmission factor** K_a applied to the free-field overpressure:

design surface pressure on a buried element = $K_a \times p_{so}$
IS 4991 Cl. 7.2 and Table 3

For **saturated** soil, $K_a = 1.0$ — water is nearly incompressible and transmits the pressure essentially undiminished. This project's founding horizon is permanently below the design water table, and the governing case is therefore the saturated one:

	Value	Class
K_a , saturated	1.0	[C] IS 4991 Cl. 7.2 / Table 3
K_a , dry compacted berm fill	≈ 0.5	[A] — deliberately not relied on

The single most consequential choice in the loading. Because $K_a = 1.0$, **2.0 m of engineered cover buys no reduction in blast pressure at all.** The roof of the buried box is designed for the same 383 kPa a bare roof would take. Every metre of that cover was bought for something else — radiation attenuation — and 2.5 names the price.

The same logic was applied to the headhouse walls, which stand behind ~ 2.5 m of berm. The earlier calculation credited the berm and used the drag pressure, 113 kPa. **Since K_a for a dry berm cannot be verified, this project designs those walls for the full 383 kPa envelope** (conflict C10, Part 7.5). The cost of that decision is one link cage; the utilisation goes from 17 % to 59 %.

2.2 From a pressure history to a design load: structural dynamics

2.2.1 The single-degree-of-freedom idealisation

A blast-loaded slab is a distributed system, but its response is dominated by its first mode. The standard idealisation replaces it with a single-degree-of-freedom (SDOF) oscillator of equivalent mass and stiffness carrying a time-varying force. The response then depends on **one ratio**: the load duration to the natural period, t_d / T .

$t_d / T < \sim 0.1$	IMPULSIVE	the load is over before the structure has moved. Only the IMPULSE matters.
~ 0.1 to ~ 10	DYNAMIC	genuine transient interaction; the full SDOF solution is needed.
$t_d / T > \sim 10$	QUASI-STATIC	the structure completes many cycles while the load is still applied. The peak response approaches that of a step load.

2.2.2 Which regime this structure is in — and the calculation that settles it

The governing element is the 900 mm pressure slab spanning 5.0 m clear. Its fundamental frequency is computed in Part 9.4 from the cracked flexural stiffness and the participating mass, including a share of the soil cover:

$m = 0.900 \times 2500 + 0.25 \times 2.0 \times 2000$	$= 3250$ kg/m ²
$0.5 EI_g = 0.5 \times 2.958e10 \times 0.900^3 / 12$	$= 8.985e8$ N.m ² /m
$f_1 = (22.373 / 2\pi) \cdot \sqrt{EI / (m \cdot L^4)}$	$= 74.9$ Hz
$T = 1 / f_1$	$= 13.4$ ms
$t_d / T = 130$ ms / 13.4 ms to 1330 ms / 13.4 ms	$= 10$ to 100

$t_d / T = 10$ to 100 **places this structure firmly in the QUASI-STATIC regime.** That is the single most important dynamic result in the project, and it is what makes a static analysis with a dynamic load factor an honest way to obtain the demand. It is not an approximation of convenience — it follows from a thick slab on a short span being extremely stiff, and a nuclear air blast being a comparatively long-duration load.

The coefficient **22.373** is the first eigenvalue constant for a uniform beam **fixed at both ends** ($\beta_L = 4.730$, $\beta_L^2 = 22.373$). Using the fixed-fixed value is consistent with the plastic mechanism adopted for strength in 2.2.4.

2.2.3 The dynamic load factor, and the ductility it is bought with

For a perfectly elastic SDOF system under a **step load**, the classical result is a dynamic amplification of exactly **2.0**: the mass overshoots the static deflection by as much again before the restoring force brings it back.

A structure that is allowed to yield does not have to store that energy elastically. Equating the work done by the load to the energy absorbed by an elastic-plastic resistance function, and defining the **ductility ratio** μ as the ratio of maximum to yield displacement, gives the form codified in IS 4991 Cl. 10.3.3:

```
DLF = mu / (mu - 0.5)

VALIDATION      mu = 1 (no ductility at all) -> DLF = 1/0.5 = 2.00
                  which is exactly the elastic step-load factor. The
                  expression reduces correctly at its own limit.

ADOPTED          mu = 5    "moderate, repairable damage"    IS 4991 Cl. 10.3.3
                  DLF = 5 / 4.5                                = 1.111

DESIGN BLAST PRESSURE = 344.7 x 1.111                        = 383 kPa
```

A ductility ratio is a promise about detailing, not a discount. Taking $\mu = 5$ asserts that the section can reach five times its yield rotation without losing capacity. That promise is kept in two places in this project and both are checkable:

First, the neutral axis depth. For the roof slab the design gives $x_u/d = 0.139$ against the Fe500 limit of 0.46 (IS 456 Cl. 38.1(f)). A section with a neutral axis at 14 % of the effective depth is very lightly reinforced in flexural terms, the steel strain at failure is large, and the section is unambiguously under-reinforced and ductile. **That number is the proof that $\mu = 5$ is defensible**, and it is why it is printed next to the result in Part 9.4.

Second, the shear detailing. Shear failure is brittle, and a brittle failure mode makes a ductility claim meaningless. Every blast-governed element in this project carries designed shear links even where nominal steel would satisfy IS 456, and the mat's link grid was added for exactly this reason (conflict C6, Part 9.3).

What is NOT demonstrated is the support rotation that $\mu = 5$ implies — $\theta \leq 2^\circ$ corresponds to $\delta \leq 87 \text{ mm} = \text{span}/57$ and requires a **non-linear SDOF check** (IS 4991 Fig. 6, or Biggs). **That is Phase 3 work and it is not claimed here.** See Part 17.

2.2.4 Why the plastic hinge mechanism, and why $wL^2/16$

Flexural design of the blast-governed elements uses the plastic mechanism moment for a fixed-fixed one-way strip:

$$M_p = w \cdot L_n^2 / 16$$

This is the collapse mechanism value, not the elastic value. An elastic fixed-fixed strip carries $wL^2/12$ at the supports and $wL^2/24$ at midspan. Once the support sections yield, they rotate at constant moment and the span redistributes until the midspan section also reaches its capacity; setting external work equal to internal work for the three-hinge mechanism gives a required capacity of $wL^2/16$ at every hinge, which is **less than the elastic peak of $wL^2/12$** .

This is only legitimate because the section is ductile enough to redistribute — the same promise $\mu = 5$ makes. The two are one assumption, checked once, and used consistently: the mechanism supplies the demand and $x_u/d = 0.139$ supplies the licence.

2.2.5 Strain-rate effects: why 25 % is added, and where it is not

Concrete and reinforcing steel are both **rate-sensitive**. Loaded in milliseconds rather than minutes, both show higher apparent strength — concrete because micro-crack growth is a time-dependent process that cannot keep pace, steel because dislocation motion at the yield plateau is similarly rate-limited. IS 4991 Cl. 10.3.1 codifies this as a flat 25 % increase for the blast case only:

```
BLAST CASE ONLY          IS 4991 Cl. 10.3.1
    fck,dyn = 1.25 x 35 = 43.75 N/mm2
    fy,dyn  = 1.25 x 500 = 625 N/mm2

NO DYNAMIC INCREASE ON SHEAR    IS 4991 Cl. 10.3.1.1
    every tau_c in this project is read at the STATIC M35 value
BOND    the clause permits +25 % (Ld 40phi -> 32phi). NOT TAKEN.
        all detailing uses the static 40phi
```

Two deliberate refusals, and both are conservative.

Shear gets no bonus, by the code's own rule. Diagonal tension failure is brittle and aggregate-interlock dependent, and the rate sensitivity of that mechanism is not the same as that of the compression block. The consequence is real and visible: the mat's one-way shear check, run at the static τ_c for the first time in this project, is what produced the requirement for a **T12 link grid at 250 × 250 throughout the mat** (conflict C6).

The 25 % bond bonus is available and is refused. Every development and lap length in this project is the static value. A 25 % shorter anchorage is not worth the argument at a construction joint that also has to be watertight and electrically continuous.

2.2.6 Why $\gamma = 1.0$ on the blast combination

Blast appears in combination 103 with a load factor of **1.0**, not 1.5. That is not a relaxation:

The blast case is an **extreme event** checked against **ultimate** capacity computed with **dynamic** material strengths. Applying a 1.5 partial factor to the load *while also* taking the 25 % material bonus would double-count the same conservatism from both ends. The project takes the material bonus and drops the load factor, states so, and does not mix the two.

IS 4991 Cl. 11.1 forbids combining wind or earthquake with blast, and Cl. 11.2 forbids live load on the roof at the time of blast. Both exclusions are applied (Part 7.9).

2.3 Ground shock, soil–structure interaction and arching

A buried structure and the soil around it are one system. Three effects matter, and this project's position on each is stated rather than assumed away:

Transmitted overpressure. Handled by K_a (2.1.4), taken at the saturated upper bound of 1.0.

Arching. A buried structure stiffer than the surrounding soil attracts load; one more flexible sheds it to the soil either side. Real buried boxes usually benefit from **positive arching**, which would reduce the roof pressure below the free-field value. **This project takes no arching credit at all.** That is conservative and it is declared, because the credit depends on the relative stiffness and on the compaction actually achieved, neither of which is measurable in advance.

Transient soil–structure interaction — participating mass, interface pressure ratios, the phasing between the ground shock arriving at the roof and at the walls — is **not demonstrated** and is listed in Part 17 as such. A static analysis with an SDOF-derived load factor gives the correct *demand*; it is not a transient interaction analysis and is never presented as one.

2.4 Nuclear radiation, and what 2.0 m of cover actually buys

2.4.1 Two different radiation problems

	Prompt radiation	Fallout
What it is	Gamma rays and neutrons emitted during the detonation itself	Gamma emission from activated debris deposited afterwards
When	Seconds	Hours to days
Spectrum	Hard; neutrons are the hard part to stop	Softer gamma
What stops it	Mass , and for neutrons hydrogen	Mass

2.4.2 Attenuation is exponential in mass, not in thickness

For a narrow beam through a uniform absorber, intensity falls as

```

I = I0 . exp( -mu . x )          mu = linear attenuation coefficient
                                x  = thickness

More useful in shielding practice, the same law written with MASS per unit
area, because it is areal density that does the work:

I = I0 . exp( -(mu/rho) . rho.x )

TENTH VALUE LAYER  TVL = ln(10) / mu  -- the thickness that cuts the dose
                                by a factor of 10.
```

The practical consequence for a buried shelter is that **what protects the occupants is the mass above them, and it has to be above the slab.** The same tonnage placed against a wall does nothing for the roof.

For neutrons the mechanism is different and it matters: fast neutrons are slowed most efficiently by **elastic collision with nuclei of similar mass**, which means hydrogen. Moisture in soil and the chemically bound water in concrete are genuinely useful; dense dry rock is less effective per unit mass against neutrons than against gamma.

2.4.3 Why 2.0 m, and what each metre is for

Requirement	Does it need cover?	Verdict
Blast	No. A buried roof takes the full p_{so} regardless of depth — IS 4991 Cl. 7.2, $K_a = 1.0$	Cover buys nothing here
Fallout	1.0 m already gives a protection factor of $\approx 2\,200$ against a requirement of $\sim 1\,000$ [C]	The first metre covers it, twice over
Prompt neutron and gamma	Needs mass, above the slab	This is what the second metre is for

Stated plainly: the second metre of cover is bought entirely for prompt neutron and gamma attenuation. Reducing the cover from 4.0 m to 2.0 m in an earlier revision saved 2 m of rock excavation, 2 m of shaft, one stair flight and 2 m of headroom, and cost nothing in blast capacity because there was nothing to lose.

And the honest caveat. The prompt-radiation requirement scales with the yield, which is [N] and deliberately so (2.1.3). **If the design basis threat is ever stated and is larger, the cover depth is the dimension that changes** — along with the burster slab. The project records this rather than pretending the depth is threat-independent.

2.4.4 The burster slab

The 200 mm M30 slab inside the cover, reinforced T12 @ 150 both ways, is not there to stop anything. It is there to **break up a penetrating item** so that its energy is dispersed in the 500 mm of crushed basalt rubble below it rather than delivered to the roof as a point load.

It breaks up a penetrating item; it does not defeat one. No direct hit is designed for, and that is now an explicit owner ruling (ruling 5, RC4): *the design stands as it is*. If a direct hit ever becomes a requirement, both the cover depth and the burster design change.

2.5 Groundwater, buoyancy and the compensated foundation

2.5.1 Archimedes, applied to a buried box

A structure below the water table displaces water and is pushed up by the weight of the water it displaces. The uplift is the water pressure at the underside integrated over the plan area:

$$\begin{array}{llll}
 u & = & \gamma_w \times h_w & = 9.81 \times 4.700 & = 46.11 \text{ kPa} \\
 A & = & 22.000 \times 6.200 & & = 136.40 \text{ m}^2 \\
 U & = & 46.11 \times 136.40 & & = 6289 \text{ kN}
 \end{array}$$

Two points are easy to get wrong and are got right here:

1. **The uplift acts on the real underside, not on a model's mid-surface area.** A plate model of the mat has its elements at mid-depth; using that area under-states the force.
2. **Spring supports in a linear analysis take tension.** In the elastic-foundation model the mat therefore **never lifts, whatever the load case**. Flotation cannot be demonstrated — or refuted — by that model. It is a hand check on real dimensions and real weights (Part 9.3).

Anyone who says "the analysis shows no uplift" has misunderstood their own model.

2.5.2 Why the construction stage governs

Buoyancy competes with weight, and the weight arrives late. Before the roof is on and the cover is placed, the box is a light, closed vessel in a wet hole:

Stage	Weight	Uplift	FoS at GWT (–)2.000
1 — mat cast only	2 009 kN	6 175 kN	0.33 FAIL
2 — mat + walls, no roof	4 802 kN	6 175 kN	0.78 FAIL
3 — box complete, no backfill	7 528 kN	6 175 kN	1.22 MARGINAL
4 — backfilled and covered	12 453 kN	6 175 kN	2.02 OK

(The absolute figures are at the pre-M1 21.600 m box; both uplift and resistance scale with length, so every factor of safety is unchanged. The hand check uses the current $136.40 \text{ m}^2 / 6\,289 \text{ kN}$ — Part 9.3.)

Flotation, not bearing and not settlement, is the governing foundation problem — and it governs during construction, not in service. That is why the mitigation in Part 9.3 is a **design output** with four mandatory items, not a note passed to the contractor.

2.5.3 The compensated foundation, and why "settlement negligible" is true

The excavation removes more ground than the structure puts back:

overburden removed, 1.13 m soil at 19.5 kN/m ³	= 22.10 kPa
overburden removed, 5.67 m basalt at 25.0 kN/m ³	= 141.67 kPa
TOTAL GROSS PRESSURE REMOVED AT FORMATION	= 163.77 kPa
replaced by, in service	= 58.40 kPa
NET CHANGE IN PRESSURE AT FORMATION	= -105.37 kPa

In service the structure is lighter than the ground it replaces, by about 105 kPa. It is a fully compensated foundation and then some. **This is the physical reason "settlement negligible" is true rather than merely hoped for** — and it is the same fact, seen from the other side, that makes flotation the governing case. One physical observation explains both conclusions.

2.5.4 Effective stress, and why water is two-thirds of the wall load

Below the water table, soil and water carry the lateral load separately. The soil skeleton transmits its share through grain contacts at the **submerged** unit weight; the water transmits its own hydrostatic pressure, in full, in every direction:

$\gamma' = \gamma_{sat} - \gamma_w = 21 - 9.81$	= 11.19 kN/m ³
$K_0 = 1 - \sin(\phi)$ (Jaky), $\phi \sim 30$ deg	= 0.50
lateral gradient = $K_0 \cdot \gamma' + \gamma_w$	= 0.50 × 11.19 + 9.81
	= 15.41 kPa/m
of which SOIL = 5.60 kPa/m	WATER = 9.81 kPa/m

Water is 64 % of the lateral gradient and soil is 36 %. That single split is why the design groundwater table is called the most important number in the project, and why the master tags it [ASSUMED] in bold. K_0 uses **Jaky's** relation for a normally consolidated soil at rest — at-rest, not active, because a stiff buried box does not move enough to mobilise active pressure.

2.6 The modulus of subgrade reaction, and why both bounds must be run

A mat on ground is modelled as a plate on a **Winkler** foundation: a bed of independent springs of stiffness k_s (kN/m³ — pressure per unit settlement). It is a crude idealisation — real soil couples adjacent points, which springs do not — but it is the standard one, and its weakness is specific and known:

k_s **does not change the total load; it changes how the load is distributed.** A soft bed lets the mat dish and drives moment into the span. A stiff bed concentrates reaction under the stiff lines — walls, corners — and drives moment there instead. **The two bounds do not bracket a small error; they can put the peak moment in different places.**

	Value	Class
k_s	100 000 to 500 000 kN/m ³	[A] — run the mat model at BOTH bounds

The project has built the upper-bound model (Underground_Shelter_ks500000.std, differing from the reference model **only** in the subgrade line and the four corner K_FY values, verified by diff). **Neither bound has been run, because STAAD.Pro is not available in this environment.** k_s remains [ASSUMED] at both bounds, and a plate load test is the only thing that closes it.

2.7 CBRN: keeping the outside outside

2.7.1 Overpressure is the primary barrier

A gas-tight envelope leaks. The defence is not perfect sealing but **positive pressure**: hold the inside above ambient, and any leak flows outward.

	Value	Class
Operating overpressure	+50 to +100 Pa	[C]
Leak-test pressure	+300 Pa	[C]
Permitted leakage	≤ 0.15 vol/h at +300 Pa = 32.5 m ³ /h = 11 % of one train	[C] / [R]
Cascade, clean to dirty	0 → +10 → +20 → +35 → +50 Pa	[C]

The cascade maps exactly onto the three decon airlock stages, which is what makes a three-stage airlock work: at every door, air moves from cleaner to dirtier and never the reverse.

A consequence that is easy to miss: every water trap inside the envelope is a pressure boundary. A seal shallower than the overpressure blows through and the shelter vents to the drain. Part 14.2 shows the sizing — 75 mm deep seals hold **736 Pa**, which is **7.4 : 1** against the operating pressure and **2.5 : 1** against the test. **And an unused gully evaporates dry**, after which it leaks air in both directions; priming is therefore a written maintenance task, not a nicety.

2.7.2 Filtration: three different physical problems

Threat	Physical form	What removes it
Radioactive particulate, biological agents	Aerosol, ~0.1–10 µm	HEPA, EN 1822 class H14 , ≥ 99.995 % at MPPS
Chemical warfare agents	Vapour	Impregnated activated carbon (ASZM-TEDA)
Coarse dust, the blast pulse	—	Pre-filter; blast valves

A HEPA filter does not remove a vapour and a carbon bed does not remove a particle. They are in series because they solve different problems. HEPA performance is quoted at the **most penetrating particle size**, which is *not* the smallest: below about 0.3 µm, diffusion capture improves as particles get smaller, while above it interception and impaction take over. The minimum in between is the number the standard quotes, which is why an H14 rating is a stronger statement than "99.995 % of dust".

Carbon beds self-heat. Adsorption is exothermic and a loaded bed is a genuine fire hazard in a protected ventilation system — which is why the filter bay appears in Part 14.5's fire-load schedule rather than being treated as inert plant.

2.7.3 Blast valves: protecting the filters from the shock

The filter train cannot take 383 kPa. **Blast valves** are fast-acting closures on each envelope penetration that shut on the pressure rise itself and reopen afterwards. This project carries **five**, all confirmed on the issued services sheet, recessed to IS 4991 Cl. 6.2.1, each with a manual quarter-turn gas-tight damper inboard of it, and with a **hand crank on each filter fan** so that ventilation does not depend on electrical power at all.

2.7.4 Closed mode: the arithmetic of a sealed box

With every valve shut there is no fresh air. Two consumables then run the clock:

```

CO2 ACCUMULATION
allowable rise 0.04 % -> 1.0 %           = 0.0096 volume fraction
occupied volume, airlock shut             = 184.96 m3
9 occupants x 0.020 m3/h/person           = 0.18 m3/h of CO2
time to 1.0 % = 0.0096 x 184.96 / 0.18   = 9.9 h [C]
over 96 h closed                          = 17.28 m3 of CO2

OXYGEN
2 x 50 L at 150 bar = 15 m3 at 4.5 m3/day = 80 h [C]

SODA LIME (CO2 scrubbing) 40 kg store     = 48 h [C]
```

HV-F1 — the soda lime, not the oxygen, limits closed mode. Unscrubbed CO₂ gives 9.9 h; the soda-lime store gives 48 h; the oxygen store gives 80 h. **The binding consumable is the one to count and the one to write on the drill card.** Closed mode covers 48 of the 96 h endurance — **the other 48 h require filtration.**

And the scrubber is therefore not optional. Without it the envelope reaches 1.0 % CO₂ in 9.9 hours against a 96-hour design occupancy. That ratio is the whole argument. Owner ruling 25 (RC7) sized the recirculation loop at **75 m³/h** — chosen at the worst cell of the efficiency / target matrix because designing to it costs almost nothing (Part 14.3).

2.7.5 The decon airlock, and the manning constraint hiding in it

The three-stage airlock has to be purged between users:

```

volume      5 x (2.0 x 2.0 x 3.2)         = 64 m3
purge at the design flow 300 m3/h         = 12.8 min [R]
-> 4-5 persons per hour
```

That is an operational constraint, not a ventilation figure. Four to five people per hour is the maximum rate at which anyone can enter the shelter through a working decon procedure, and it belongs on the drill card. The master carries it as a confirmed item *requiring attention* for exactly that reason.

2.8 EMP: why concrete and rebar are not a shield

2.8.1 What HEMP is, and what it is not

A high-altitude nuclear detonation produces three distinct electromagnetic components:

Component	Time scale	What it couples to
E1	nanoseconds, very fast rise	Short conductors — cables, boxes, PCBs. The component that destroys electronics
E2	microseconds to milliseconds	Similar to lightning; conventional surge protection is relevant
E3	seconds	Long conductors — power lines, pipelines. Quasi-DC induced currents

HEMP is not a personnel hazard. The occupants are protected from blast, CBRN and fallout by the box. **EMP protection exists so that the shelter can still function afterwards.** It protects equipment. That is why the answer is an enclosure and not a room, and why nobody should expect to shelter inside it.

2.8.2 The requirement, and the aperture arithmetic that defeats the concrete

REQUIREMENT MIL-STD-188-125-1: 80 dB, 10 kHz to 1 GHz [C]
that is FIVE DECADES of frequency

A conducting screen pierced by a regular array of apertures behaves as a high-pass filter. For apertures of pitch s the classical single-aperture estimate is

$$SE = 20 \cdot \log_{10}(\lambda / 2s) \quad \lambda = c / f$$

Applied to this project's reinforcement cage, whose bar spacing is **150 mm both curtains both ways** — itself an EMP requirement, stricter than IS 456 Cl. 26.3.3 needs:

Frequency	SE of the cage
10 kHz	99.99 dB
100 kHz	79.99 dB
1 MHz	59.99 dB
100 MHz	19.99 dB
1 GHz	0.00 dB
Mesh cutoff $c / 2s$	999.31 MHz
Highest frequency meeting 80 dB	99.93 kHz
Fraction of the required band met	one decade of five — 20 %

And a coincidence worth saying out loud: $2s = 300$ mm, and the wavelength at 1 GHz is 299.79 mm. The cage stops shielding at almost exactly the frequency at which the standard stops asking. That is arithmetic, not design.

So the cage is a genuine low-frequency measure and must never be described as a MIL-STD-188-125-1 boundary. It earns its keep where it can: E3, the slow component that drives long conductors, lives at the bottom of the band where the cage is strongest, and 99.99 dB at 10 kHz is not nothing.

The second, independent reason is simply the holes: a 900 mm slab with a **2 800 × 3 160 stair void** in it — 8.85 m² — is not a shield at any frequency, and neither are two 1 400 mm escape shafts.

2.8.3 Waveguide below cutoff — the one place geometry works for you

A conducting tube of diameter d will not propagate below its TE₁₁ cutoff:

$$f_c(\text{TE}_{11}) = 1.8412 \cdot c / (\pi \cdot d)$$

and BELOW cutoff the attenuation is roughly

$$A = 32 \cdot L / d \quad \text{dB} \quad (L = \text{tube length, } d = \text{diameter})$$

A penetration bounded by metal and long relative to its bore is therefore a very effective attenuator. This is why a small pipe through a thick wall is not a problem and a large one is:

Penetration	Bore	Wall	Bounded by	Cutoff	WBC	Verdict
BV-1 / BV-2	100 Ø	600	steel	1 757 MHz	192 dB	PASS
BV-3	100 Ø	400	steel	1 757 MHz	128 dB	PASS

Penetration	Bore	Wall	Bounded by	Cutoff	WBC	Verdict
BV-4 / BV-5	350 Ø	600	steel	502 MHz	54.8 dB	FAIL, both criteria
PD-05	50 Ø	600	steel	3 514 MHz	384 dB	PASS
ESC 1 / ESC 2	1 400 Ø	3 050 / 3 600	concrete	126 MHz	none	FAIL
Stair void	3 160	900	concrete	47.4 MHz	none	FAIL

Read the three-figure numbers correctly. No practical penetration achieves 192 or 384 dB. Real assemblies flatten out around 100–120 dB and what limits them is **workmanship** — the bond at the frame, the gasket, the one sleeve nobody welded. A large number means *"the bore is not the problem here."*

A depth credit is taken only where the bore is bounded by metal. Concrete is a lossy dielectric, not a waveguide wall. The escape shafts' 250 mm collars carry five T25 bars each side — five bars is not a conducting tube, so as built they earn nothing.

Lining a 1 400 mm shaft does not fix it: above 125.5 MHz it propagates however it is lined. The treatment that works is a **bonded conducting hatch at the head**, which *terminates* the shaft instead of trying to attenuate down it.

2.8.4 Bonding inductance — where EMP design actually lives

A bond strap is not a resistor; at EMP frequencies it is an inductor, and its impedance rises linearly with frequency:

$$L = 2e-7 \cdot l \cdot [\ln(2l / (w+t)) + 0.5 + 0.2235(w+t)/l] \quad \text{henries}$$

$$X = 2 \cdot \pi \cdot f \cdot L$$

Strap	L	X at 1 MHz	X at 10 MHz	X at 100 MHz
600 mm × 25 × 3	512.2 nH	3.22 Ω	32.2 Ω	321.8 Ω
100 mm × 50 × 3	38.9 nH	0.24 Ω	2.45 Ω	24.5 Ω

A 600 mm strap is 322 Ω at 100 MHz. That is not a bond; it is a resistor with a nice green sleeve on it. Shortening it to 100 mm and widening it to 50 mm takes the inductance from 512 nH to 39 nH — a factor of **13.2** — and it costs nothing.

The five rules that follow are the ones that get built wrong: every bond ≤ 100 mm long; width-to-length at least 5 : 1; **flat strap only** — never a round wire, never a pigtail, never *"loop it round to the nearest stud"*; clean bare metal both ends, protected after making off; and **the shield bonds to the structure at ONE place**, because a second bond is a loop and a loop is an antenna.

2.8.5 Earthing — and why 5 Ω is the wrong target to chase

Electrode	ρ = 1 000 Ω·m	ρ = 10 000 Ω·m
One 3 m × 16 mm rod	335 Ω	3 349 Ω
Rods needed for 5 Ω, <i>ignoring interaction</i>	67	670
The structure itself — 136.4 m ² , r _{eq} 6.589 m, p/4r	38 Ω	379 Ω

Deccan basalt is **1 000 – 10 000 Ω·m** [C].

≤ 5 Ω is not achievable with rods here, and it is not an EMP number in any case. It is a power-safety and lightning requirement from IS 3043 / IEEE 142; it is real and it still applies. **But an EMP shield works by being equipotential, and equipotential is decided by bonding inductance, not by earth resistance.** The mat and its cage are already a large concrete-encased electrode, an order of magnitude better than a rod and costing nothing because it is already built. **Bond to the structure; do not chase rods.**

2.9 Seismic response, and why a buried box is the easy case

Seismic design force follows from a design spectrum reduced by a response reduction factor:

$$A_h = (Z/2) \cdot (I/R) \cdot (S_a/g)$$

$$V_b = A_h \cdot W$$

IS 1893 (Pt 1) Cl. 6.4.2
Cl. 7.6.3

	Underground box	Sentry post
Zone factor Z (Zone III)	0.16	0.16
Importance I	1.5	1.5
Response reduction R	4.0	3.0
S _a /g	2.5	2.5

	Underground box	Sentry post
A_h	0.075	0.100
Seismic weight W	14 002 kN	731.80 kN (model)
Base shear V_b	1 050 kN	73.18 kN

For the buried box, seismic is negligible and the calculation says so numerically, not by assertion: 525 kN per long wall gives $\tau = 0.063$ N/mm², and IS 13920 Cl. 10.4 boundary elements are checked and **not triggered**. A buried box moves with the ground; there is no tall, flexible mass to excite.

For the sentry post seismic governs over wind by **2.4 : 1** (73.18 kN against 29.9 kN), and the choice of $R = 3.0$ is the conservative one:

$R = 3.0$ is taken **because the infill panels are not positively separated from the frame**. $R = 5.0$ would require a special moment-resisting frame with separated infill, which this project does not claim. The consequence is quantified: $R = 5.0$ would give $A_h = 0.060$ against 0.100, so the design base shear of **73.18 kN is about 1.67 × what a separated SMRF would need**. Owner ruling 21 (RC6) keeps $R = 3.0$ on exactly that basis.

2.10 Fire in a volume that cannot be ventilated

Every conventional fire strategy assumes smoke can be got out. **Here it cannot.**

Internal volume	332.8 m³ gross
Design air supply	300 m³/h
Air changes	≈ 0.9 per hour
Smoke extract	NONE — and there cannot be one. Every opening in the envelope is a blast valve or a blast door; a smoke vent would be a hole in the protective boundary

In **Mode 3 CLOSED** the shelter has **zero** air exchange, and opening a valve to clear smoke breaks the protection the closed mode exists to provide.

So the strategy is not the usual one. It is: prevent, detect early, extinguish while it is small — and if it cannot be extinguished, LEAVE, because you cannot wait it out. Part 14.5 sets out the routes and the decision rule, and records the five findings a fire plan in this situation is obliged to raise.

2.11 The thermal problem of a sealed box

Nine people, lighting, plant and a dehumidifier in a sealed 332.8 m³ volume put heat in and have nowhere to put it. The 96-hour balance was computed under owner ruling 10 (RC4):

```

electrical load dissipated INSIDE the envelope      = 5.733 kW
9 occupants at 70 W sensible                       = 0.630 kW
TOTAL SENSIBLE GAIN                                = 6.363 kW
over 96 h = 6.363 x 345600 s                        = 2.199 GJ

BOUND 1  adiabatic, AIR ONLY: 216.96 m3 x 1.2 x 1.005 = 261.7 kJ/K
          dT = 8404 K -- PHYSICALLY MEANINGLESS.
          THE AIR HOLDS NOTHING; THE STRUCTURE IS THE ENTIRE STORY.
BOUND 2  air + the concrete the heat can REACH in 96 h
          thermal penetration sqrt(alpha.t) = sqrt(5.5e-7 x 345600) = 0.436 m
          414.38 m2 of wetted concrete surface, responding depth 0.30 m [A]
          124.3 m3 x 2500 x 0.88                  = 273 493 kJ/K
          dT                                       = 8.03 K
BOUND 3  the air-to-surface film, h = 3.0 W/m2K [A]
          dT_film = 6363 / (3.0 x 414.38)          = 5.12 K

AIR TEMPERATURE RISE OVER 96 h = 8.03 + 5.12      = 13.2 K
from a ground temperature of 26 degC at 6 m depth [A]
-> about 39 degC by hour 96, AND STILL RISING, because nothing has
    reached equilibrium

COOLING DUTY TO HOLD 30 degC
the structure can absorb 273 493 x 4 K             = 1.094 GJ
heat to be rejected 2.199 - 1.094                 = 1.105 GJ
mean sensible duty 1 105 079 / 345 600            = 3.20 kW
plus latent                                         = 0.405 kW

```

TOTAL MEAN DUTY 3.60 kW = 1.02 TR
 ADOPT 4 kW (1.14 TR) as the closed-mode cooling duty [R]

Ruling 17 (RC5) then directed that the heat be rejected **to the ventilation air in open mode only, adding no new penetration** — and the arithmetic of that ruling is the finding:

ventilation 300 m³/h → 0.1005 kW/K → 63.3 K needed to reject 6.363 kW
 at a realistic 5 K difference, 300 m³/h → 0.50 kW = 7.9 % of the gain

RC5-F1 — **300 m³/h is a CONTAMINANT rate, not a heat-rejection rate**, and it is about an order of magnitude short. The second train is standby, not simultaneous, so 600 m³/h is not a case the design contemplates. **And in Pune it can be worse than nothing**: whenever ambient exceeds the internal temperature, ventilating *adds* sensible heat. **No ambient design temperature exists anywhere in this project** [N], so the number of hours cannot be stated — only the direction.

The ruling is therefore recorded as an acceptance of the ≈ 39 °C condition, not a solution to it. The structure and the surrounding rock carry essentially the whole load in both modes. RC4-V1 is narrowed, not closed: from "*no rejection path exists*" to "**the rejection path is the ground through the structure, and it has not been modelled.**" A transient soil-structure **thermal** model is the missing piece — the sibling of the transient soil-structure **interaction** already deferred to Phase 3.

2.12 What the science section does not supply

Not supplied	Why, and where it is recorded
Weapon yield	[N], and ruled to stay unstated — the overpressure and duration are the design inputs. Ruling 6, RC4
Negative-phase and rebound load case	Not designed as a separate case; handled physically at the hatches. Part 17
Support rotation / non-linear SDOF	Phase 3. Part 17
Shock propagation down the entry shaft, and the resulting door loading	Needs CFD or a shock tube. Vendor / specialist. Part 17
Transient soil-structure interaction	Not demonstrated. Part 17
Transient soil-structure thermal model	RC4-V1, open
Ambient design temperature	[N] — nowhere in the project
Incident EMP field, waveform, E1/E2/E3 decomposition	[N]. The EMP work is designed to the 80 dB performance requirement , which needs none of them
Concrete permittivity / conductivity; whether rebar crossings are tied or welded	[N] — and the second materially changes 2.8.2

PART 3 — CODES, STANDARDS AND REFERENCES

3.1 The governing caveat, stated before anything else

IS 4991:1968 Cl. 1.1 expressly EXCLUDES nuclear explosions from its scope.

It is used in this project **for its blast loading rules and its dynamic material strengths only**, as a documented and conservative extrapolation. **Every structural capacity in this project is computed to IS 456:2000**, with IS 13920:2016 for the ductile detailing of the sentry post frame.

A conventional static STAAD.Pro analysis gives the correct DEMAND. It is not, and must never be presented as, proof of blast resistance.

This caveat appears on every deliverable in the project, and it appears here first because a reader who does not have it cannot correctly weigh anything in Parts 7 to 11.

3.2 Code register — with the clauses actually relied upon

No clause number in this register was invented. Where a clause could not be confirmed from the material available it is not listed. A code named **by title only** is one the project uses for its existence, not for a quotation; **no clause is ever quoted from a document that is not in the workspace.**

Code / reference	Used for	Clauses relied upon
IS 456:2000	All reinforced concrete design	Cl. 6.2.3.1 (E _c); Cl. 13.4 (construction joints); Cl. 22.2(a) (effective span); Cl. 23.2.1 + Fig. 4 (deflection); Cl. 24.4, 24.5 + Fig. 7 (two-way slabs, load to beams); Cl. 25.1.2 (short column); Cl. 25.4 (minimum eccentricity); Cl. 26.2.1, 26.2.1.1, 26.2.5.1 (L _d , laps); Cl. 26.3.3 (bar spacing); Cl. 26.4.2, 26.4.2.1, 26.4.2.2 + Table 16 (cover); Cl. 26.5.1.1, 26.5.1.2, 26.5.1.5, 26.5.1.6 (beam steel, links); Cl. 26.5.2.1, 26.5.2.2 (slab steel); Cl. 26.5.3.1, 26.5.3.2 (column steel, ties); Cl. 31.6, 31.6.1, 31.6.3.1 (punching); Cl. 32.2, 32.5(a)(b)(c) (walls); Cl. 33.1(b), 33.2 (stairs); Cl. 34.2.4.1 (footing shear); Cl. 36.4.2 (γ _m); Cl. 38.1, 38.1(c), 38.1(f) (limit state, stress block, x _{u,max}); Cl. 39.1, 39.3, 39.6 (columns, biaxial); Cl. 40.1, 40.2.1.1, 40.2.3, 40.4(a) (shear); Tables 3, 5, 16, 18, 19, 20, 26, 27, 28; Annex D (D-1.1 to D-1.8, D-2); Annex G-1.1(b), G-1.1(c)
IS 875 (Part 1):1987	Dead loads, unit weights	Table 1
IS 875 (Part 2):1987	Imposed loads	Plant / storage 5.0 kPa; office / observation post 3.0; accessible roof 1.5
IS 875 (Part 3):2015	Wind — sentry post only	Cl. 6.3, 6.3.4, 7.2, 7.2.1, 7.3.3.13, 7.4; Tables 1, 2, 4, 26; Annex A (V _b = 39 m/s)
IS 875 (Part 5):1987	Load combinations	with IS 456 Table 18
IS 1893 (Part 1):2016	Seismic	Cl. 6.3.1.2, 6.3.2.2 (combinations, directional); Cl. 6.4.2 + Fig. 2 (A _h , S _a /g); Cl. 7.2.3 (I), 7.2.6 (R), 7.3.1, 7.3.2 (seismic weight); Cl. 7.6.1, 7.6.2, 7.6.2(c), 7.6.3 (period, base shear, distribution); Cl. 7.11.1 (drift); Table 3 (Z)
IS 13920:2016	Ductile detailing — sentry post	Cl. 6.1.1, 6.1.2, 6.1.3 (beam geometry); Cl. 6.2.1(b), 6.2.2, 6.2.3, 6.2.4 (beam steel); Cl. 6.3.3, 6.3.4 , 6.3.5.1, 6.3.5.2 (capacity-design shear, hoops); Cl. 7.1, 7.2.1 (strong column-weak beam), 7.3, 7.4 (columns); Cl. 8.1, 8.2 (special confining reinforcement); Cl. 10.4 (boundary elements — checked, not triggered)
IS 3370 (Parts 1, 2):2021	Crack width, surface-zone steel	0.2 mm limit; 0.35 % surface-zone steel over a 250 mm zone
IS 1786:2008	Fe500D reinforcement	D grade — guaranteed minimum elongation
IS 1904:1986	Presumptive bearing capacity	Table 1 (hard rock, 3 240 kPa)
IS 2950 (Part 1):1981	Raft foundations	General provisions
IS 12070	Rock foundations	Cl. 6 (settlement on sound rock); Table 2 (broken bedrock, 10 kg/cm ² — the SEMT report's own basis)
IS 1498:1970	Soil classification	CH, GM, GP, SC; free-swell-index bands — the SEMT report's classification basis
IS 2720 Pts IV, VIII, X, XIII, XL	Soil testing	Classification (Pt IV); compaction, OMC-MDD (Pt VIII); UCS (Pt X); direct shear (Pt XIII); free swell (Pt XL) — the SEMT report's own test methods, cited as it cites them
IS 2720 (Part 28)	In-situ density	Backfill compaction testing, WBS A7015
IS 1121 (Part I)	Rock strength	Compressive strength of rock cores — the SEMT report's Appendix C basis. (<i>The report prints the year as "1874"; that is a typographical error, transcribed as printed, not corrected</i>)
IS 2470 (Part 1):1985	Septic tank	Cl. 6.2 (24 h detention), 6.3 (30 L/person/yr sludge), 6.5 (L:B 2–4), 6.6 (min width 750, min depth 1.0 m), 6.9 (vent); Table 1 (≤ 10 users: 1.5 × 0.75 × 1.0)

Code / reference	Used for	Clauses relied upon
IS 2470 (Part 2):1985	Soak pit / dispersion	Cl. 4 (percolation test — mandatory), Cl. 5 (dispersion trench)
IS 4991:1968	Blast loading rules and dynamic material strengths ONLY	Cl. 1.1 (EXCLUDES NUCLEAR — quoted in every deliverable) ; Cl. 6.2.1 (reflected pressure, recessing); Cl. 7.2 + Table 3 (buried roof / walls, K_2); Cl. 7.4 (drag on bermed faces); Cl. 10.3.1 (dynamic f_{ck}, f_y); Cl. 10.3.1.1 (no dynamic increase on shear; +25 % bond) ; Cl. 10.3.3 (DLF = $\mu/(\mu-0.5)$); Cl. 11.1 (no wind / EQ with blast); Cl. 11.2 (no live load on the roof at blast); Fig. 6 (SDOF — Phase 3)
IS 1077	Common burnt clay bricks	Sentry post masonry infill, 190 × 90 × 90 modular, SP-B1
IS 732:2019	Electrical wiring installations	EL1
IS 3043:2018 / IEEE 142	Earthing	≤ 5 Ω target — a power-safety and lightning number, not an EMP one
IS 694:2010 · IS 1554 (Pt 1):1988	Cables	Final circuits; sub-mains and generator cabling
IS 13416	Construction-phase fire safety	In the register for exactly that
SP 16:1980	Design aids	Column interaction charts — cross-checked only ; the P-M values in Part 11.5 were computed from first principles, not read off
SP 34:1987	Detailing	Shape codes; Cl. 5.5 (opening-corner detailing)
NBC 2016 Part 4	Fire and life safety	Stair geometry, 1 100 guarding, means of escape
BS 8666	Bar shape codes	Used alongside SP 34 in every bar bending schedule
UFC 3-340-02	US criteria, cited as US criteria	§4-27 (opening trimmers); §4-30 (direct shear)
UFC 3-350-04AN	US criteria	Referenced in the Phase 1 report
MIL-STD-188-125-1	EMP	§5.4 (access), §5.5 (waveguide below cutoff), §5.7.2.1 (power PCI), §5.7.4.1 (fibre), §5.7.6 (RF); 80 dB, 10 kHz – 1 GHz
IEEE Std 299	EMP verification	Shielding effectiveness survey
FEMA 453	Shelter ventilation	0.25 cfm/ft ² ; carbon adsorber performance
EN 1822	HEPA classification	H14, ≥ 99.995 % at MPPS
Glasstone & Dolan	Weapons effects	Unclassified, public — blast and radiation data only
Biggs, <i>Structural Dynamics</i>	SDOF method	Referenced for the Phase 3 support-rotation check

3.3 Codes named by title only

These are named because the work touches their subject, and **no clause is quoted from any of them**, because no copy is in the workspace:

IS 1742 · IS 5329 · NBC 2016 Part 9 (drainage) · IS 3103 and the ISHRAE / ASHRAE handbooks (ventilation) · IS 3696 (Part 2) (ladders) · IS 3764 (excavation safety and slopes) · IS 2502 (bar bending — **explicitly NOT held and NOT cited**, which is why the schedules take no bend deduction and say so).

3.4 The hierarchy used to resolve a conflict

Ten conflicts were closed in this project and a further set of rulings settled everything that could be settled on the evidence. The precedence that produced those rulings is stated here because it is itself part of the design record:

1. **A drawing beats an older document.** Precedent **C1**: the drawings are Rev F, the design report is Rev D; where they disagree, the drawing governs.
2. **Arithmetic beats a transcription.** Precedent **C18**: $(-)8.000 - (-)7.600 = 0.400$, so the sump base is 400 thick and the "300" in a sheet note is a transcription error.
3. **The .std file beats a screenshot of it.** Part D of the master was written from nineteen screen captures; six .std files are now in the workspace, and the files are primary.
4. **A value reproduced from independent lines beats a lone outlier.** Precedent **C21**: nine statements of 300 m³/h against one of 250, with 250 failing the sheet's own FEMA criterion.
5. **Where nothing in the project can decide it, it stays open.** No amount of choosing between recorded values supplies a datum, a test result or a client decision.

PART 4 — SITE SELECTION AND GEOTECHNICS

4.1 The site, and what is and is not known about it

Item	Value	Class
Installation	CTW / College of Military Engineering campus, Pune, Maharashtra	[C]
Plot	Undeveloped rectangle east of the two CTW blocks , west of the perimeter track and nullah line, CBRN Live Training Area to the south	[C]
Latitude / longitude	NOT ESTABLISHED	[N]
Plot boundary and area	NOT DIMENSIONED ANYWHERE. No scale bar, grid, boundary dimension or coordinate appears on any supplied site image	[N] — master gap D3
Site benchmark	NONE. Nothing ties any supplied level to the project datum	[N]
Project datum	finished grade = 0.000 , local	[C]
Site orientation	+X = EAST, +Y = NORTH	[A] — fixed for the first time at SG2

"No site plan" is not closed and cannot be closed by imagery. Until a boundary, a dimension, a benchmark and a coordinate exist, the berm, the hardstanding, the access, the five external drainage runs, the concealment layout and the cut-and-fill cannot be drawn. What the site package added is that the **setting** is now recorded, which it was not.

4.1.1 The selection, and what it got right

The site was chosen at the P1 stage. Its recorded SWOT, verbatim: **Strengths** — proximity to FCBRNP · availability of electric lines · good road connectivity · good water supply · **no pipelines**. **Weakness** — *located near perimeter fence*. **Opportunities** — training value. **Threat** — interference with the CTW schedule.

Three of those strengths are load-bearing for the design that followed, and none had been written down as such:

1. **"No pipelines."** For a 22 × 6.2 m box excavated to (–)6.800 with a continuous gas-tight, EMP-bonded envelope, a buried main crossing the footprint would have been a first-order problem: **every envelope penetration is a blast, gas and EMP discontinuity**. A clear plot is worth more here than on an ordinary building.
2. **"Availability of electric lines."** The only other record of an incoming mains supply anywhere in the project is a line in the owner's own construction schedule. This slide is earlier. It gives no capacity — **EL-V4** — but it confirms the supply exists.
3. **"Good road connectivity."** The works programme carries **994 m³ of rock excavation by hydraulic breaker with no blasting** and tipper haulage throughout. That is a haul-route requirement and the site meets it.

4.1.2 What the selection did not consider

Not considered	What it costs, quantified
Ground conditions on the plot itself	The selection rests on a soil report for three <i>other</i> buildings. The confirmatory investigation A1075 (12 d, critical path) must be located on this plot — SG-V1
Depth of investigation	Nothing is known below about 1.5 m; the structure founds at (–)6.800 — SG-V2, the governing item
Ground slope	The plot is on the edge of a slope falling east. No cut-and-fill item exists in the bill because no site plan exists
"Near perimeter fence"	It is also a concealment weakness, and the distance has never been dimensioned — SG2-V2, CAM-V2

4.2 The sub-soil investigation, and the one fact that governs it

#	Document	What it is	Class
1	SEMT/67/15 — <i>Report on Sub-Soil Investigation for CTW Ph-III ACCN Project at CME Pune</i> , Soil Engineering & Material Testing Wing, College of Military Engineering. Raised on CTW letter 8722/Ph-III/CTW/116/Q dated 20 May 2015	A real investigation: 11 trial pits at 3 locations , Appendices A (classification), B (soil test results), C (rock cores)	[C]
2	P1 presentation deck — <i>Construction of CBRN Hardened Underground Ops Room</i> , 32 slides	The project's own first-phase presentation: location, contour, watershed, pipelines, elevation profile, SWOT, wind, precipitation, temperature, seismic, soil	[C] as a record of what was presented

4.2.1 It is not an investigation of this plot

The SEMT report investigates **G Building (TP-1 to TP-4)**, **H Building (TP-5 to TP-8)** and the **Mess Building (TP-9 to TP-11)**. The project plot is a separate undeveloped area east of the CTW blocks. The report may be carried across only on the strength of its own paragraph 5:

"The area under investigation is underlain by the Deccan Trap formation. These are volcanic lava flows, which were poured out between the late cretaceous and early Eocene times. These basaltic flows are horizontally bedded and more or less uniform in character over a wide area."

That carry-across is an assumption [A], not a finding of the report. And the report itself shows how much variation "uniform over a wide area" permits: across three locations a few hundred metres apart, rockhead moved from **0.9 m to 1.5 m** and the surface stratum changed from a 1.0 m high-plasticity clay to a 0.2 m clayey-sand murrum.

4.2.2 SG-F2 — the governing finding

	Level
Deepest stratum boundary recorded anywhere in the report	(-)1.500
Sentry footing F1, on in-situ basalt	(-)2.000
Underground box, internal floor	(-)6.100
Mat soffit	(-)6.700
FORMATION — what the mat actually bears on	(-)6.800
Sump SU-01 base	(-)8.000

The investigation reached about 1.5 m. The structure founds at 6.8 m.

There are **5.3 m of completely unlogged ground** between the bottom of the deepest trial pit and the formation the 600 mm mat bears on, and **6.5 m** to the sump base. **Only the sentry post footing F1 at (-)2.000 lies inside the investigated horizon, and only just.**

Nothing the report says about rockhead continuity, groundwater, jointing, red-bole seams or modulus of subgrade reaction can be read as covering the founding horizon of the shelter. SG-V2, **and it is the item a reviewer should look at first.**

4.2.3 The report is internally consistent, and that was checked before it was used

Chain	Check	Result
Soil	$SBC = q_{ult} / 2.5$, the report's own stated factor of safety	6 of 6 rows reproduce
Rock, step 1	$UCS = \text{max load} / \text{specimen area}$	6 of 6 rows reproduce
Rock, step 2	$SBC = UCS / 25$, rounded — the factor the report applies but never prints	6 of 6 rows reproduce

SG-F1. **The SEMT report is internally consistent and is usable.** That is worth establishing before anything is built on it.

4.3 Ground parameters — the design set

Parameter	Value	Class
Ground	Deccan basalt (trap), with red-bole / vesicular seams at flow contacts	[A]
Rockhead	(-)1.500 to (-)2.000	[A]
Presumptive safe bearing capacity	3 240 kPa (IS 1904:1986 Table 1, hard rock)	[A]
Bulk unit weight γ	20 kN/m ³	[A]
Saturated unit weight γ_{sat}	21 kN/m ³	[A]
Submerged $\gamma' = 21 - 9.81$	11.19 kN/m³	[C] derived
At-rest coefficient $K_0 = 1 - \sin \phi$	0.50 ($\phi \approx 30^\circ$)	[A]
Design groundwater table	(-)2.000 (monsoon)	[A] — the single most important number to confirm
Modulus of subgrade reaction k_s	100 000 to 500 000 kN/m³	[A] — run the mat model at BOTH bounds
Settlement	Negligible on sound basalt (IS 12070)	[A]
Soil transmission factor K_a , saturated	1.0 (IS 4991 Cl. 7.2 + Table 3)	[C]
K_a , dry compacted berm fill	≈ 0.5	[A] — deliberately not relied on

The hazard is not the basalt — it is the flow contacts. A single red-bole seam under the mat produces the differential-support case that **SIZES the mat** (Part 9.3, Case 2, 84 % utilised). Horizontally bedded flows are precisely the geometry that produces flow contacts at intervals, and **the report neither finds nor excludes one, because it never reached the founding horizon.** Over-excavate any red-bole or vesicular seam at founding level and replace with M15 lean concrete.

4.3.1 What the evidence changed, and what it did not

Not one parameter above is changed by the supplied documents. What each gained is a stated provenance and a quantified margin:

Row	What the evidence says	Effect
Ground	Deccan Trap confirmed — <i>"horizontally bedded and more or less uniform"</i>	Corroborated. The flow-contact hazard is neither found nor excluded
Rockhead	0.9–1.5 m at three locations	The report's band is entirely at or above the assumed 1.5–2.0 m. Conservative for founding depth, unconservative for rock excavation quantity: +118 m³, ≈ 2 days (SG-F5)
SBC 3 240	Measured basalt 1 961–2 059 kPa soaked , 2 942–3 530 unsoaked	The founding horizon is 4.8 m below the design GWT , so soaked governs . Worst utilisation rises 12.5 % → 20.6 % ; every element still passes with a factor of 4.8 in hand. 3 240 stands — it is an IS 1904 presumptive value and is declared as one (SG-F3)
$\gamma_{20} / \gamma_{\text{sat } 21}$	95 % MDD at OMC gives 19.12–19.61 ; γ_{sat} back-figures to 21.26	Bulk 20 is 2–5 % conservative for lateral load; γ_{sat} agrees within 1.2 %
K_0 0.50	Measured ϕ 27–35° → K_0 0.426–0.546	Conservative against all three granular murrum samples. Worst case moves the wall design load under 1 % — the walls are blast-governed (SG-F9)
GWT (–)2.000	<i>"Water table was not encountered in any trial pit" — in pits ~1.5 m deep</i>	The provenance, recorded for the first time: no water was found, so 2 m was CHOSEN. <i>Not encountered here means not reached</i> (SG-F6)
k_s	Nothing — no plate load test	Untouched. Both bounds still outstanding
Settlement	Net pressure at formation ≈ –105 kPa	This is <i>why</i> "negligible" is true — and the same fact that makes flotation the governing problem (SG-F4)

4.3.2 Bearing: it does not govern anything

Check	Demand	/ 3 240 [A]	/ 1 961 soaked	Verdict
Mat, service	58.4 kPa	1.80 %	2.98 %	PASS
Mat, blast	404.9 kPa	12.50 %	20.64 %	PASS
Sentry footing F1	157.6 kPa	4.86 %	8.04 %	PASS

Even at the **broken** basalt value of 981 kPa — the horizon 5 m *above* the formation — the mat under full blast would be at **41 %**. **The bearing case is nowhere near governing anything.**

4.4 Groundwater — the central question

"Water table was not encountered in any trial pit. However, water table may raise during/after rainy season." — SEMT/67/15 para 13 [C]

"Proposed structure safe for water table at depth of 2 m below GL" — P1 deck slide 29 [C]

The provenance of (–)2.000, recorded for the first time: no water was found, so a depth of 2 m below ground level was CHOSEN, and the structure was then designed to be safe for it. That is an assumption adopted in the absence of data. It is a defensible one — it puts the table near the surface, which is the conservative direction for uplift and for lateral load. **It is not a measurement, and nothing in this project turns it into one.**

Three reasons the observation does not reach the design question:

1. **Depth.** The pits reached about 1.5 m. The design GWT is at (–)2.000 — **already below the deepest pit** — and the founding horizon is at (–)6.800. **Here, not encountered means not reached.**
2. **Season.** The report itself says the table *"may raise during/after rainy season"*. The request is dated 20 May 2015; if the field work followed promptly it was the **pre-monsoon minimum**. The report nowhere states the field-work date, so this is inference and is tagged [U], not asserted.
3. **Ground type.** Deccan Trap groundwater is not a simple water table in a porous medium. It sits in the vesicular and jointed zones **at the flow contacts** — the same feature that sizes the mat. Such water is commonly perched and strongly seasonal, and **an open trial pit in the dry season is close to the worst available instrument for finding it.**

And it is the number that matters most. Of the 15.41 kPa/m lateral gradient, **9.81 is water**. The hydrostatic uplift on the mat is **46.11 kPa = 6 289 kN over 136.4 m²**. What closes it is a **standpipe piezometer read through a full monsoon** — and the programme's own monitoring window, 12 November to 4 December, **is not in the monsoon**.

4.5 Black cotton soil — confirmed, and it reaches two elements

New, and recorded nowhere in this project before the site package: the surface soil is BLACK COTTON, CH, free swell index 60–65 % — the top band of the IS 1498 scale. [C], by measurement, from SEMT/67/15 Appendix A.

Nothing structural founds in it. The mat is at (–)6.700 and footing F1 at (–)2.000, both in basalt — and Part 11.6's insistence that F1 bears *"on in-situ rock, never on backfill"* is now backed by a measurement rather than by instinct. **Two elements are exposed anyway:**

Element	Exposure	Status
The covered entry stairwell's stepped raft	Its top founds at about (–)0.300, inside the 0.18–1.0 m CH horizon . This is heave , not bearing — and the stairwell is expendable against <i>blast</i> , which heave is not	RULED — strip and replace. Ruling 12, RC4: strip the CH horizon to 1.000 m below existing ground over 4.410 × 4.000 m, replace with granular fill at FSI ≤ 20 % (IS 2720 Pt XL) in 200 mm layers to ≥ 95 % MDD , extending 1.000 m beyond every raft edge . BOQ item added: 8 m ³ , rate [A]
The 300 mm turf concealment layer	Re-laid <i>from the site's own stockpile</i> over the 150 mm granular filter. If the site's own topsoil is this CH clay it cracks in the dry season — a concealment defect — and the wet/dry cycles pump fines into the filter, which is the one thing the filter exists to stop	LEFT OPEN by owner ruling 16 (RC4). SG–V7. The load is unaffected: 300 at 18 kN/m ³ = 5.40 kPa is right for a black cotton soil

The two collide, and the collision is recorded rather than resolved. The strip-and-replace ruling produces a stockpile of **exactly the material** SG–V7 **warns against re-laying** — and the excavation-face ruling below produces 76–118 m³ more of it.

4.6 The excavation face — ruling 22 (RC6)

The project had measured **1.000 m working space and vertical unbenched faces for the full 6.800 m**. In basalt that is reasonable. But rockhead is at 0.9–1.5 m with a CH horizon at FSI 60–65 % on top of it:

So the face is about **1.0–1.5 m of soil — including very-high-swelling clay — standing vertically on 5.3–5.9 m of rock, with the rock excavation undercutting it**. Under the flotation acceptance (Part 18.4) it also stands open **across a monsoon**. That soil cap is the one part of this excavation that has no business being vertical.

```
ADOPTED  batter 1 : 1 MINIMUM over the soil cap, grade to rockhead, all four
          sides;  VERTICAL FACE RETAINED IN ROCK below rockhead

          extra excavation    1 : 1    1.500 m offset    76.4 m3
                               1.25 : 1    1.875 m offset    96.7 m3
                               1.5 : 1    2.250 m offset   117.6 m3
          (a 1.000 m bench at rockhead instead would be    96.6 m3)

          WORKING SPACE UNCHANGED -- the 1.000 m is at FORMATION, 5.3 m below
          the battered zone, so no BOQ working-space quantity moves
```

1 : 1 is a MINIMUM, not the answer. A batter in FSI 60–65 % clay, undercut and wetted across a monsoon, may need to be considerably flatter. IS 3764 is named **by title only**. **The angle and its seasonal variation are the geotechnical engineer's, and the quantity moves with the angle** [A]. WM–V9 **closes as a measurement basis** — the project now states a face treatment instead of an unexamined vertical — **but the slope-stability assessment it always asked for is still outstanding**.

4.7 Rockhead, and what it costs

Location	Sound basalt below	On the project datum
G Building (TP-1 to TP-4)	1.500 m	(–)1.500
H Building (TP-5 to TP-8)	0.900 m	(–)0.900
Mess Building (TP-9 to TP-11)	1.000 m	(–)1.000
Report band		(–)0.900 to (–)1.500 [C]
Assumed band		(–)1.500 to (–)2.000 [A]

The two bands touch at exactly one point, (-)1.500.

Rockhead	Depth of rock cut	Rock volume m ³	vs BOQ
(-)0.900	5.900 m	1 161.12	+118.12
(-)1.000	5.800 m	1 141.44	+98.44
(-)1.500	5.300 m	1 043.04	+0.04
(-)1.750 ← BOQ E-02b	5.050 m	993.84	-49.16
(-)2.000	4.800 m	944.64	-98.36

Overrun at the report's shallow rockhead: +118.1 m³ (+11.3 %), ≈ 2 extra days at the programme's 60 m³/day output. **Total excavation does not change** — soil falls by the same volume as rock rises — so the money at stake is the **rate difference** over about 118 m³, not a rock rate over 118 m³. **No BOQ quantity is changed**; the owner's own bill governs and its risk register already carries the item, saying *"roughly one extra day"*. **What this adds is the number: it is about two.**

One consequence is favourable and is recorded because it is free: the sliding argument rests on the box being *"socketed ~4.8 m into basalt"*. On the report's band the socket is **5.3 m to 5.9 m. The argument gets stronger.**

PART 5 — GEOMETRY AND CONFIGURATION

5.1 Coordinate system

ORIGIN = south-west EXTERNAL corner of the underground box, at ground level
 X = EAST (along the length, 0 to 22000)
 Y (plan) = NORTH (across the width, 0 to 6200)
 LEVELS = metres relative to finished site grade 0.000, negative downwards
 UNITS = millimetres in all DXF geometry; metres in all level annotation

Site orientation +X = EAST, +Y = NORTH **was fixed for the first time at SG2** and is [A]. Every drawing and every calculation in this project uses this system. **The STAAD underground model does not:** it is built on mid-surface geometry with its origin at the **underside of the mat**, so project level = model Y - 6.700. Part 8.2.

5.2 Level schedule

Level	Value	Description
Grade	0.000	Finished site level, crowned, falls 1:50 away
Top of pressure slab	(-)2.000	= headhouse floor level
Roof soffit	(-)2.900	900 mm slab
Internal floor / top of mat	(-)6.100	3 200 mm clear height
Underside of mat	(-)6.700	600 mm mat
Formation / underside of PCC	(-)6.800	100 mm blinding
Design groundwater table	(-)2.000	monsoon [A]
Rockhead	(-)1.500 to (-)2.000	[A]
Stair landing L1	(-)4.7333	8 risers up from the floor
Stair landing L2	(-)3.3667	16 risers up from the floor
Headhouse roof soffit	+0.400	2 400 mm clear internally
Headhouse roof top	+0.900	no earth cover; berm graded to this level
Entry stairwell roof at head	+2.450	soffit +2.200
Sentry post GF FFL	+0.450	= base of the STAAD model
Sentry post first floor	+3.650	storey height 3 200
Sentry post roof	+6.700	storey height 3 050
Sentry post parapet top	+7.000	
Sump pit invert	(-)7.600	pit base slab (-)8.000

5.3 The main box

Item	Value	Was	Class
External length	22 000 mm	21 600 (pre-M1)	[C]
External width	6 200 mm	unchanged	[C]
Internal length	20 800 mm	20 400	[C]
Internal width (clear span of the roof)	5 000 mm	unchanged — do not widen	[C]
Perimeter wall thickness	600 mm	unchanged	[C]
Roof (pressure) slab	900 mm	unchanged	[C]
Mat foundation	600 mm	unchanged	[C]
Blinding / PCC	100 mm M15	unchanged	[C]
Internal clear height	3 200 mm	unchanged	[C]
Engineered cover over roof	2 000 mm layered	reduced from 4 000 in an earlier revision	[C]

Why lengthening the box is structurally free, and widening it is not. The roof aspect ratio is $20.8 / 5.0 = 4.2$, and IS 456 Table 26 is tabulated only to $l_y/l_x = 2.0$, so **two-way action cannot be claimed**. The roof therefore spans one-way across the 5 m width, and its moment depends on that width alone. **Adding length adds no roof moment; widening 5.0 → 6.0 m would increase it by 44 %.** That single fact is the structural basis of Modification M1.

5.4 Bay schedule, and internal walls

Bay	X range (mm)	Clear width	Use
1	600 – 3500	2900	Emergency stores, 1 000 L potable tank, ESC 1
2	3610 – 5410	1800	Lavatory (1800×2000) + medical (1800×3000)
3	5520 – 9020	3500	Ops room and hazard plotting; EMP Zone 2 enclosure
4	9130 – 10930	1800	Berthing, 3 × 3-tier bunks, 9 berths
5	11040 – 12600	1560	CBRN plant: 2 × 300 m³/h filter trains, CO ₂ /O ₂ , dehumidifier, clean sump
6	12800 – 14800	2000	Decon airlock, 3 stages (2000×2000, 2000×1500, 2000×1500)
7	15200 – 18000	2800	Stair shaft (shifted +200 by M1)
8	18400 – 21400	3000	Generator 15 kVA (grey zone), ESC 2 at X 19 900

Mark	X range (mm)	Thickness	Notes
W8 partitions ×4	3500–3610, 5410–5520, 9020–9130, 10930–11040	110	Each with a permanent 900 mm gap at Y 2500–3400
W5	12600 – 12800	200	Bay 5/6, fire and gas-tight only , no pressure differential
W6	14800 – 15200	400	Bay 6/7 — M1, was 200. Protective boundary. Blast Door 1 opening Y 600–1800
W7	18000 – 18400	400	Bay 7/8 — M1, was 200. Protective boundary. Blast Door 2 opening Y 600–1800

Modification M1 is not a uniform shift, and it must never be applied as one. Nothing west of X 14 800 moves. W6 and W7 thicken **about their own faces**. The shaft moves +200; Bay 8 and the east end move +400; the headhouse and the covered stairwell move +200 **as whole structures. Re-derive per element, never translate wholesale.**

5.5 Stair shaft, roof void and cantilever pad

```

Stair shaft (Bay 7) clear      X 15200 – 18000   Y 600 – 5600   = 2800 x 5000
Void in the pressure slab     X 15200 – 18000   Y 600 – 3760   = 2800 x 3160
Cantilever pad (remaining slab) X 15200 – 18000   Y 3760 – 5600   = 2800 x 1840
Flight A (flights 1 and 3, stacked) X 15300 – 16500   (1200 wide)
Well                          X 16500 – 16700   (200)
Flight B (flight 2)           X 16700 – 17900   (1200 wide)
Landing L1 (–)4.7333          Y 3760 – 4960, full 2800 width
Landing L2 (–)3.3667          Y 600 – 1800, full 2800 width (over the arrival landing)
Arrival landing (–)6.100 = the mat surface, Y 600 – 1800
Store under landing L1       Y 4960 – 5600
MAIN STAIR 24R @ 170.8333, tread 280, 3 flights x 8R, rise 4100, headroom 2533

```

5.6 Escape shafts, and the 750 mm rule

	ESC 1	ESC 2
Centre	(2050, 2050)	(19900, 2050) — shifted +400 by M1
Clear opening	1 400 mm dia	1 400 mm dia
Collar	250 mm RC, OD 1 900	250 mm RC, OD 1 900
Clearance, opening edge to wall face	750 mm all round	800 / 800 mm
Head level	+0.150	+0.700
Climb from (–)6.100	6.250 m	6.800 m
Escape route	R2	R3 — shares Bay 8 with the generator

THE 750 mm CLEARANCE RULE. Opening edge to structural wall face ≥ 750 mm = 250 collar + ~400 trimmer band + 100 tolerance. **A bay must therefore be at least $1\,400 + 2 \times 750 = 2\,900$ mm wide to hold an escape shaft.** That rule drove the Rev C lengthening, and it is why **Modification M1 could not take the 400 mm it needed out of Bay 8.**

5.7 Headhouse

External X 13600 - 18400 Y 200 - 6000 = 4800 x 5800
 Internal X 14000 - 18000 Y 600 - 5600 = 4000 x 5000
 Walls 400 thk (HW1 south, HW2 north, HW3 west, HW4 east)
 Roof 500 thk, soffit +0.400, top +0.900, NO earth cover
 Floor = the top of the 900 pressure slab, (-)2.000
 Clear internal height 2400
 Inner security door 900 x 2100 in HW2 (north), X 14450 - 15350. NOT blast rated.
 Berm graded against all four walls to +0.900 at 1.5 : 1

Wall	Runs	Length	Sits over
HW1 south	in X	4 000	Box south perimeter wall — direct
HW2 north	in X	4 000	Box north perimeter wall — direct
HW3 west	in Y	5 000	Bay 6, mid-slab. NO wall below — a line load on the pressure slab, checked at 26 % two-way / 41 % on the one-way bound (Part 10.4)
HW4 east	in Y	5 000	Wall W7 (18000-18400). M1 made these align exactly

5.8 Covered entry stairwell

External X 9250 - 16050 Y 5750 - 7750 = 6800 x 2000
 Internal X 9500 - 15800 Y 6000 - 7500 = 6300 x 1500
 Walls 250 RC (both sides, headwall, east wall)
 Top landing X 9500 - 11000 at 0.000, 250 thk
 Flight X 11000 - 14300, 12R @ 166.6667, going 300, 11 x 300 = 3300, waist 250
 Platform X 14300 - 15800 at (-)2.000, 1500 x 1500, 250 thk
 Roof 250 RC raking, soffit 2200 above the flight; IT STAYS 250 OVER THE PLATFORM
 Entry door 1000 x 2100 at grade in the headwall (X 9250 - 9500), opens OUTWARD
 300 mm channel + grating, full 1500 width, at X 8950 - 9250
 Stepped RC raft 300 thk on compacted fill; MOVEMENT JOINT where it meets the headhouse
 1.0 m3 external sump at the platform, own soakaway
 Berm 1.5:1 against the walls to +0.900, toe at grade over 1350

The roof stays 250 mm over the platform — conflict C16, ruled and closed. The clause used to read *"over the platform it becomes the 500 headhouse roof"*, which **no other part of the project supported**: the structural design designs a 250 roof, the load register loads one, the reinforcement register registers one, and **the headhouse footprint Y 200-6000 does not reach the platform at Y 6000-7500 at all**. One clause was the only place in the project saying 500.

5.9 Sentry post

External plan 4000 x 5000 Internal 3600 x 4600
 Column grid A-B 3650 c/c (X) . 1-2 4650 c/c (Y)
 Grid coordinates A: x = 175 B: x = 3825 | 1: z = 175 2: z = 4825
 Columns C1 350 x 350, 4 No., both storeys
 Beams B1 250 x 450, spanning A-B (3650 c/c), on grids 1 and 2
 Beams B2 250 x 450, spanning 1-2 (4650 c/c), on grids A and B
 Slab S1 150 thk two-way, 3650 x 4650 c/c, both floors
 Plinth beam PB 250 x 400 at +0.450
 Footings F1 1500 x 1500 x 600, 4 No., on in-situ basalt at (-)2.000
 Ground storey infill 190 BRICK MASONRY in the 200 structural zone (SP-B1)
 First storey infill armoured vision panels, 1200 wide
 Spiral stair external, 1000 R, 250 dia central pole
 Door D1 900
 Storey heights ground 3200 (+0.450 -> +3.650), first 3050 (+3.650 -> +6.700)
 Siting >= 10 m clear of the shelter excavation
 Site position X 32000 - 36000, Y 600 - 5600 [A] -- RC4, EAST ruling

5.9.1 SP-B1 — the walls are brick masonry, not RC ballistic panels

The Rev F drawings call the ground-storey infill *"200 RC ballistic panels"*. By instruction that became **190 mm one-brick modular brickwork to IS 1077, laid in CM 1:6, built inside the unchanged 200 mm structural zone**, with the residual 10 mm taken up at the internal face in the plaster.

190 and not 230 because 230 conventional brickwork would project 30 mm past the column faces and change the confirmed 4 000 × 5 000 external envelope. **Every other value in the block above is unchanged.** Quantity: **12.20 m³ / 64.19 m² face**, ≈ 6 400 bricks.

Four consequences were recorded rather than buried:

1. **Lintels became necessary.** The RC panels needed none. Designed at SP-B2 — Part 11.7.
2. **A wall tie detail became necessary.** None existed. Designed at SP-B2.
3. **The seismic weight falls.** Brick at ≈ 20 kN/m³ over 0.190 × 2.600 gives ≈ **9.88 kN/m** against the modelled **13.000 kN/m**, so w_{fb} falls, and **every member designed to 73.18 kN is over-designed.** *That is a direction, not a verification* — WM-V6, **left open by owner ruling 19 (RC5): wait for STAAD.** The modelled 13.000 kN/m is retained because it is conservative.
4. **The ballistic function is lost.** Brick masonry does not give the protection the Rev F panels were named for. **RULED — ruling 7 (RC4): ballistic protection is NOT required. Brick stands.**

5.9.2 The sentry post's position — one assumed position instead of two, and it still fails the rule as written

Ruling 8 (RC4): the EAST position is adopted. The elevation's own note sets **X 32 000 – 36 000** — 4 000 wide, which is the post's actual external dimension. *(An earlier record of X 31 700 – 36 300 is 4 600 wide and matches nothing; the drawing is the primary source and is corrected.)* **An elevation cannot give a Y, so Y 600 – 5600, centred on the box longitudinal centreline Y 3100, is set by the ruling.**

RC4-F3 — **and it fails the rule as that rule is written.** "≥ 10 m clear of the shelter EXCAVATION": X 32 000 is **10.00 m** clear of the **box face** at X 22 000, but the **excavation** face with its 1 000 mm working space is at X 23 000 — giving **9.00 m**. The drawing's note cites the excavation rule while applying the box face. Satisfying it as written moves the post to X 33 000 – 37 000, which passes every check. **BOTH readings are recorded and NEITHER is adopted over the other: the EAST position is a drawing convention, not a survey coordinate, and U4 STAYS [ASSUMED].**

And it costs the post its stated function. One of the site package's five reasons for +X = east was *"the sentry post covers the approach from the north"*. With the post 10 m beyond the **far** end of the box it covers nothing — the approach is at the **west** end. **No stated rule fails, but a reviewer will ask.**

5.10 Every opening through the protective boundary

Mark	Where	Leaf	Level	Rating	Class
Blast Door 1	W6 , X 14800–15200 (400 thk), opening Y 600–1800 — stair shaft → Bay 6	1200 × 2100	(–)6.100	≥ 7 bar, gas-tight, rebound-rated	[C] geometry · the door itself is vendor data
Blast Door 2	W7 , X 18000–18400 (400 thk), opening Y 600–1800 — stair shaft → Bay 8	1200 × 2100	(–)6.100	≥ 7 bar	[C] · vendor
Inner security door	HW2 , headhouse north wall, X 14450–15350	900 × 2100	(–)2.000	NOT blast rated	[C]
Entry door	Entry stairwell headwall, X 9250–9500, opens outward	1000 × 2100	0.000	NOT blast rated — outside the boundary, expendable	[C]
D-05, the W5 gas-tight door	W5 , X 12600–12800	—	(–)6.100	—	RULED — ruling 9 (RC4): design it now. W5 was designated <i>fire and gas-tight</i> with no door anywhere in the project , and escape route R1 has to cross it
W8 partition gaps ×4	3500–3610, 5410–5520, 9020–9130, 10930–11040	permanent 900 gap, Y 2500–3400	(–)6.100	no doors, by design	[C]
ESC 1 / ESC 2 heads	Bays 1 and 8, through the pressure slab	1 400 dia bore	+0.150 / +0.700	1.54 m² hole in the protective boundary each	See Part 10.5

Both blast doors ARE the protective boundary. They sit in the same plane as W6/W7, and Finding F1 is why those walls are 400 thick: the shaft equalises to full p_{so} , so the doors and the walls beside them take the same **383 kPa**. Each frame is a **cast-in steel frame anchored into and welded to the cage**, which is both the blast fixing and the only EMP continuity the opening has.

PART 6 — MATERIALS, DURABILITY AND DETAILING RULES

6.1 Concrete and reinforcement

Item	Shelter / stairs / headhouse	Sentry post	Source
Concrete	M35	M30	IS 456 Table 5 — M35 is the minimum for very severe exposure , Table 3
Free water / cement ratio	≤ 0.45	—	IS 456 Table 5
Minimum cement content	≥ 340 kg/m ³	—	IS 456 Table 5
Admixture	Integral crystalline waterproofing	—	[R]
Blinding	M15, 100 thk	—	
Burster slab (in the cover)	M30, 200 thk	—	
Reinforcement	Fe500D to IS 1786:2008	Fe500	
$E_c = 5000\sqrt{f_{ck}}$	29 580 N/mm²	27 386 N/mm²	IS 456 Cl. 6.2.3.1
Poisson's ratio	0.20	0.20	[A] — standard; closed by ruling 23 (RC6)
Unit weight, reinforced concrete	25 kN/m ³	25 kN/m ³	IS 875 (Pt 1) Table 1
γ_m concrete / steel	1.5 / 1.15	same	IS 456 Cl. 36.4.2

Why Fe500D and not plain Fe500. The **D** designation carries a guaranteed minimum elongation. A ductility ratio of 5 is claimed in the blast case (Part 2.2.3); claiming it on steel with no guaranteed elongation would be claiming it on nothing. The sentry post, which is not blast designed, uses Fe500.

6.2 Cover — IS 456 Cl. 26.4.2 / Table 16

Face	Cover
Cast against blinding / trimmed rock	75 mm
Formed earth face	50 mm
Internal faces	40 mm
Stair shaft faces (wet / dirty zone)	30 mm
Sentry post beams and slabs	30 mm
Sentry post columns	40 mm — Cl. 26.4.2.2, never less than 40 nor the bar diameter
Footings cast against earth	50 mm — Cl. 26.4.2.1
Maximum bar spacing, both curtains	150 mm — an EMP requirement , stricter than IS 456 Cl. 26.3.3

The 5 mm cover reduction IS 456 Table 16 permits for M35 and above is deliberately not taken. This structure is permanently below the design water table on a very severe exposure; 5 mm is not worth the argument.

And the 150 mm bar spacing is an EMP decision, recorded as one. It is stricter than IS 456 needs and it is carried in the master, in the structural CAD package, in eight bar bending schedules and on the drawings. Part 2.8.2 is the arithmetic behind it, and it is the reason the cage gives 99.99 dB at 10 kHz.

6.3 Development and lap lengths — IS 456 Cl. 26.2.1

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Id = phi . sigma_s / (4 . tau_bd),      sigma_s = 0.87 fy = 435 N/mm2
tau_bd (Cl. 26.2.1.1):  M30 -> 1.5      M35 -> 1.7
      x 1.6 for deformed bars in tension
      / 0.8 (i.e. x 1.25) for bars in compression

```

Grade	$\tau_{bd} \times 1.6$	L_d tension	L_d compression
M35	2.72	40 ϕ	32 ϕ
M30	2.40	46 ϕ	37 ϕ

Bar	L_d M35	Lap M35	L_d M30	$L_{d,c}$ M30
T8	320	400	368	296
T10	400	500	460	370
T12	480	600	552	444
T16	640	800	736	592
T20	800	1000	920	740
T25	1000	1250	1150	925

Lap policy. IS 456 Cl. 26.2.5.1(c) gives lap = L_d or 30ϕ , whichever is greater = 40ϕ . Cl. 26.2.5.1 requires $\times 1.4$ if more than 50 % of bars are lapped at one section. **All laps are staggered so that ≤ 50 % are spliced at any section, and the lap is specified as 50ϕ — 25 % above requirement.**

Blast bond bonus: refused. IS 4991 Cl. 10.3.1.1 permits +25 % on bond, which would take L_d to 32ϕ . **Not taken. All detailing uses the static 40ϕ** (Part 2.2.5).

6.4 Waterproofing and the tank

The structure is a **tanked box, not a drained one**. That single decision sets the entire water strategy, and four rules follow from it:

1. **Nothing penetrates the roof.** No rainwater outlet, no vent stack, no soil stack through the 2 000 mm cover — that would breach both the radiation mass and the roof membrane. **There is no downpipe on the buried roof and none is added.**
2. **One services envelope crossing.** The service entry plate. The sump rising main uses it; no second drainage penetration is created.
3. **Three streams, never combined.** Clean (seepage and condensate) → storm soakaway. Foul (peacetime) → septic tank → soak pit. Decon effluent → 1 000 L tank → **tanker only**.
4. **Drainage is not in the safety path.** No drainage failure may block egress, breach the envelope or flood an entrance.

Element	Specification
Membrane	Continuous tank: on the blinding, turned up the external face, lapped to the roof membrane
Admixture	Integral crystalline waterproofing throughout
Construction joints (~6 m)	Reinforcement fully continuous, two waterstops, and a welded Cu / galvanised EMP strap
Movement joints	NONE inside the protective envelope. A movement joint is a guaranteed blast, gas and EMP discontinuity
Floor falls	Formed in the screed, never cut into the 600 mm mat — cutting them would reduce the section and the 75 mm cover to the bottom curtain
Pipe crossings	Puddle flange welded to the sleeve, membrane dressed and clamped to it

The box is never drained to relieve groundwater. A relief drain under the mat would defeat the tank and would not reduce uplift by design intent — the structure is held down by mass and by the staged construction sequence, not by pressure relief. **The only temporary relief permitted is the six construction-stage knock-out plugs, grouted after backfill** (Part 9.3).

PART 7 — LOADING

7.1 Blast

DESIGN BASIS THREAT	nuclear air blast, $p_{so} = 344.7$ kPa (50 psi)	[C]
Positive phase duration	$t_d = 0.13$ to 1.33 s	[C]
Reflected pressure	$p_r = 1366$ kPa	[C]
Dynamic pressure	$q = 282$ kPa	[C]
Ductility ratio	$\mu = 5$ (moderate, repairable damage)	IS 4991 Cl. 10.3.3
DLF = $\mu / (\mu - 0.5)$	$5 / 4.5 = 1.111$	
validation: $\mu = 1 \rightarrow DLF = 2.00 =$ the elastic step-load factor		
*** DESIGN BLAST PRESSURE	$344.7 \times 1.111 = 383$ kPa ***	
applied to the ROOF and to the WALLS ($K_a = 1.0$, IS 4991 Cl. 7.2)		
Roof natural period T	13.4 ms $\rightarrow t_d/T = 10$ to $100 \rightarrow$ QUASI-STATIC	
Dynamic material strengths, BLAST CASE ONLY	IS 4991 Cl. 10.3.1	
$f_{ck,dyn} = 1.25 \times 35 = 43.75$ N/mm ²	$f_{y,dyn} = 1.25 \times 500 = 625$ N/mm ²	
NO dynamic increase on shear	IS 4991 Cl. 10.3.1.1	

7.2 Gravity, soil and water

Load	Value	Applied to	Source
Self weight, reinforced concrete	25 kN/m ³	all	IS 875 (Pt 1) Table 1
Roof self, 0.900×25	22.50 kPa	roof	
Mat self, 0.600×25	15.00 kPa	mat	
Engineered cover, 2 000 layered	40.65 kPa	roof	[C] — Part 7.3
Superimposed dead (services, finishes)	2.0 kPa roof / 1.0 kPa mat		[R]
Live load, internal floor	5.0 kPa — plant and storage, not 2.0 residential	floor	IS 875 (Pt 2)
Earth + water lateral gradient	$K_0 \gamma' + \gamma_w = 0.50 \times 11.19 + 9.81 =$ 15.41 kPa/m	walls	[C] derived
— at roof soffit (–)2.900	33.9 kPa	walls	
— at floor (–)6.100	83.2 kPa	walls	
Hydrostatic uplift on the mat	$4.700 \times 9.81 =$ 46.11 kPa ; total 6 289 kN over 136.4 m ²	mat	[C]
Construction surcharge	20 kPa vertical / 10 kPa lateral		[A]
Staircase, smeared on the two shaft walls	2.947 kPa	W6 / W7	[C]

Of the 15.41 kPa/m lateral gradient, **9.81 is water and only 5.60 is soil**. Water is nearly two-thirds of the lateral load. **This is why the groundwater level is the most important number the site investigation must confirm.**

7.3 The engineered cover — 2 000 mm, 40.65 kPa

From the top	Thickness	Unit weight	Load kPa	Function
Topsoil / turf	300	18	5.40	Concealment , erosion, sheds rain
Granular filter	150	19	2.85	Stops fines clogging
RC burster slab M30, T12 @ 150 B/W	200	25	5.00	Breaks up a penetrating item
Crushed basalt rubble 25–75 mm	500	17	8.50	Scatters burster energy
Compacted engineered fill @ 95 % MDD	750	20	15.00	Radiation mass
Protection screed over the membrane	100	24	2.40	Protects the waterproofing
Sum of the six layers	2 000		39.15	
Declared allowance, held	—		+1.50	
TOTAL — DESIGN VALUE	2 000		40.65	

Conflict C17, ruled and closed. The six layers sum to **39.15 kPa**; the table has always stated **40.65**, and the two never agreed. **40.65 is held as the design value** and the table now shows why: it is the layer sum **plus a declared allowance of 1.50 kPa**. That is the only choice that costs nothing and changes nothing — 40.65 is the number in the roof total, in the final register and in **DL2 in every underground .std file**, it is the **larger** of the two, and adopting 39.15 instead would lighten the blast combination by 1.50 kPa and require a re-analysis that cannot be run here. **The allowance is not fabricated evidence: it is the difference the project has always carried, now stated instead of hidden.**

7.3.1 BS1 — the burster slab is laid to falls

By instruction the 200 mm burster slab is **no longer laid flat**. It is laid to a **1:50 crossfall, crowned on the box longitudinal centreline (Y = 3100) and falling each way to the box edges** — that is, **parallel to the finished grade**, which is already recorded as *crowned, falling 1:50 away*. Over the 3 100 mm half-width the slab drops **62 mm** from crown to box edge.

Why. The granular filter sits directly on the burster slab, and the drainage basis already states that water infiltrating the topsoil is *"intercepted by the granular filter and dispersed at the berm toe"*. **On a flat slab it cannot be** — it ponds on the slab and finds the construction joints. The crossfall gives the filter layer the gradient that claim depends on. **No pipe is introduced anywhere in the cover.**

Thicknesses and load — unchanged. The fall is taken up entirely in the **compacted engineered fill** below the rubble, which is **750 nominal at the crown thinning to 688 at the box edge**. Every other layer keeps its nominal thickness. **The cover therefore weighs exactly what it weighs today at the crown, and less toward the edges: the cover load does not increase anywhere.**

7.4 Total roof load — combination 103

Component	kPa
Blast, 344.7×1.111	383.00
Engineered cover	40.65
Superimposed dead	2.00
Self weight	22.50
TOTAL w	448.15

Live load is excluded — IS 4991 Cl. 11.2: *"No live load shall be considered on roof at the time of blast."*

Static ULS on the roof (COMB 101)
 $1.5 \times (40.65 + 2.0 + 22.5 + 20) = 127.7 \text{ kPa}$
 BLAST GOVERNS $448.15 / 127.7 = 3.51 : 1$

7.5 Headhouse loads

Element	Load	Basis
Roof (flush horizontal at the berm crest)	396.5 kPa = 383 blast + 12.5 self + 1.0 SIDL	IS 4991 Cl. 7.2
Walls — CURRENT	383 kPa, acting EITHER FACE	Conflict C10
Walls — earlier basis (superseded)	113 kPa = $C_d \cdot q = 0.4 \times 282$	IS 4991 Cl. 7.4

Conflict C10 and why the upgrade was taken. The walls sit behind ~2.5 m of berm and **earth transmits pressure**. K_a for dry compacted fill is an assumption that cannot be verified, so the walls are designed for **the full 383 kPa upper bound**. **Cost of the upgrade: one link cage, T12 4-leg @ 250. Utilisation rises 17 % → 59 %.** That is what removing an unverifiable assumption from the critical path is worth, and it is cheap. Owner ruling 23 (RC6) confirms the choice and closes assumption **A6**.

And the pressure acts from inside too. The inner security door is not blast rated and the entry stairwell is expected to be lost, so **the headhouse fills and the walls are pushed outwards**. They are reinforced symmetrically for 383 kPa either way — *a stated requirement, not an accident of detailing*.

7.6 Entry stairwell loads — static only, NOT blast rated

Element	Load
Flight, waist 250	waist 7.150 + steps 2.083 + finishes 1.000 + LL 5.000 = 15.233 → $w_u = 22.85 \text{ kPa}$
Side walls 250	$K_0 0.50 \times 20 \times 2.9 + 0.5 \times 10$ surcharge = 34 kPa at base

Element	Load
Raking roof 250	self 6.25 + waterproofing 2.0 + earth lap 5.4 + imposed 20 = 33.65 → $w_u = 50.5 \text{ kPa}$
Top landing 250	18.4 + flight reaction 36.5 = 54.9 kPa

The 20 kPa imposed load on the raking roof is deliberate: it is there for a stray vehicle on the berm. It is not a code minimum and it is not an accident.

7.7 Sentry post loads

Item	Value	Verification
Slab dead, first floor	4.750 kPa	$0.150 \times 25 + 1.0 \text{ finish}$
Slab dead, roof	5.250 kPa	$3.75 + 1.5 \text{ screed / waterproofing}$
Imposed, first floor	3.000 kPa	IS 875 (Pt 2), observation post
Imposed, roof	1.500 kPa	IS 875 (Pt 2), accessible
Peak two-way intensity, DL floor	8.669 kN/m	4.75×1.825
Peak two-way intensity, DL roof	9.581 kN/m	5.25×1.825
Peak two-way intensity, LL floor	5.475 kN/m	3.00×1.825
Peak two-way intensity, LL roof	2.737 kN/m	1.50×1.825
Infill on the first-floor beams	13.000 kN/m	$0.200 \times 2.600 \times 25$ — retained after SP-B1 because brick is lighter (≈ 9.88) and the modelled value is therefore conservative
Roof projection + parapet	4.162 kN/m	[C] as drawn — half re-derived, see below
— parapet term, $0.300 \times 0.150 \times 25$	1.125 kN/m	[C], reproduces exactly
— roof projection term	3.037 kN/m	[N] — the projection dimension is recorded nowhere
$w_{u \text{ floor}} = 1.5(4.75 + 3.00)$	11.625 kPa	governs the slab
$w_{u \text{ roof}} = 1.5(5.25 + 1.50)$	10.125 kPa	

U8-F1 — the parapet confirmation is necessary but not sufficient. Ruling 15 (RC4) confirmed the parapet at 300×150 , and $0.300 \times 0.150 \times 25 = 1.125 \text{ kN/m}$ reproduces exactly. The residual **3.037 kN/m** is the roof projection, and **the projection dimension is recorded nowhere in this project**. Back-solving gives 810 mm (slab alone) or 578 mm (slab + finish); **neither is round, neither is drawn, neither is adopted. 4.162 continues to be used as given — no member force changes.**

Load path to the beams — IS 456 Cl. 24.5, 45° yield lines:

```

Peak intensity on any beam = w x short span / 2 = w x 3.650/2 = w x 1.825 m
SHORT beams B1 (3650 long) -> TRIANGLE
LONG beams B2 (4650 long) -> TRAPEZOID
    w_eq = w_peak x [ 1 - 1/(3 r^2) ],    r = 4650/3650 = 1.2740
    1 - 1/(3 x 1.2740^2) = 0.79463 -> 0.7946 adopted

```

The trapezoid factor is **exact**; using the same factor for both bending moment and fixed-end moment is the simplification, and it is worth $\approx 2\%$ on B2's support moment, inside its 89 % utilisation. **Closed by owner ruling 24 (RC6), accepted on the margin — assumption A12.**

7.8 Seismic and wind

7.8.1 Underground box — IS 1893 (Part 1):2016

```

Z = 0.16 (Zone III)      I = 1.5      R = 4.0      Sa/g = 2.5
Ah = (Z/2) (I/R) (Sa/g) = 0.08 x 0.375 x 2.5      = 0.075
W = 14 002 kN      ->      Vb = 0.075 x 14 002      = 1050 kN
Per long wall 525 kN
    tau = 525e3 / (600 x 0.8 x 21600)      = 0.063 N/mm2
                                NEGLIGIBLE
IS 13920 Cl. 10.4 boundary elements CHECKED and NOT TRIGGERED

```

7.8.2 Sentry post — IS 1893 (Part 1):2016

```

Z = 0.16      I = 1.5      R = 3.0      Sa/g = 2.5 (rock, T on the plateau)
R = 3.0 because the infill panels are NOT separated from the frame.
R = 5.0 would require a special moment-resisting frame with positively
separated infill, which this project does not claim.
Ah = 0.08 x 0.50 x 2.5      = 0.100

Fundamental period -- all three estimates land inside 0.10-0.40 s,
so Sa/g = 2.5 regardless of which is used:
bare RC MRF, Cl 7.6.2      Ta = 0.075 h^0.75
                        = 0.075 x 6.250^0.75      = 0.297 s
with infill, Cl 7.6.2(c)      Ta = 0.09 h / sqrt(d)
                        X 0.281 s      Z 0.252 s

HAND-CALCULATED seismic weight
Roof +6.700 : slab 89.1 + parapet/projection 69.1 + beams 31.1
              + half-columns 18.7 + half first-storey infill 107.9 = 315.9 kN
Floor +3.650 : slab 80.6 + beams 31.1 + columns 38.3
              + half ground-storey infill 114.1 + 25 % LL 12.7      = 276.8 kN
TOTAL W = 592.7 kN      ->      Vb,hand = 0.100 x 592.7      = 59.3 kN

*** STAAD MODEL VALUE, WHICH GOVERNS DESIGN: Vb = 73.18 kN ***
stated in the load-case titles and CONFIRMED by the reactions:
Fx = 18.295 kN per column x 4      = 73.18 kN

```

Conflict C9 — and it is now fully explained rather than merely adopted. The model prints its own seismic-weight summary: roof 331.46 + floor 400.34 = **W = 731.80 kN**, and $V_b = 0.100 \times 731.80 = 73.18$ kN returns $A_h = \mathbf{0.1000}$ exactly. Rebuilding both storey weights from the model's own load blocks reproduces 331.46 and 400.34 to the kilonewton, **and the 139.1 kN gap to the hand check resolves completely:**

- **107.9 kN** is half the first-storey infill, which the hand check's floor line does not allocate; - **31.1 kN** is the model taking the full 450 mm beam depth in SELFWEIGHT while also applying the full slab pressure, where the hand check nets the beam to 300 mm.

Nothing is unexplained, and every member is already designed to the higher value.

7.8.3 Wind on the sentry post — IS 875 (Part 3):2015

```

Vb = 39 m/s (Pune)      k1 = 1.08 (100-yr, Table 1)      k2 = 1.00 (Cat 2, 10 m)
k3 = k4 = 1.00
Vz = 39 x 1.08      = 42.12 m/s
pz = 0.6 Vz^2      Cl 7.2      = 1.065 kPa
pd = pz x Kd 0.90 x Ka 0.897 x Kc 1.0      = 0.859 kPa
F = Cf 1.3 x Ae (4.0 x 6.7 = 26.8 m2) x pd      = 29.9 kN

SEISMIC 73.18 kN      vs      WIND 29.9 kN      ->      SEISMIC GOVERNS 2.4 : 1

```

$k_1 = 1.08$ assumes a 100-year design life [A]. **Closed by owner ruling 23 (RC6), accepted:** a 100-year life on a load path that seismic already beats 2.4 : 1 costs nothing.

7.9 Load combinations

7.9.1 Underground box — five combinations, matching the STAAD model

No.	Title	Factors
101	ULS static	1.5 (DL + SIDL + LL + SOIL + UPLIFT)
102	ULS uplift	0.9 DL + 1.5 UPLIFT
103	BLAST	1.0 (DL + SIDL + SOIL + UPLIFT + BLAST) ← governs every element
104	SLS crack width	1.0 (DL + SIDL + LL + SOIL + UPLIFT), IS 3370 Pt 2
105	Construction	1.5 (DL + SOIL + UPLIFT + SURCHARGE)

$\gamma = 1.0$ on blast because it is an extreme event checked against **ultimate** capacity with **dynamic** material strengths (IS 4991 Cl. 10.3.1). Applying 1.5 while also taking the 25 % material bonus would be inconsistent (Part 2.2.6).

Wind and earthquake are absent from 103 — IS 4991 Cl. 11.1 forbids combining them with blast.

7.9.2 Sentry post — sixteen combinations, read from the .std file

No.	Combination
101	1.5 DL + 1.5 LL
102-105	1.2 DL + 1.2 LL + 1.2 EQ ($\pm X$, $\pm Z$)
106-109	1.5 DL + 1.5 EQ ($\pm X$, $\pm Z$)
110-113	0.9 DL + 1.5 EQ ($\pm X$, $\pm Z$)
201	1.0 DL + 1.0 LL — SERVICE
202 · 203	1.0 DL + 1.0 EQ+X · 1.0 DL + 1.0 EQ+Z — DRIFT CHECK, IS 1893 Cl. 7.11

A correction worth recording. This table used to read "*15 combinations*" and stopped at 202, on the authority of the screen captures. **The .std file itself carries sixteen LOAD COMB cases — 203, the second drift check, was missing from the master.** No combination factor, member force or bar changes: 203 is a serviceability drift check that was already in the model being designed to. **The file is primary; the screenshots were not.**

PART 8 — ANALYSIS MODELS

8.1 What the models are, and what they are not

STAAD.Pro is not available in this environment. No analysis has been run in this project and no result — moment, displacement, shear or reaction — exists for the underground box or for the three mesh models. Editing, reconciling and validating a .std file is **not** running an analysis and is never reported as one.

Every underground design value in Parts 9 and 10 was produced by hand calculation, to IS 456 with IS 4991 loading rules. The model supplies the demand distribution and the visual check that loads are applied to the correct faces. **For the sentry post, only the forces were taken from the model** — the base shear and the seismic axial couples — and no STAAD concrete design output was relied upon anywhere.

8.2 Model inventory

Model	File in current/staad/	Role	Status
Underground box — reference	Underground_Structure_WITH_LOADS_worked_example (4).STD	The reconciled Phase 2 Rev A + M1 plate model. Every box value in Part 9 is checked against it	[C]
Sentry post	Sentry_Post_Framed_Seismic.std	Two-storey RC frame, seismic	[C]
Entry stairwell	Entry_Stairwell.std	Covered approach stairwell, static only, not blast rated	[C]
Underground box — COARSE	Underground_Shelter_Mesh_Coarse.std	Mesh study — 324 joints, 336 plates	[C]
Underground box — MEDIUM	Underground_Shelter_Mesh_Medium.std	Reference mesh — 1 113 joints, 1 138 plates	[C]
Underground box — FINE	Underground_Shelter_Mesh_Fine.std	h/2 refinement — 4 500 joints, 4 552 plates	[C]
Underground box — k_s upper bound	Underground_Shelter_ks500000.std	The reference model at $k_s = 500\ 000$; ELASTIC MAT SUBGRADE and the four corner KFY ($\times 5$) are the only lines that differ , verified by diff	[C]

Units: metres and kilonewtons throughout. **Coordinate system:** STAAD global — **Y is vertical**, X and Z horizontal.

The underground model's datum is not the project datum. It uses **mid-surface** geometry and its origin $Y = 0$ is at the **underside of the mat**, real level (-6.700) .

project level = model $Y - 6.700$. Apply the offset before comparing any model output to a drawing. $Y = 0.300$ is the mid-plane of the 600 mat; $Z = 0.300$ and $Z = 5.900$ are the mid-planes of the 600 perimeter walls; $X = 0.300$ is the mid-plane of the west end wall.

8.3 The underground box model

Form. A **plate + beam mid-surface model** of the whole box — mat, four perimeter walls, internal walls and the pressure slab — meshed on a regular grid over the full 22×6.2 m footprint, with support symbols at every mat node.

Shell elements	1 138 , in four thickness groups: mat 600 / roof 900 / perimeter 600 / internal 200–400
Supports	ELASTIC MAT DIRECT Y SUBGRADE 100000 — vertical springs on every mat node, generated from k_s
Corner joints	Explicit KFY restated per corner from the post-M1 tributary areas: 6 891 / 8 269 / 7 219 / 8 663 kN/m
Analysis command	PERFORM ANALYSIS PRINT STATICS CHECK — no P-Delta

Load cases, read verbatim from the file:

#	Title	Value
1	DL1 SELF WEIGHT (SELFWEIGHT Y -1)	auto
2	DL2 EARTH COVER ON ROOF 40.65 KN/M2	40.65 kPa
3	DL3 SIDL SERVICES AND FINISHES	2.0 kPa
4	LL1 LIVE LOAD ON INTERNAL FLOOR	5.0 kPa
5	DL4 STAIRCASE ON THE TWO SHAFT WALLS	2.947 kPa

#	Title	Value
6	EP1 EARTH + WATER ON EXTERNAL WALLS	33.9 → 83.2 kPa
7	HY1 HYDROSTATIC UPLIFT ON MAT	46.11 kPa
8	BL1 BLAST ON ROOF 383 KN/M2	383 kPa
9	BL2 BLAST ON EXTERNAL WALLS 383 KN/M2	383 kPa
10	LL2 CONSTRUCTION SURCHARGE	20.0 kPa
11	BL3 BLAST IN THE STAIR SHAFT ON W6 AND W7 383 KN/M2	383 kPa, acting out of the shaft on both faces

Load case 11 is Modification M1, expressed as a load. The model previously applied blast to the **external** walls only. BL3 is the demand Finding F1 identified — the stair shaft equalises to full p_{so} and therefore loads W6 and W7 from inside — and it is included in combination 103. **M1 exists to carry this case.**

8.3.1 The modelling warning that must not be lost

Spring supports in STAAD take tension unless explicitly declared compression-only. In this model the mat therefore **never lifts**, in any load case. **The flotation check of Part 9.3 cannot be performed in STAAD and is not.** It is a hand calculation on real dimensions and real weights. *Anyone who says "the analysis shows no uplift" has misunderstood their own model.*

8.4 The sentry post model

Joints	Twelve — base 1, 2, 11, 12 at Y = +0.450; first floor +100; roof +200
Members	16 — 1–8 columns PRIS YD 0.35 ZD 0.35; 9–16 beams PRIS YD 0.45 ZD 0.25
Supports	1 2 11 12 FIXED — all four column bases fully fixed
Member releases	NONE. The command appears nowhere in the file — every beam–column joint is fully continuous, which is what a moment frame requires and what the portal-method forces in Part 11.1 assume
Infill	Modelled as MEMBER LOAD only , never as a strut
Slab S1	Not modelled as plates — the slab load is applied to the beams as triangular / trapezoidal loading
Analysis command	PERFORM ANALYSIS PRINT STATICS CHECK — no P-Delta

Fixed bases are justified, not assumed: the columns sit on isolated footings **on in-situ rock**. Rotational restraint is real. **On backfill it would not be defensible** — which is precisely why the post is sited ≥ 10 m clear of the shelter excavation.

The statics check passes in every case — difference between loads and reactions = 0.000.

L/C	ΣF_x	ΣF_y	ΣF_z	ΣM_x	ΣM_y
1 (DL)	0.000	-624.548	0.000	1561.369	0.000
2 (LL)	0.000	-76.376	0.000	190.941	0.000
3 (EQ+X)	73.180	0.000	0.000	0.000	182.950
5 (EQ+Z)	0.000	0.000	73.180	436.631	-146.360

Two confirmations taken from the reaction table and used directly in design:

- 1 $F_x = 18.295$ kN per column $\times 4 = 73.18$ kN
-- the portal-method distribution used in design is EXACTLY what the model produces.
- 2 $F_y = \pm 36.127$ kN (EQ+X) and ± 27.891 kN (EQ+Z)
-- these ARE the seismic axial couples used in the column design.
They were READ from the model, not calculated by hand.

And one difference, stated rather than smoothed over. Back-calculating the model's own vertical distribution from its load resultants gives $Q_{roof} = 55.57$ kN and $Q_{floor} = 17.61$ kN, against the hand wh^2 distribution's 59.51 and 13.67. **The design used the hand distribution at the model's total of 73.18 kN, which gives the LARGER ground-storey column moment. Conservative, and recorded.**

8.5 Mesh sensitivity — MS1

Three mesh-density variants of the reconciled M1 model were built: **COARSE** ($\sim 1.9 \times$ the reference element size), **MEDIUM** (an unmodified copy of the reference mesh) and **FINE** (an exact $h/2$ refinement).

Geometry — including every M1 value — thicknesses, materials, supports and soil springs, loads, load cases and combinations are **identical in all three**; only the plate mesh density differs. The opening boundaries and the footprint are pixel-identical across all three: **net roof area 104.048 m² and hole area 15.792 m² match exactly**. Corner spring K_{FY} values and the $EP1$ earth-plus-water row pressures were **re-derived per mesh from the same physical formulas the reference model itself encodes** —

```
KFY = k_s x A_trib
EP1 p = 15.4071 x depth - 10.8198 kN/m2, depth = 6.700 - Y
fitted from the reference model's own six data points, residual <= 0.01
```

— not re-guessed.

The convergence conclusion is OUTSTANDING. The three models are built and validated; **STAAD.Pro is not available, so no result exists for any of them.** Saying the mesh converges would be inventing the study's conclusion.

8.6 MS2 — the input-file corrections, and why they matter

The project owner ran the COARSE model in STAAD.Pro — **the first time any model in this project had been opened in STAAD** — and it reported errors. All four box models were then re-read line by line.

#	Defect	Fix
MS2-1	Three <code>LOAD 6</code> south-wall lines were 80 columns against the file's own <code>INPUT WIDTH 79</code> . STAAD reads 79 columns and discards the rest, so it read <code>-77.6 / -57.3 / -37.0</code> where the file says <code>-77.64 / -57.36 / -37.07</code> . The north wall (79 columns) was read correctly, so <code>LOAD 6</code> no longer balanced — 4 848.4 kN in +Z against 4 843.6 kN in -Z, an out-of-balance of 4.8 kN on a case that is self-equilibrating by construction , and the file asks for exactly the statics check that reports it	Split on the STAAD – continuation character. Every data line in all four box models is now ≤ 79 columns
MS2-2	<code>LOAD 10</code> was defined after <code>LOAD 11</code> in all four models, so the primary case numbers ran 1...9, 11, 10	Reordered. Physically neutral — combinations reference cases by number and 101–105 are byte-identical before and after
MS2-3	Corner spring K_{FY} at joint 1 was 27 562 where the model's own formula gives 27 562.5 and the study documentation gives 27 563 , while the sibling half-value was rounded up	27 563 . A 0.5 kN/m difference on one spring; the point is that the file now agrees with its own documentation
MS2-4	A load title and one <code>JOB CLIENT</code> line also exceeded 79 columns (titles only, no numeric consequence)	Shortened

Verified after the edits, on all four box models: the 336 / 1 138 / 4 552-plate topology is unchanged and clean — every element resolves to defined joints, no repeated node, no degenerate or non-planar quadrilateral, no duplicate coordinate or element, no unused joint, no gap in the numbering; every joint still satisfies the model's own numbering rule; and, **re-read as STAAD reads them at 79 columns, COARSE, MEDIUM and FINE now produce identical panel-by-panel load resultants for all eleven load cases** — mat, roof, each of the four external walls, W6 and W7 — **which they did not before**.

MS2 changed no analysis model. No geometry, thickness, material, support stiffness, load magnitude, load case number or combination factor moved. **It is a lesson about input files, and it is the kind of defect that only shows up when someone actually opens the model.**

8.7 What must be obtained before the STAAD work can be extended

1. **A STAAD.Pro run.** Everything below waits on it.
2. The underground model's **post-processing output** — plate M_x/M_y , support reactions, deflections.
3. The **mesh convergence table**, which is the conclusion MS1 exists to produce.
4. **Both k_s bounds run** — the upper-bound model exists and has never been executed.
5. The **sentry post re-run after SP-B1** ($WM-V6$) — confirmation, not risk: brick is lighter than the modelled infill, so every member is over-designed.

PART 9 — STRUCTURAL DESIGN CALCULATIONS: THE UNDERGROUND BOX

Every value in Parts 9 to 11 has been computed numerically in this project, and the drawings carry exactly this reinforcement. No drawing shows an arrangement that differs from these calculations.

The flexure equations used throughout — IS 456 Annex G-1.1(b) and Cl. 38.1:

$$\begin{aligned} \mu &= 0.87 f_y \cdot A_{st} \cdot d \cdot [1 - (A_{st} \cdot f_y) / (b \cdot d \cdot f_{ck})] \\ x_u &= 0.87 f_y A_{st} / (0.36 f_{ck} b) && \text{IS 456 Cl. 38.1} \\ x_{u,max} / d &= 0.46 \text{ for Fe500} && \text{IS 456 Cl. 38.1(f)} \\ \mu_{lim} &= 0.133 f_{ck} b d^2 && \text{IS 456 Annex G-1.1(c)} \end{aligned}$$

Notation: T16 @ 150 EF EW = 16 mm Fe500D deformed bars at 150 mm centres, Each Face, Each Way. 4L = four-legged links.

9.1 Perimeter walls W1-W4 — 600 thk — 68 %

INPUTS	t 600, clear height 3200, cover 75 earth / 40 internal, T16 d = 600 - 75 - 8 = 517 mm Blast 383 kPa (Ka 1.0); static earth+water 83.2 kPa at base -> BLAST GOVERNS 4.6 : 1
FORMULA	$M_p = w \cdot L_n^2 / 16$ (fixed-fixed plastic mechanism)
SUBSTITUTE	$M_p = 383 \times 3.200^2 / 16$
RESULT	$M_p = 245.1 \text{ kNm/m}$
CHECK	$\mu_{lim} = 0.133 \times 43.75 \times 1000 \times 517^2 = 1556 \text{ kNm/m}$ -> singly reinforced OK $A_{st,req} (f_y, dyn 625, f_{ck}, dyn 43.75) = 894 \text{ mm}^2/\text{m}$ IS 456 Cl. 32.5(a) min vertical $0.0012 \times 600 = 720 \text{ mm}^2/\text{m}$ IS 456 Cl. 32.5(b) min horizontal $0.0020 \times 600 = 1200 \text{ mm}^2/\text{m}$ IS 3370 Pt 2 surface zone 0.35 % x 250 each face = 875 mm ² /m/face
ADOPTED	T16 @ 150 c/c EACH FACE EACH WAY = 1340 mm ² /m $\mu = 362.8 \text{ kNm/m}$ -> UTILISATION 68 % $x_u = 46.3$ -> $x/d = 0.089 < 0.46$
SHEAR	V at d = 383 (1.600 - 0.517) = 414.8 kN/m $\tau_{av} = 414.8e3 / (1000 \times 517) \text{ Cl. 40.1} = 0.802 \text{ N/mm}^2$ $\rho_t = 0.259 \% \rightarrow \tau_{uc} \text{ (Table 19, M35)} = 0.375 \text{ N/mm}^2$ $\tau_{uc,max} \text{ (Table 20, M35)} 3.70 > 0.802 \text{ OK Cl. 40.2.3}$ $V_{us} = (0.802 - 0.375) \times 517 = 220.8 \text{ kN/m}$ $A_{sv}/s_v = 220800 / (0.87 \times 500 \times 517) \text{ Cl. 40.4(a)} = 0.982 \text{ mm}^2/\text{mm}$ -> T12 2-leg at 230 Cl. 26.5.1.5 limit min(0.75d = 388, 300) = 300
ADOPTED	T12 CLOSED LINKS @ 200 c/c (Asv/sv = 1.131 OK)
AXIAL	$N = 448.15 \times 2.50 + 3.2 \times 0.6 \times 25 = 1168 \text{ kN/m}$ $f = 1.95 \text{ N/mm}^2 = 11 \% \text{ of } 0.4 f_{ck}, dyn (17.5)$ -> P-M not critical
CRACK	IS 3370 Pt 2, limit 0.2 mm; surface steel 875 < 1340 OK

Why 600 mm, stated honestly. The wall is **not strength-governed** — it is 68 % utilised. 600 mm is set by five things at once, and none of them is bending:

1. **75 mm cover** against blinding and trimmed rock; 2. **congestion** — two curtains, closed links, waterstops and cast-in frames all in one section; 3. **IS 3370 crack control** under sustained hydrostatic load; 4. **the EMP double curtain at 150 mm** (Part 2.8.2); 5. **the 1 168 kN/m axial path** from the roof.

Saying so matters: an optimiser who sees 68 % and thins the wall loses four of the five reasons it is that thick.

9.2 Walls W6 / W7 — 400 thk — Modification M1 — 69 %

9.2.1 Finding F1 — why these two walls are different

FINDING F1 The stair shaft is open to atmosphere through the 2800 x 3160 roof void, the headhouse and the entry stairwell.

Fill time $V / (A \cdot c) = 105 / (3.3 \times 340) = 0.094 \text{ s}$
 against the positive phase duration $t_d \quad 0.13 - 1.33 \text{ s}$

-> THE SHAFT EQUALISES TO FULL p_{so} , and it does so in well under one t_d .

W6 and W7 therefore sit in the SAME PLANE as the 7-bar blast doors, separating a PRESSURISED shaft from Bay 6 / Bay 8 at AMBIENT.

That is the whole of Modification M1 in one paragraph. The protective boundary was not continuous: two 200 mm walls were standing where 7-bar doors stand. **The design does not pretend the shaft stays at ambient; it accepts that it fills and carries the consequence.**

9.2.2 Why 200 mm is impossible — not a detailing problem

$d = 200 - 40 - 6 = 154 \text{ mm}$
 $\mu_{u,lim} = 0.133 \times 43.75 \times 1000 \times 154^2 = 138.0 \text{ kNm/m}$
 Demand $M_p = 383 \times 3.200^2 / 16 = 245.1 \text{ kNm/m}$

$138 < 245 \rightarrow$ NO STEEL RATIO MAKES IT WORK

Two-way action was CHECKED BEFORE REJECTION:
 panel 3200×3800 , fixed on three edges, yield line
 $\sim 0.7 \times \text{one-way} = 172 \text{ kNm/m}$ -- still above 138.

The section is over its singly-reinforced capacity before any steel is chosen. This is the one place in the project where the answer was not "add bars" but "change the geometry" — and it is why M1 is a **modification**, not a detailing revision.

And the 400 mm could not be found anywhere convenient. It could not come out of Bay 8, because the 750 mm escape-shaft clearance rule (Part 5.6) fixes Bay 8's minimum width at 2 900 mm. It could not come out of the shaft, because $2\,800 = 2 \times 1\,200$ flights + 200 well exactly. **So the box got 400 mm longer**, which the one-way roof makes structurally free (Part 5.3).

9.2.3 Design of the 400 mm wall

$d = 400 - 50 \text{ (shaft face)} - 8 = 342 \text{ mm}$
 $M_p = 383 \times 3.200^2 / 16 = 245.1 \text{ kNm/m}$
 $\mu_{u,lim} = 0.133 \times 43.75 \times 1000 \times 342^2 = 680.6 \text{ kNm/m}$ OK
 $A_{st,req} = 1400 \text{ mm}^2/\text{m}$

ADOPTED T20 @ 150 EF EW $= 2094 \text{ mm}^2/\text{m}$
 $\mu_u = 355.3 \text{ kNm/m} \rightarrow$ UTILISATION 69 %, $x/d = 0.211$

SHEAR V at $d = 383 \text{ (1.600 - 0.342)} = 481.8 \text{ kN/m}$
 $\tau_{u,v} 1.409 \mid \tau_{u,c} 0.540 \text{ (pt } 0.612 \%) \mid \tau_{u,c,max} 3.70 \text{ OK}$
 $V_{us} 297.2 \text{ kN/m} ; A_{sv}/s_v 1.998 \rightarrow$ T12 4-leg at 226
 $Cl. 26.5.1.5 \text{ limit } \min(0.75d = 257, 300) = 257$

ADOPTED T12 4-LEGGED LINKS @ 200 c/c

BLAST DOOR OPENING 1200×2100
 Interrupted steel $2094 \times 1.2 = 2513 \text{ mm}^2/\text{face}$
 $\rightarrow 1256 \text{ each jamb}$

ADOPTED 4 No. T20 EACH JAMB EACH FACE (1257 mm^2), anchored L_d 800 beyond

HEADER 400×1100 over 1200 clear
 $w = 383 \times 2.1/2 = 402 \text{ kN/m}$
 $M = w L^2 / 12 = 48.2 \text{ kNm}$

$V = 241 \text{ kN} ; \tau_u = 0.578 \text{ N/mm}^2$
 ADOPTED 4-T20 TOP + 4-T20 BOTTOM, T12 4-leg links @ 150

FRAME Cast-in steel frame anchored into and WELDED to the cage (EMP).
 Door: proprietary, $\geq 7 \text{ bar}$, rebound-rated, gas-tight [vendor]

9.3 Mat foundation — 600 thk — 84 %, the most heavily worked element

9.3.1 Bearing governs nothing

BEARING Service $58.4 \text{ kPa} = 1.8 \%$ of 3240 kPa
 Blast $404.9 \text{ kPa} = 12.5 \%$ (20.6 % on the measured soaked value -- Part 4.3.2)
 BEARING GOVERNS NOTHING.

9.3.2 Two flexural cases, and the second one sizes the mat

```

CASE 1 -- NET UPLIFT (COMB 102)
    u = 4.700 x 9.81                = 46.11 kPa
    self 0.600 x 25                 = 15.00 kPa
    net 31.1 kPa UP over the 5000 clear span
    Mp = 31.1 x 5.0^2 / 16          = 48.6 kNm/m
                                      NOMINAL

CASE 2 -- SOFT / RED-BOLE ZONE      *** GOVERNS ***
    A 3.0 m band of red-bole or vesicular material removed,
    at the WORST POSITION.
    q = 404.9 kPa (blast bearing) over a 3.0 m unsupported span
    M = q . L^2 / 12 = 404.9 x 9 / 12    = 303.7 kNm/m
    d = 600 - 75 - 8                  = 517 mm
    Ast,req                          = 1115 mm2/m
ADOPTED    T16 @ 150 EF EW = 1340 mm2/m
    Mu = 362.8 kNm/m  ->  UTILISATION 84 %

THICKNESS STUDY
    500 -> Ast,req 1407 = 105 % of provided    FAIL
    600 -> 1115 = 84 %                        ADOPTED
    700 -> 926 = 70 %
    800 (as once drawn) ~ 57 % = 33 % unused reserve

```

Case 2 is the geology, expressed as a load case. The flow contacts in a Deccan Trap sequence are exactly where a soft seam sits, and a single red-bole band under the mat removes support over its width. **That case, not bearing and not uplift, is what sets the 600 mm mat and its T16 @ 150 EF EW.** It is also why Part 4.2.2's unlogged 5.3 m matters so much: **the case that sizes the mat is a case nobody has looked for at the founding horizon.**

9.3.3 One-way shear — conflict C6, and the link grid it produced

```

ONE-WAY SHEAR -- NOT CHECKED IN THE PHASE 1 REPORT (CONFLICT C6)
    V at d from the soft-band edge
    = 404.9 (1.500 - 0.517)          = 398.0 kN/m
    tau_v 0.770 | tau_c 0.375 (pt 0.259 %) | tau_c,max 3.70    OK
    *** tau_c read at the STATIC M35 value --
        IS 4991 Cl. 10.3.1.1 gives NO dynamic increase on shear ***
    Vus = 204.2 kN/m ; Asv/sv        = 0.908 mm2/mm

ADOPTED    T12 CLOSED LINKS ON A 250 x 250 GRID THROUGHOUT
    supplied Asv/sv = (4 x 113.1) / 250 = 1.810 = 2.0 x required
    -- and the grid DOUBLES AS THE SPACER SYSTEM between the two curtains

```

9.3.4 The checks that do not apply, declared rather than ignored

Check	Verdict
Punching , IS 456 Cl. 31.6	NOT APPLICABLE. No column or pedestal bears on the mat; the headhouse walls bear on the roof . <i>Declared, not ignored</i>
Sliding	The box is socketed ~4.8 m into basalt (5.3–5.9 m on the measured rockhead) with the 300 mm annulus backfilled in M15 lean concrete. Not a credible mechanism — the reasoning is stated and no spurious factor of safety is computed
Overturning	Buried box 22.0 × 6.2 × 4.7 m under a symmetrical lateral load. NOT APPLICABLE
Settlement	Negligible on sound basalt, IS 12070 Cl. 6 — and Part 2.5.3 gives the physical reason. The hazard is the flow contacts, not the settlement

9.3.5 Flotation — the construction stage governs

Stage	Weight	Uplift	FoS @ (-)2.000	FoS flooded
1 mat cast only	2 009 kN	6 175 kN	0.33 FAIL	0.23 FAIL
2 mat + walls, no roof	4 802 kN	6 175 kN	0.78 FAIL	0.55 FAIL
3 box complete, no backfill	7 528 kN	6 175 kN	1.22 MARGINAL	0.86 FLOATS
4 backfilled + cover	12 453 kN	6 175 kN	2.02 OK	1.41 OK

Require FoS >= 1.2.
 ULS COMB 102 = 0.9 x 12453 / (1.5 x 6175) = 1.21

Uplift acts on the REAL UNDERSIDE, not on the plate model's mid-surface area.

*** THE ABSOLUTE FIGURES ABOVE ARE AT THE PRE-M1 21.600 BOX ***
 21.600 x 6.200 = 133.92 m2 -> 46.11 x 133.92 = 6175 kN (table)
 22.000 x 6.200 = 136.40 m2 -> 46.11 x 136.40 = 6289 kN (CURRENT)

Both uplift and resistance scale with length ($\times 22.0/21.6$),
 so EVERY FoS IS UNCHANGED: $0.325 / 0.778 / 1.219 / 2.017$,
 and COMB 102 stays $0.9 \times 12684 / (1.5 \times 6289) = 1.21$
 THE HAND CHECK USES $136.40 \text{ m}^2 / 6289 \text{ kN}$.

Mandatory mitigation — a design output, not a contractor's problem:

1. **Continuous dewatering** from the start of excavation until backfill and cover are complete.
2. **Temporary pressure-relief valves / knock-out plugs in the mat, 6 No., grouted up only after backfill.**
3. **Programme the sub-structure to complete before the monsoon**, or bund and positively drain with standby pumping and generator back-up.
4. **Backfill SYMMETRICALLY, in layers.**

And mitigation 1 is NOT met by the current programme — the departure is recorded, in writing. The owner's schedule ends dewatering on 11-05-27; side backfill runs 20-07-27 to 30-07-27 and the burster slab is not cast until 21-08-27. **The gap spans the whole 2027 monsoon with the box at stage 3 — FoS 1.22 at the design water table and 0.86 flooded to grade.**

Owner ruling 1 (RC4): the residual flotation risk is ACCEPTED IN WRITING. No date moves. B.3 mitigations 2, 3 and 4 stay mandatory. This is a departure from mitigation 1, recorded as one — which is the only honest way to carry it.

9.4 Pressure (roof) slab — 900 thk — THE GOVERNING ELEMENT — 51 %

9.4.1 One-way, and why that is the licence for M1

ONE-WAY ARGUMENT Aspect ratio $20.8 / 5.0 = 4.2$
 IS 456 Table 26 is tabulated only to $l_y/l_x = 2.0$,
 so TWO-WAY ACTION CANNOT BE CLAIMED.
 -> M_p depends on L_n of the 5 m WIDTH only.
 -> LENGTHENING THE BOX IS STRUCTURALLY FREE (basis of M1)
 Widening 5.0 -> 6.0 m would raise the roof moment 44 %.

9.4.2 Natural period — the calculation that puts this in the quasi-static regime

$m = 0.900 \times 2500 + 0.25 \times 2.0 \times 2000 = 3250 \text{ kg/m}^2$
 $0.5 \text{ EIG} = 0.5 \times 2.958 \times 10^8 \times 0.900^3 / 12 = 8.985 \times 10^8 \text{ N.m}^2/\text{m}$
 $f_1 = (22.373 / 2 \pi) \cdot \sqrt{EI / (m \cdot L_n^4)} = 74.9 \text{ Hz}$
 $T = 13.4 \text{ ms}$
 $t_d / T = 10 \text{ to } 100 \rightarrow \text{QUASI-STATIC}$

9.4.3 Flexure

FLEXURE $d = 900 - 75 - 12.5 = 812.5 \text{ mm}$
 $M_p = w \cdot L_n^2 / 16 = 448.15 \times 25 / 16 = 700.2 \text{ kNm/m}$
 $M_{u, \text{lim}} = 0.133 \times 43.75 \times 1000 \times 812.5^2 = 3841 \text{ kNm/m}$ OK
 $A_{st, \text{req}} (\text{pt } 0.20 \%) = 1632 \text{ mm}^2/\text{m}$
 Cl. 26.5.2.1 minimum $0.12 \% \times 900 = 1080 \text{ mm}^2/\text{m}$
 Cl. 26.5.2.2 max bar dia $D/8 = 112$ OK for T25

 ADOPTED T25 @ 150 EF EW = 3272 mm^2/m
 $M_u = 1362.4 \text{ kNm/m} \rightarrow \text{UTILISATION } 51 \%$
 $x_u = 113.0 \rightarrow x/d = 0.139 < 0.46$

 *** $x/d = 0.139$ IS THE PROOF THAT $\mu = 5$ IS DEFENSIBLE ***

9.4.4 Shear, including the direct-shear check

SHEAR V at the support face $= 448.15 \times 2.50 = 1120 \text{ kN/m}$
 V at d from the face $= 448.15 \times (2.50 - 0.8125) = 756.3 \text{ kN/m}$
 $\tau_{u,v} 0.931 \mid \tau_{u,c} 0.450 (\text{pt } 0.403 \%) \mid \tau_{u,c, \text{max}} 3.70$ OK
 $V_{us} 390.8 \text{ kN/m} ; A_{sv}/s_v 1.106 \rightarrow \text{T12 4-leg at } 409$
 Cl. 26.5.1.5 limit 300 GOVERNS

 ADOPTED T12 4-LEG @ 250 IN THE END 1500 EACH SIDE
 T12 2-LEG @ 300 ELSEWHERE

 DIRECT SHEAR $V_{d, \text{max}} = 0.16 f_{ck, \text{dyn}} b d = 5688 \text{ kN/m}$
 against 1120 applied = 19.7 %
 [UFC 3-340-02 Sec. 4-30, cited as US criteria, NOT IS 456]

Direct shear is a blast-specific failure mode that IS 456 does not address. At a support, a very short, very high shear pulse can shear the section off cleanly before flexural response develops. It is checked here against a US criterion, **and the criterion is named as a US criterion** rather than blended into the Indian Standard work.

9.4.5 Why 900 mm and not 800

t	d	Mp	Ast,req	tau_v	verdict
600	512.5	689	2673	1.71	links at 164, congested
800	712.5	696	1868	1.12	the STRUCTURAL OPTIMUM
900	812.5	700	1633	0.93	ADOPTED
1000	912.5	704	1453	0.78	11 % over

900 is adopted over the structural optimum of 800 for three stated reasons:

1. **Reserve at the two 1 400 mm collars**, where the slab is locally interrupted;
2. **Headroom against a raised design basis threat while the yield is unconfirmed** — the honest consequence of Part 2.1.3's [N];
3. **Support rotation kept inside 2° without resorting to UFC lacing reinforcement.**

9.5 Roof openings

9.5.1 The two 1 400 dia escape-shaft openings

```

Interrupted steel = 3272 x 1.400                = 4581 mm2/face/direction
Trimmers each side = 4581 / 2                  = 2291 mm2

ADOPTED  5 No. T25 EACH SIDE, EACH FACE, EACH DIRECTION (2454 mm2)
          anchored Ld = 1000 beyond the opening
          Collar 250 RC; T16 @ 150 hoops (3 layers) + T16 @ 150 radials
          Slab thickened 900 -> 1200 over a 600 mm annulus

```

Circular is the right shape: hoop action, and no re-entrant stress concentration. This detail is what SETS the 750 mm clearance rule (Part 5.6) — 250 collar + ~400 trimmer band + 100 tolerance — and that rule then propagates all the way back into the bay widths and into what Modification M1 was allowed to do.

9.5.2 The 2 800 × 3 160 stair void — Finding F2

The void leaves an 1840 mm CANTILEVER of 900 slab, 2800 wide, at 448.15 kPa.

```

M(root) = 448.15 x 1.840^2 / 2                  = 758.6 kNm/m
*** THIS EXCEEDS THE 700.2 kNm/m MIDSPAN Mp ***

```

```

Ast,req TOP = 1772 mm2/m < 3272 provided -> UTILISATION 56 % OK
V(root) = 448.15 x 1.840                      = 824.6 kN/m
tau_v 1.015 > tau_c 0.450 ; Vus 459.4 kN/m ; Asv/sv 1.300
-> T12 4-leg at 348 ; Cl. 26.5.1.5 limit 300

```

```

ADOPTED  T12 4-LEG LINKS @ 250 THROUGHOUT THE PAD          (NEW requirement)
          plus Free edge thickened 900 -> 1200 over 600, with 6-T25 top +
          6-T25 bottom and T12 closed links @ 150
          plus 6 No. T25 each face, top and bottom, in a 900 band over W6 and W7
          plus Diagonal trimmers 4 No. T25 each face at 45 deg at BOTH re-entrant
          corners, 2000 long each way, anchored Ld beyond
          plus 1100 mm guarding to the free edge             (NBC 2016 Pt 4)

```

Finding F2: the cantilever root moment exceeds the midspan moment of the element it hangs off. The flexural steel already provided happens to cover it at 56 % — but **the shear did not**, and the links through the pad are a new requirement that the Phase 1 report did not carry. **A 1.84 m cantilever of 900 mm slab at 448 kPa is not a detail; it is a structural element, and it is now designed as one.**

PART 10 — STAIRS, HEADHOUSE AND APPURTENANCES

10.1 Main staircase, Bay 7 — FROZEN GEOMETRY

STATUS: INSIDE the protective envelope. NOT a blast element. The shaft pressurises, but that pressure acts on the shaft walls and on the blast doors — **not as a net load on a slab that is open on both faces.** The stair is therefore designed to IS 456 with normal partial factors, **not** with IS 4991 dynamic strengths. That is stated explicitly on the drawing.

Geometry check — NBC 2016 Part 4:

Item	Provided	Limit	Verdict
Riser	170.833 mm	≤ 190	OK
Tread / going	280 mm	≥ 250	OK
Flight width	1 200 mm	$\geq 1\ 000$	OK — stretcher-capable
Risers per flight	8	≤ 12 preferred	OK
Headroom	2 533 mm	$\geq 2\ 200$	OK
Total rise	4 100 = 24 × 170.833	(-)6.100 → (-)2.000	closes exactly

```

FLIGHT -- waist 200
  theta = arctan(170.833 / 280) = 31.4 deg,  cos theta = 0.8535
  waist self    0.200 x 25 / 0.8535          = 5.858 kPa
  steps         0.5 x 0.170833 x 25   IS 456 Cl. 33.2 = 2.135 kPa
  finishes      = 1.000 kPa
  live          IS 875 (Pt 2), plant + escape route = 5.000 kPa
  total 13.99  ->  wu = 1.5 x 13.99          = 21.0 kPa

  Leff = going + min(landing/2, 1000) each end  IS 456 Cl. 33.1(b)
        = 1960 + 600 + 600                      = 3160 mm
  d = 200 - 30 - 6                                = 164 mm
  M = wu . Leff^2 / 8                              = 26.2 kNm/m
  Ast,req 380 ;  Cl. 26.5.2.1 min 0.12 % x 200 = 240

ADOPTED  T12 @ 150 MAIN (754)  ->  Mu 50.3,  UTILISATION 52 %
         T10 @ 200 DISTRIBUTION (393)

DEFLECTION  Leff/d = 19.3 ; basic 20 ; fs = 0.58 x 500 x 380/754 = 146
             pt 0.46 % ; MF ~ 1.9 -> permissible 38 > 19.3      OK
SHEAR  V = 33.2 kN/m ; tau_v 0.202 ;
        tau_c 0.479 x k 1.20 (Cl. 40.2.1.1)          = 0.575  OK, no links
TOP STEEL  T12 @ 150 for 0.25 Leff = 800 into the flight, anchored Ld 480

LANDINGS L1 / L2 / ARRIVAL -- 200 thk, spanning ACROSS the shaft
  self 7.50 + finishes 1.50 + live 7.50 + flight reaction 27.70 = 44.20 kPa
  Leff = 2800 clear + 200 bearing      Cl. 22.2(a)              = 3000 mm
  M = 44.2 x 3.000^2 / 8                  = 49.7 kNm/m
  Ast,req 745 ;  T12 @ 150 gives 754 = 100 %  -- TOO TIGHT

ADOPTED  T12 @ 125 BOTTOM (905)  ->  Mu 59.5,  UTILISATION 84 %
         T12 @ 125 TOP for 900 from each support
         T10 @ 200 distribution

DEFLECTION  3000/164 = 18.3 ; fs 239 ; pt 0.552 % ; MF ~ 1.35
             -> 27 > 18.3                                     OK
SHEAR  V 66.3 kN/m ; tau_v 0.404 < tau_c 0.519 x 1.20 = 0.622  OK

```

T12 @ 150 on the landings gives exactly 100 % utilisation, and 100 % is not a design. It was tightened to @ 125. That is the kind of decision that never appears in a summary and always appears in a site query.

Compatibility notes. The arrival landing at (-)6.100 is the mat surface — there is no separate slab. Flight 3 lands directly on the 900 mm pressure-slab pad at (-)2.000. The landings prop W6 and W7 at (-)4.733 and (-)3.367, **but only over their 1 200 mm depth — and the 400 mm wall design in Part 9.2 does NOT rely on that prop.**

10.2 Entry (approach) stairwell

STATUS: OUTSIDE the protective boundary, NOT blast rated, and expected to be LOST in the design event. Designed to IS 456 with normal partial factors.

```

FLIGHT -- 12R @ 166.6667 / 300, 1500 wide, waist 250
  theta = 29.05 deg,  cos theta = 0.8742
  waist 7.150 + steps 2.083 + finishes 1.000 + live 5.000 = 15.233
                                     -> wu      = 22.85 kPa
  Leff = 3300 + 750 + 750           IS 456 Cl. 33.1(b) = 4800 mm
  d = 250 - 30 - 8                  = 214 mm
  M = 22.85 x 4.800^2 / 8            = 65.8 kNm/m
  Ast,req 744 ; min 0.12 % x 250 = 300

ADOPTED  T16 @ 200 MAIN (1005) -> Mu 87.3, UTIL 75 %, x/d 0.162
         T10 @ 200 DISTRIBUTION (393)
         T16 @ 200 TOP for 1200 into the flight, anchored Ld 640

DEFLECTION  4800/214 = 22.4 ; basic 20 ; fs 215 ; pt 0.470 %
            MF ~ 1.5 -> 30                                OK
SHEAR  V 54.8 kN/m ; tau_v 0.256 ; tau_c 0.484 x k 1.10 = 0.532 OK

SIDE WALLS 250 -- retained 2.9 m, propped by the roof and the raft
  sigma_h at base = 0.5 x 20 x 2.900 + 0.5 x 10          = 34 kPa
  M ~ w L^2 / 12 on an equivalent 20 kPa UDL             = 14.0 kNm/m
  d = 250 - 50 - 6 = 194 ; Ast,req 168
  Cl. 32.5(a) min 0.0012 x 250 = 300 ; Cl. 32.5(b) 0.0020 x 250 = 500
  Cl. 32.5(c) two curtains required (t > 200)           OK
ADOPTED  T12 @ 200 EF EW (565/face)
         -- MINIMUM STEEL GOVERNS, 31 % utilised
SHEAR  V ~ 49 kN/m ; tau_v 0.253 < tau_c 0.484         OK

RAKING ROOF 250 -- 1500 clear
  self 6.25 + waterproofing/screed 2.0 + earth lap 5.4
  + IMPOSED 20
                                     -> wu      = 50.5 kPa
  (20 kPa imposed taken DELIBERATELY, for a stray vehicle on the berm)
  M = w L^2 / 12                                     = 9.5 kNm/m
  d = 250 - 30 - 6 = 204 ; Ast,req 108 ; min 300 GOVERNS
ADOPTED  T12 @ 200 EF EW -> Mu 48.2, 20 % utilised
SHEAR  V 38 kN/m ; tau_v 0.186                        OK

TOP LANDING 250 (0.000)
  wu 54.9 kPa ; Leff 1750 ; M 21.0 kNm/m ; Ast,req 229
ADOPTED  T12 @ 200 EF EW (565) -> Mu 50.6, 41 % utilised

PLATFORM 250 ((-)2.000, on fill)   ADOPTED  T12 @ 200 EF EW, nominal
HEADWALL 250                       ADOPTED  T12 @ 200 EF EW
  Door lintel over 1000 clear: 250 x 350, 3-T12 top + 3-T12 bottom, T8 @ 150
STEPPED RAFT 300 on compacted fill ADOPTED  T12 @ 200 EF EW

```

10.2.1 The opening corner — the detail most often got wrong

At the TOP of the flight the tension face turns through 209 degrees.

A bar bent round that corner has its bend resultant directed OUT of the concrete: it spalls the cover and the bar loses its anchorage.

MAIN BARS SHALL NOT BE BENT ROUND IT.

Each layer is continued STRAIGHT, CROSSED, and anchored Ld = 640 into the OPPOSITE face, plus a U-bar T16 @ 200 across the corner.
[SP 34:1987 Cl. 5.5 and standard detailing practice]

At the FOOT of the flight the corner CLOSES (151 degrees) -- bars may turn normally.

This is a geometry problem, not a preference. An obtuse re-entrant corner in a tension face puts the resultant of the bend into the cover, and the cover has nothing to push against. Two corners, two different answers, and the drawing shows both.

10.3 Entry headhouse — roof, 500 thk — 90 %, the highest utilisation in the project

```

w = 383 (blast) + 12.5 (self) + 1.0 (SIDL)              = 396.5 kPa
Panel 4000 x 5000 clear, monolithic with 400 walls on all four
sides,  ly/lx = 1.25
d = 500 - 75 - 10                                       = 415 mm

```

```

Mu,lim = 0.133 x 43.75 x 1000 x 415^2 = 1002 kNm/m

THREE ANALYSES, AND THE MOST CONSERVATIVE IS ADOPTED
| # | Method | M |
| 1 | One-way fixed-fixed strip, Mp = w . Ln^2 / 16 | 396.5 | <- ADOPTED
| 2 | IS 456 Table 26 Case 1, alpha_x- = 0.045 at 1.25 | 285.5 |
| 3 | Yield line, FIXED on 4 edges, isotropic | 162.2 |

Method 3 formula: wu = 48 m / [ a^2 ( sqrt(3 + (a/b)^2) - a/b )^2 ]
VALIDATION: as b -> infinity, 48/3 = 16 = the fixed-fixed strip w L^2/16
(the 24-coefficient version gives 8 = SIMPLY SUPPORTED w L^2/8)

Ast,req (one-way basis) = 1879 mm2/m
Cl. 26.5.2.1 minimum 0.12 % x 500 = 600 mm2/m

ADOPTED T20 @ 150 EF EW = 2094 mm2/m -> Mu = 438.5 kNm/m
UTILISATION 90 % one-way | 65 % Table 26 | 37 % yield line
xu 72.3 -> x/d = 0.174 << 0.46

SHEAR *** NEW AND MANDATORY -- NOT IN THE PHASE 1 REPORT ***
V at d = 396.5 (2.000 - 0.415) = 628.4 kN/m
tau_v 1.514 | tau_c 0.502 (pt 0.505 %) | tau_c,max 3.70 OK
(tau_c read at the STATIC M35 value -- IS 4991 Cl. 10.3.1.1)
Vus 420.0 kN/m ; Asv/sv 2.327 -> T12 4-leg at 194
Cl. 26.5.1.5 limit min(0.75d = 311, 300) = 300
ADOPTED T12 4-LEG @ 175 IN THE END 1200 EACH SIDE; @ 250 ELSEWHERE

DETAILING
No opening in the headhouse roof (the stair void is in the FLOOR).
Corner torsion steel (Cl. D-1.8) NOT required -- all four edges carry
full T20 @ 150 hogging steel continuous into the walls.
Haunch 300 x 300 at all four wall-roof junctions, diagonal T16 @ 150.
Top steel continuous over every support, anchored Ld = 800 into the wall.

```

Three analyses, and the reason the crudest one is adopted. The one-way strip is the most conservative of the three by a factor of 2.4 against the yield-line value. A yield-line solution is only as good as the assumption that all four edges are genuinely fixed and that the panel can develop a full mechanism; under a blast load on a structure whose neighbour is expendable, that is not a bet worth taking for a saving of one bar size. **The project takes the strip, prints all three, and says which one it used.**

The corrected yield-line coefficient, recorded as an error. The yield-line value was once computed as 324.4 kNm/m using the **simply-supported** coefficient 24 instead of the **fixed-four-edges** coefficient 48. It is **162.2**. The validation that catches it is in the formula itself: as $b \rightarrow \infty$, $48/3 = 16 = wL^2/16$, which is the fixed-fixed strip. **No reinforcement changed, because the one-way value was the one adopted and nothing depended on the yield-line number** — but the error is recorded rather than quietly fixed.

And the shear is new. $\tau_v = 1.514$ against $\tau_c = 0.502$ is a factor of three. The Phase 1 report did not check it. **This is conflict C2, and the links are mandatory.**

10.4 Headhouse walls HW1–HW4 — 400 thk — conflict C10 — 59 %

	CASE A (report)	CASE B (ADOPTED)
Lateral pressure	113 kPa drag	383 kPa full envelope
$M_p = w \times 2.400^2 / 16$	40.7 kNm/m	137.9 kNm/m
$A_{st,req}$	221 mm ² /m	766 mm ² /m
Cl. 32.5(a) min vertical	480 mm ² /m -- governs	480
Cl. 32.5(b) min horizontal	800 mm ² /m -- governs	800
Utilisation on T16@150 EF EW	17 %	59 %
Shear V at d	97.0 kN/m	328.6 kN/m
τ_v	0.283	0.961
τ_c (pt 0.392 %)	0.444	0.444
Verdict	NO LINKS	LINKS REQUIRED

$d = 400 - 50 - 8 = 342$; $M_{u,lim} = 681$ kNm/m

ADOPTED T16 @ 150 EF EW = 1340 mm²/m -> M_u 235.2, UTIL 59 %, x/d 0.135
 Vus 176.9 kN/m ; Asv/sv 1.189 -> T12 4-leg at 381
 Cl. 26.5.1.5 limit min(0.75d = 256, 300) = 256 GOVERNS

ADOPTED T12 4-LEGGED LINKS @ 250 c/c THROUGHOUT ALL FOUR WALLS

Two-way check (information only, fixed on 4 edges):
 HW1/HW2 4000 x 2400 -> $m = 69.9$ kNm/m (one-way strip 137.9)
 HW3/HW4 5000 x 2400 -> $m = 79.8$ kNm/m (one-way strip 137.9)

AXIAL Roof reaction, two-way (Cl. 24.5):
 HW3/HW4 trapezoid 2379 kN / 5.0 m = 476 kN/m
 HW1/HW2 triangle 1586 kN / 4.0 m = 396 kN/m
 One-way bound: $396.5 \times 2.0 = 793$ kN/m ; self 29 kN/m

Worst $N = 822 \text{ kN/m} \rightarrow f = 2.06 \text{ N/mm}^2$
 $= 12 \% \text{ of } 0.4 f_{ck, \text{dyn}}$ OK

*** PRESSURE ACTS FROM INSIDE TOO. The inner security door is NOT blast rated and the entry stairwell is expected to be lost, so the headhouse FILLS and the walls are pushed OUTWARDS. THE WALLS ARE REINFORCED SYMMETRICALLY FOR 383 kPa EITHER WAY -- a stated requirement, not an accident of detailing. ***

DOOR OPENING 900 x 2100 IN HW2 -- and a geometric trap
 Interrupted steel 1340 x 0.900 = 1206 mm²/face $\rightarrow$ 603 each jamb
 ADOPTED 2 No. T20 EACH JAMB EACH FACE (628), anchored Ld 800 beyond

Door head at $(-)2.000 + 2.100 = +0.100$; roof soffit +0.400
 $\rightarrow$ ONLY 300 mm OF WALL ABOVE THE DOOR -- too shallow for a lintel.

The 300 wall and the 500 roof act TOGETHER as an 800 mm deep edge band:
 $w = 383 \times 2.100/2 = 402 \text{ kN/m}$; $M = w L^2/12 = 27.1 \text{ kNm}$; $V = 181 \text{ kN}$
 $d = 800 - 40 - 8 - 10 = 742$; $\tau_{av} = 0.610 \text{ N/mm}^2$
 ADOPTED 400 x 800 EDGE BAND: 4-T20 TOP + 4-T20 BOTTOM,
 T12 4-LEG @ 150, over the opening and 600 each side

Door frame cast in and WELDED to the cage (EMP), even though the leaf is a security door and is not blast rated.

10.4.1 HW3 — the wall that has nothing under it

The pressure slab spans one-way in Y over 5000 clear. HW3 runs IN the Y direction, so its load is carried by a slab strip acting as a beam.

$b_{\text{eff}} = 0.400 + 2 \times 1.05$ (45 deg spread through the 900 slab) = 2.5 m [A]

CASE 1 -- two-way roof distribution (IS 456 Cl. 24.5, correct)

Line load 476 (trapezoid) + 29 (self) = 505 kN/m
 Equivalent UDL = $505/2.5 + 22.5 + 2.0$ = 226 kPa
 $M = w \cdot L_n^2 / 16 = 226 \times 25 / 16$ = 354 kNm/m
 Against the slab's 1362 kNm/m capacity = 26 % OK

CASE 2 -- one-way roof bound

Line load 793 + 29 = 822 kN/m $\rightarrow$ 353 kPa $\rightarrow$ $M = 552 \text{ kNm/m}$ = 41 % OK

ADOPTED No slab change needed. DETAILED AS A LINE LOAD, NOT AS A SUPPORT.

Robustness: 4 No. T25 ADDITIONAL TOP AND BOTTOM in a 1200 mm band beneath HW3, lapped 1250 (50 phi) into the main mesh.

$b_{\text{eff}} = 2.5 \text{ m}$ is an assumption, and it is closed on the margin. Owner ruling 24 (RC6) accepts it: the strip is at 26 % on the correct two-way distribution and 41 % on the conservative one-way bound, so **more than half the capacity is spare** and b_{eff} would have to be badly wrong to matter — assumption A11.

10.5 Escape shaft head hatches — RC5-F2, the finding nobody had asked for

The protective boundary is defined as the blast doors plus the perimeter walls, the mat and THE PRESSURE SLAB. ESC 1 and ESC 2 are 1 400 mm dia bores straight through the pressure slab. They begin in bay 1 — inside the gas-tight envelope — and in bay 8, and they finish at grade.

Each shaft head is therefore a 1.54 m² hole in the protective boundary, and the only thing that can close it is a hatch that did not exist. The EMP penetration register had listed both as FAIL and prescribed a bonded conducting hatch — but it prescribed it for EMP. Nothing anywhere in this project stated what the head has to resist STRUCTURALLY.

The design was silent, so owner ruling 18 (RC5) took the only defensible reading: the head is part of the boundary and takes the full 383 kPa design blast.

total force on the leaf $383 \times \pi \times 0.700^2$ = 589.6 kN
 $M = w a^2 (3 + \nu) / 16$ = 38.71 kNm/m
 $V \text{ at the seating} = w a / 2$ = 134.1 kN/m

flat Fe250 plate $t = \sqrt{6M / \sigma} = 30.5 \rightarrow$ say 32 mm
 mass of a 1600 dia x 32 leaf = 505 kg

505 kg is the point, not the thickness. Half a tonne cannot be lifted by a person escaping up a 6.8 m ladder in the dark. That is why the answer is not a flat plate.

ADOPTED: a ribbed steel weldment — 1 600 dia, 12 mm face, 8 No. radial ribs 150 × 10 and a 150 × 12 perimeter ring — on a steel seating ring cast into the 250 mm collar, bearing 150 mm all round, with **four quarter-turn dogs so the leaf resists UPLIFT as well as downward pressure** (the negative phase and the rebound both lift it), and **counterbalanced or spring-assisted, openable from inside by one person without a key or a tool**. Indicative leaf mass ≈ 322 kg — still a mechanically assisted item.

Not designed and not invented [N]: the rib proportioning (an orthotropic plate problem — the flat-plate demand above is exact and is **the brief**), the counterbalance mechanism, and every EMP figure.

And the consequence of splitting it the way the ruling splits it, stated plainly. The EMP treatment for this opening is a **bonded conducting hatch**. Bonding is not a finish applied later — it is continuity between the leaf, the seating ring and the collar reinforcement, and it has to be designed **into** the weldment and the seat. **Ruling the structure and the bonding into different packages is exactly how these two shaft heads came to be undesigned for four revisions.** EM-V6 stays open for its EMP half.

10.6 The escape shaft ladders

Owner ruling 2 (RC4): ladder only; fall-arrest deferred. Before it, the project held **no position at all** on how either shaft was climbed — no ladder, no rung, no fall-arrest anywhere in any document, against a **6.250 m** climb out of Bay 1 and a **6.800 m** climb out of Bay 8.

One ladder type serves both shafts:

```
20 mm dia galvanised MS rungs, 400 clear width, EQUAL PITCH within each shaft
  ESC 1   297.6 mm over 21 spaces
  ESC 2   295.7 mm over 23 spaces
>= 200 behind the rung .    >= 750 clear climbing space in front
2 No. 50 x 10 galvanised flat stringers
cast-in lugs to the 250 collar and the roof-slab bore,
expansion-anchored brackets at 1.5 m over the lower 3.200 m
grab rails 1100 above the head
```

IS 3696 (Part 2) and NBC 2016 Part 4 are named **by title only** — neither is in the workspace and no unconfirmed clause is cited.

FS-V7 **DOES NOT CLOSE. IT CHANGES** — and three things are still missing, and the ruling knows it:

- **no fall-arrest** — deferred by the ruling; - **no rest platform** — a 1 400 mm bore cannot take one without blocking the escape; - **the injured-person question** — **a vertical ladder cannot pass a stretcher**, and whether a casualty is expected to use a shaft at all is a client question.

PART 11 — THE SENTRY POST

11.1 Why it is not blast designed, and what that decision cost

The sentry post is NOT designed for the blast, and that is a RECORDED decision, not an omission.

Reflected pressure on the 4.0 x 6.7 m face
 $p_r \times A_e = 1366 \text{ kPa} \times 26.8 \text{ m}^2 = 36\,600 \text{ kN}$ (about 3730 t)
 Hardening would need roughly 700 mm of RC on all four above-ground faces.

Not justified for a peacetime observation post. The post is declared expendable, and the shelter's protection does not depend on it in any way.

All member design uses the STAAD base shear $V_b = 73.18 \text{ kN}$, not the hand-calculated 59.3 kN (Part 7.8.2).

11.1.1 Seismic member forces — portal method at $V_b = 73.18 \text{ kN}$

```
Q_roof = 73.18 x (315.9 x 6.25^2) / Sigma = 59.51 kN
Q_floor = 73.18 x (276.8 x 3.20^2) / Sigma = 13.67 kN

Column shear, ground storey = 73.18 / 4 (= STAAD Fx 18.295) = 18.30 kN
Column shear, first storey = 59.51 / 4 = 14.88 kN
Column moment, ground = 18.30 x 3.200 / 2 = 29.27 kNm
Column moment, first = 14.88 x 3.050 / 2 = 22.69 kNm

Beam moment, first-floor joint (corner joint, one beam)
= 29.27 + 22.69 = 51.96 kNm
Beam moment, roof joint = 22.69 kNm

Axial from overturning -- READ DIRECTLY FROM THE STAAD REACTIONS:
EQ+X Fy = +/- 36.127 kN EQ+Z Fy = +/- 27.891 kN
-> 36.127 GOVERNS
```

11.1.2 The frame analysis assumption, stated rather than hidden

Single-bay portal in each direction. Beams are NOT fully fixed; the joint rotates. Slope-deflection with symmetric loading:

```
I_beam = 250 x 450^3 / 12 = 1.8984e9 mm4
I_col = 350^4 / 12 = 1.2505e9 mm4
Kc (first-floor joint) = 4EI/3200 + 4EI/3050 = 3.203e6 . E
M_support = -FEM + (2 E I_b / L_b) . theta,
theta = FEM / (2 E I_b / L_b + Kc)
```

Beam	FEM	2EI/L	Reduction	M_support	M_midspan
B1, 1.5(DL+LL)	39.49	1.040e6 E	x 0.755	29.81	30.90
B1, DL only	22.53	--	x 0.755	17.01	--
B2, 1.5(DL+LL)	70.60	0.8165e6 E	x 0.797	56.26	49.64
B2, DL only	39.21	--	x 0.797	31.25	--

11.2 Slab S1 — 150 thk two-way

```
Cover 30 (Table 16, moderate) ; T8 bars
dx = 150 - 30 - 4 = 116 mm
dy = 150 - 30 - 8 - 4 = 108 mm
Clear spans 3400 and 4400

Effective span, IS 456 Cl. 22.2(a) -- the LESSER of clear + d, or c/c:
lx = min(3400 + 116, 3650) = 3516
ly = min(4400 + 108, 4650) = 4508
ly / lx = 1.282 < 2 -> TWO-WAY Cl. 24.4 / Annex D

MOMENT COEFFICIENTS -- IS 456 Table 26 (Annex D-1.1) Case 9: four edges
discontinuous but RESTRAINED, with corner torsion steel per Cl. D-1.8
-> Table 26, NOT Table 27.
alpha_x = 0.0777 (interpolated at 1.282) alpha_y = 0.056

wu = 1.5 (4.750 + 3.000) = 11.625 kPa
(the FIRST FLOOR governs; the roof is 10.125)

Mx = 0.0777 x 11.625 x 3.516^2 = 11.17 kNm/m
```

```

My = 0.0560 x 11.625 x 3.516^2          = 8.05 kNm/m
Ast,req x 229 | y 176
Cl. 26.5.2.1 min 0.12 % x 150          = 180 mm2/m
Cl. 26.5.2.2 max bar dia D/8 = 18.75    OK
Cl. 26.3.3(b) max spacing 3d or 300     OK

ADOPTED T8 @ 150 c/c BOTH WAYS BOTTOM   = 335 mm2/m
Mu = 16.09 (x) / 14.92 (y) kNm/m -> UTILISATION 69 % / 54 %
xu = 13.5 -> x/d = 0.116 / 0.125

DEFLECTION lx/dx = 3516/116 = 30.3 ; basic taken as SIMPLY SUPPORTED = 20
fs = 0.58 x 500 x 229/335 = 198 ; pt = 0.289 % ; MF (Fig. 4) ~ 1.68
Permissible 20 x 1.68 = 33.6 > 30.3      OK
(as continuous, basic = 26 and the margin doubles -- it passes on the
conservative assumption, so that assumption is the one used)

SHEAR Vu ~ wu . lx / 3 = 13.6 kN/m ; tau_v 0.117 ;
tau_c 0.390 x k 1.30 (D <= 150)         = 0.507 OK

```

Detailing — IS 456 Annex D:

Clause	Requirement	Adopted
D-1.2 / D-1.3	Middle strip = $\frac{3}{4}$ width, edge strips $\frac{1}{8}$ each side; Table 26 moments apply to MIDDLE STRIPS ONLY	—
D-1.4	50 % of midspan bottom steel to within 0.1 L of a discontinuous edge; remainder curtailed at 0.25 L	—
D-1.6	At every discontinuous edge, top steel = 50 % of the midspan bottom steel, extending 0.1 L = 400 mm into the span	T8 @ 300 TOP, 400 mm band, all four edges
D-1.7	Edge strips: minimum reinforcement only	—
D-1.8	Torsion at corners — both edges discontinuous at all four corners → steel = $\frac{3}{4}$ of the midspan area, in four layers (2 top + 2 bottom, each way), over $l_x/5 = 700$ mm each way	T8 @ 200, 4 LAYERS, 700 × 700 AT EACH CORNER

The torsion mats are a HOLD POINT for reinforcement inspection. Table 26 is valid **only** because they are provided. Omit them and the slab must be reassessed on Table 27, which gives larger coefficients. **This is why the drawing carries a panel headed "why Table 26 and not Table 27".**

11.3 Beam B1 — 250 × 450, span 3 650

```

IS 13920:2016 section limits
b >= 200                                250      OK
b/D = 0.556 >= 0.3                       OK
clear span / D = 3300/450 = 7.33 >= 4     OK

d = 450 - 30 - 8 - 8                     = 404 mm
Mu,lim = 0.133 x 30 x 250 x 404^2         = 162.8 kNm

| Combination                               | M_support hogging |
| 1.5(DL + LL)                             | 29.8 |
| 1.2(DL + LL + EL) = 1.2(19.87 + 51.96)   | 86.2 |
| 1.5(DL + EL) = 1.5(17.01 + 51.96)         | 103.5 <- GOVERNS |
| 0.9 DL + 1.5 EL                           | 93.2 |

Mu = 103.5 < 162.8 -> singly reinforced ; Ast,req = 661 mm2
ADOPTED 4-T16 TOP AT SUPPORTS (804) -> Mu = 122.6 kNm, UTIL 84 %, x/d 0.321

BOTTOM
Reversal 1.5(17.01 - 51.96) = -52.4 kNm sagging -> Ast,req 313
IS 13920 Cl. 6.2.3: positive steel at the joint face >= 50 % of negative
= 0.5 x 804 = 402 mm2 <- GOVERNS
Midspan sagging 30.9 kNm -> Ast,req 181
IS 13920 Cl. 6.2.1(b) rho_min = 0.24 sqrt(fck)/fy
= 0.00263 x 250 x 404 = 266
IS 456 Cl. 26.5.1.1 As,min = 0.85 b d / fy = 172
IS 13920 Cl. 6.2.2 rho_max = 0.025 = 2525 OK
Cl. 6.2.4 top or bottom >= 25 % of max at the joint face = 201 OK
ADOPTED 2-T16 BOTTOM CONTINUOUS (402) -> Mu = 66.0 kNm, x/d 0.160

SHEAR -- CAPACITY DESIGN, IS 13920 Cl. 6.3.3
Vu = 1.2(D+L) . L/2 +/- 1.4 (Mu^As + Mu^Bh) / L
Unfactored gravity total = 54.29 (UDL) + 25.81 (triangle) = 80.1 kN
1.2 x 80.1 / 2 = 48.1 kN
1.4 x (122.6 + 66.0) / 3.650 = 72.3 kN
Vu = 120.4 kN

```

```

tau_v = 1.192 N/mm2 ; tau_c,max (Table 20, M30) 3.50 OK

*** IS 13920 Cl. 6.3.4: EQ share 72.3 / 120.4 = 60 % >= 50 %, no axial
-> tau_c TAKEN AS ZERO in the plastic hinge region ***

Asv/sv = 120.4e3 / (0.87 x 500 x 404) = 0.685 -> T8 2-leg at 147
Cl. 6.3.5.1 hinge region (2d = 810 from each face):
sv <= min(d/4 = 101, 8 x 16 = 128), need not be < 100 -> 100 mm
Cl. 6.3.5.2 elsewhere: sv <= d/2 = 202 mm

ADOPTED T8 2-LEGGED HOOPS @ 100 c/c OVER 2d = 810 FROM EACH FACE;
T8 @ 150 c/c ELSEWHERE. First hoop within 50 mm of the column face.

DEFLECTION l/d = 3650 / 404 = 9.0 << 20 OK

```

Capacity design, in one sentence: the beam is designed for the shear that accompanies its own flexural failure, not for the shear from the analysis. That is what $1.4 (M_u^As + M_u^{Bh}) / L$ means — the hinges form at over-strength and the shear follows. **And when the earthquake share exceeds half the total, IS 13920 Cl. 6.3.4 requires τ_c to be taken as zero:** a hinge that has cycled has lost its aggregate interlock, so the links carry everything.

11.4 Beam B2 — 250 × 450, span 4 650 — 89 %

```

| Combination | M_support hogging |
| 1.5(DL + LL) | 56.3 |
| 1.2(DL + LL + EL) = 1.2(37.51 + 51.96) | 107.4 |
| 1.5(DL + EL) = 1.5(31.25 + 51.96) | 124.8 <- GOVERNS |
| 0.9 DL + 1.5 EL | 106.1 |

Mu = 124.8 < 162.8 OK ; Ast,req = 822 mm2
ADOPTED 3-T20 TOP AT SUPPORTS (942) -> Mu = 139.8 kNm, UTIL 89 %, x/d 0.376
*** 3-T20 CHOSEN OVER 5-T16: five T16 bars need 256 mm in a 250 mm
wide beam. That is exactly the check that becomes a site RFI
if it is skipped. ***

BOTTOM Cl. 6.2.3 -> 0.5 x 942 = 471 ; midspan 49.6 kNm -> Ast,req 297
ADOPTED 2-T20 BOTTOM CONTINUOUS (628) -> Mu = 98.9 kNm, 50 %, x/d 0.250

SHEAR Unfactored gravity (UDL-equivalent) = 26.115 x 4.650 = 121.4 kN
1.2 x 121.4 / 2 = 72.8 kN
1.4 x (139.8 + 98.9) / 4.650 = 71.9 kN
Vu = 144.7 kN
tau_v = 1.433 ; tau_c,max 3.50 OK

EQ share 71.9 / 144.7 = 49.7 % -- a hair under the Cl. 6.3.4 trigger.
*** TAKEN AS TRIGGERED (tau_c = 0). A design must not turn on the
third decimal place of a ratio. ***

Asv/sv = 0.823 -> T8 2-leg at 122 ; hinge spacing 100 governs
Outside the hinge V falls to ~124 kN and tau_c = 0.641 (pt 0.933 %)
may be used -> Asv/sv 0.337 -> T8 at 298
ADOPTED T8 @ 100 OVER 810 FROM EACH FACE ; T8 @ 150 ELSEWHERE
DEFLECTION l/d = 4650 / 404 = 11.5 << 20 OK

```

11.5 Column C1 — 350 × 350

```

SLENDERNESS
clear height, ground = 3650 - 450 - 450 (beam depth) = 2750 mm
Effective length factor 1.2 (Table 28, unbraced, both ends restrained)
lex = 3300 ; lex/D = 9.43 < 12 -> SHORT COLUMN Cl. 25.1.2 OK

MINIMUM ECCENTRICITY Cl. 25.4
emin = 1/500 + D/30 = 5.5 + 11.67 = 17.2, but >= 20 -> 20 mm
Mmin = 276.8 x 0.020 = 5.5 kNm << applied OK

AXIAL, per column, unfactored: DL 148.4 kN, LL 19.1 kN,
EQ axial +/- 36.127 kN
1.5(DL + LL) = 251.3 kN
1.5(DL + EL) = 276.8 kN <- max
1.5(DL - EL) = 168.4 kN <- min

LONGITUDINAL STEEL Cl. 26.5.3.1: min 0.8 % = 980 ; max 6 % = 7350
ADOPTED 8-T16 = 1608 mm2 -> p = 1.31 %, bars on all four faces
Cover 40 (Cl. 26.4.2.2) ; d' = 40 + 8 + 8 = 56 ; d'/D = 0.16

```

UNIAXIAL CAPACITY -- computed from the IS 456 Cl. 38.1 stress block and the Fe500 design curve (Fig. 23B), NOT read off a chart:

Pu (kN)	Mu1 (kNm)
168.4	105.8
251.3	111.7
276.8	112.9

BIAXIAL -- IS 456 Cl. 39.6
 $P_{uz} = 0.45 f_{ck} A_c + 0.75 f_y A_{sc}$
 $= 0.45 \times 30 \times 120892 + 0.75 \times 500 \times 1608 = 2235 \text{ kN}$
 $P_u / P_{uz} = 276.8 / 2235 = 0.124 < 0.2 \rightarrow \alpha_n = 1.0$
Worst case, EQ in Z with 30 % in X IS 1893 Cl. 6.3.2.2:
 $M_{ux} = 25.6 \text{ kNm}$; $M_{uy} = 66.8 \text{ kNm}$
 $25.6/112.9 + 66.8/112.9 = 0.227 + 0.592 = 0.819 \leq 1.0 \quad \text{OK}$

STRONG COLUMN - WEAK BEAM IS 13920 Cl. 7.2.1, $\Sigma M_c \geq 1.4 \Sigma M_b$

Joint	ΣM_b	$1.4 \Sigma M_b$	ΣM_c	Verdict
First floor, X (B1)	122.6	171.6	$112.9 + 101 = 213.9$	OK
First floor, Z (B2)	139.8	195.7	213.9	OK
Roof, Z (B2 roof, 2-T20)	98.9	138.5	101	OK, marginal

TRANSVERSE REINFORCEMENT
General, Cl. 26.5.3.2: tie dia $\geq \max(6, \phi/4) = 8$;
spacing $\leq \min(350, 16 \phi = 256, 300) = 256$
IS 13920 Cl. 7.4: hoops over the full length at $\leq$ half the least dimension $= 175$
ADOPTED T8 HOOPS @ 150 c/c outside the confining zone

SPECIAL CONFINING REINFORCEMENT IS 13920 Cl. 8.1
 $l_o \geq \max(350, \text{clear height}/6 = 458, 450) = 500 \text{ mm}$
spacing $\leq 0.25 \times 350 = 87.5$, but $75 \leq s \leq 100 = 85 \text{ mm}$
 $A_{sh} = 0.18 s h (f_{ck}/f_y) [(A_g/A_k) - 1]$
 $h = 350 - 2 \times 40 = 270$; $A_g = 122500$; $A_k = 270^2 = 72900$
 $= 0.18 \times 85 \times 270 \times (30/500) \times (1.680 - 1) = 168.5 \text{ mm}^2$
minimum $= 0.05 \times 85 \times 270 \times 30/500 = 68.9 \text{ mm}^2$
ADOPTED T10 HOOPS + ONE CROSS-TIE EACH WAY @ 85 c/c
(3 legs $\times 78.5 = 235.5 \text{ mm}^2$ OK)
over 500 mm from every joint face, top and bottom of every column,
AND THROUGH THE JOINT Cl. 8.2

DRIFT -- IS 1893 Cl. 7.11.1
limit $0.004 \times \text{storey height} = 12.8 \text{ mm (ground), } 12.2 \text{ mm (first)}$
Calculated drift under 3 mm. Not critical, but CHECKED.

The roof joint at 101 against 138.5 is marginal and is printed as marginal. Strong column-weak beam is the mechanism check that keeps the hinges in the beams; at the roof, where only one column continues above, the margin is naturally smallest. It passes, and it is flagged so that nobody reduces the roof beam's bottom steel without re-running it.

11.6 Isolated footing F1 — 1 500 × 1 500 × 600 on in-situ basalt at (–)2.000

Founded on IN-SITU ROCK, never on backfill. This is the whole reason the sentry post was relocated $\geq 10 \text{ m}$ clear of the shelter excavation — and Part 4.5's measured black cotton soil, CH at FSI 60–65 %, is why that insistence is now backed by a measurement rather than by instinct.

SERVICE $P = 167.5 \text{ (column)} + 36.1 \text{ (EQ axial)} + 33.75 \text{ (footing self)} = 237.4 \text{ kN}$
 $M \text{ (base, EQ)} = 29.3 \text{ kNm}$
 $e = 29.3 / 237.4 = 0.123 \text{ m} < L/6 = 0.250 \rightarrow \text{NO TENSION OK}$
 $q_{max} = P/A (1 + 6e/L) = 105.5 \times 1.494 = 157.6 \text{ kPa}$
vs presumptive SBC 3240 kPa (IS 1904 Table 1) [A] $= 4.9 \% \text{ OK}$
vs measured SOAKED basalt 1961 kPa $= 8.0 \% \text{ OK}$

ULS Governing $1.5(DL + EL)$: $P_u = 1.5(148.4 + 36.127) = 276.8 \text{ kN}$
 $M_u = 1.5 \times 29.27 = 43.9 \text{ kNm}$
 $e_u = 0.128$; $q_{u,max} = 276.8/2.25 \times (1 + 6 \times 0.128/1.5) = 190.0 \text{ kPa}$

FLEXURE Cantilever from the column face $= (1500 - 350)/2 = 575 \text{ mm}$
 $M_u = 190.0 \times 0.575^2 / 2 = 31.4 \text{ kNm/m}$
 $d = 600 - 50 \text{ (Cl. 26.4.2.1)} - 8 = 542 \text{ mm}$
 $A_{st,req} 136$; Cl. 26.5.2.1 min $0.12 \% \times 600 = 720 \text{ } < \text{GOVERNS}$

ADOPTED T12 @ 150 c/c BOTH WAYS BOTTOM (754 mm²/m), $x/d 0.056$

ONE-WAY SHEAR IS 456 Cl. 34.2.4.1, critical at d from the column face:
 $575 - 542 = 33 \text{ mm from the edge}$; $V_u = 6.3 \text{ kN/m}$; $\tau_{u,v} = 0.012 \text{ OK}$

```

PUNCHING  IS 456 Cl. 31.6.1, critical perimeter at d/2:
          b0 = 4 x (350 + 542) = 3568 ; Vu = 276.8 - 190.0 x 0.892^2 = 125.6 kN
          tau_v = 0.065 ; ks = (0.5 + beta_c) <= 1.0 = 1.0 ;
          tau_c = 0.25 sqrt(30) Cl. 31.6.3.1 = 1.369
          1.369 >> 0.065 OK

ANCHORAGE Column starters T16: Ld, compression = 37 x 16 = 592 mm
          Available 600 - 50 - 24 = 526 straight + 8 phi bend 128 = 654 mm OK

*** THE 600 mm DEPTH IS GOVERNED BY THE ANCHORAGE OF THE COLUMN STARTERS,
    NOT BY BENDING AND NOT BY SHEAR -- stated so that nobody "optimises"
    it to 350. ***

```

11.7 Lintel L1 and the wall ties — SP-B2

SP-B1 made the sentry post walls masonry but left **no lintel design and no tie detail**, so the walls as recorded **could not be built**. Both are now designed. **This adds nothing to the frame and changes no frame member.**

```

LINTEL L1 -- ONE TYPE OVER EVERY OPENING IN THE SENTRY POST

190 wide x 150 deep, M30 / Fe500, cover 30, bearing 200 each end
2-T10 bottom . 2-T8 top (hangers) . T6 two-legged links @ 150

Openings served -- ELEVEN in all:
ground D1 900 and W1 1200
first D1 900 and the EIGHT 1200-wide vision-panel openings
The 1200 opening governs; one type covers all.

```

Design basis — and the reason it does not depend on an unknown.

```

Effective span = min(clear + d, c/c bearings) = min(1315, 1400)
              IS 456 Cl. 22.2 = 1.315 m

*** THE OPENING HEIGHTS ARE NOT STATED ON ANY DRAWING -- WM-V3, still open.
L1 is deliberately designed to the bound that does not depend on the
height at all: masonry standing JUST BELOW the 60 deg arching height,
0.866 x 1.315 = 1.139 m
which is the HEAVIEST case any opening height can produce.
Above it, arching relieves the lintel; below it, there is less
masonry. THE DESIGN THEREFORE HOLDS WHATEVER WM-V3 IS RULED TO BE. ***

w = 0.190 x 20 x 1.139 = 4.33 kN/m
M = 0.935 kNm ; V = 2.85 kN
Mu = 1.403 kNm against Mu,lim = 0.133 fck b d^2 = 10.03 kNm
                                -> 14 % UTILISED

Ast required 29.5 mm2
IS 456 Cl. 26.5.1.1 minimum 0.85 b d / fy = 37.1 mm2 <- GOVERNS
2-T10 = 157 mm2 provided
tau_v = 0.195 against tau_c ~ 0.56 (M30, pt 0.72 %)
                                -> NO SHEAR STEEL REQUIRED
the T6 @ 150 links are the Cl. 26.5.1.6 nominal minimum
(maximum spacing 324)

*** THE LINTEL CARRIES MASONRY ONLY. The floor and the roof go to beams
    B1/B2 at each level, which is the whole point of the frame. ***

```

Wall ties. 6 mm dia MS ties at **every fifth course (≈ 450 mm)** up both column faces, projecting **200 mm** into the bed joint, anchored to the column by a cast-in or drilled-and-grouted 10 mm dowel. **The top course is built tight to the beam soffit and the last joint packed** — because the analysis takes **R = 3.0**, the infill is **not** separated from the frame **and must not be**. Separating it would be the R = 5.0 special moment frame case, which this project does not claim.

What SP-B2 does NOT change. No frame member, no footing, no slab, no storey height, no envelope. **Infill on the first-floor beams stays at 13.000 kN/m:** brick gives ≈ 9.88 kN/m, which is lighter, so the modelled value stays conservative. **WM-V6 is still open** — the structural discipline must re-run the seismic check, and nothing here is a substitute for that.

PART 12 — REINFORCEMENT REGISTER AND QUANTITIES

12.1 Underground box

Element	Size	Cover	d (mm)	Main reinforcement	Links	A _{st} prov	x _u /d	Util.
W1 south perimeter	600	75/50/40	517	T16 @ 150 EF EW	T12 closed @ 200	1340/face	0.089	68 %
W2 north perimeter	600	75/50/40	517	T16 @ 150 EF EW	T12 closed @ 200	1340/face	0.089	68 %
W3 west end	600	75/50/40	517	T16 @ 150 EF EW	T12 closed @ 200	1340/face	0.089	68 %
W4 east end	600	75/50/40	517	T16 @ 150 EF EW	T12 closed @ 200	1340/face	0.089	68 %
W5 Bay 5/6	200	40	154	T12 @ 150 EF EW	—	754/face	—	nominal
W6 Bay 6/7 (M1)	400	50/40	342	T20 @ 150 EF EW	T12 4L @ 200	2094/face	0.211	69 %
W7 Bay 7/8 (M1)	400	50/40	342	T20 @ 150 EF EW	T12 4L @ 200	2094/face	0.211	69 %
W8 partitions x4	110	25	—	A252 mesh both faces	—	252	—	non-structural
MAT	600	75/50	517	T16 @ 150 EF EW	T12 @ 250 × 250 grid	1340/face	0.089	84 %
ROOF midspan	900	75	812.5	T25 @ 150 EF EW	T12 4L @ 250 end 1500 / 2L @ 300 mid	3272/face	0.139	51 %
ROOF cantilever pad	900	75	812.5	T25 @ 150 top continuous	T12 4L @ 250 throughout	3272	0.139	56 %
Sump pit walls / base	300/400	50	244	T16 @ 150 EF EW	—	1340	—	nominal

Additional bars — underground box

Location	Requirement
Wall starters	T16 @ 150 EF, 900 mm horizontal leg into the mat , lap 800 (50 φ) above a 150 mm kicker
Haunches, all wall–roof and wall–mat junctions	500 × 500 , diagonal T20 @ 150
Mat edge, all free edges	T16 @ 150 U-bars closing both curtains
Sump pit opening	4-T20 trimmers each face, each side , L _d 800 beyond
Blast door jambs (W6, W7)	4-T20 each jamb, each face , L _d 800 beyond
Blast door header	400 × 1100: 4-T20 top + 4-T20 bottom, T12 4L @ 150
ESC 1 / ESC 2 collars	Collar 250 RC; 5-T25 each side, each face, each direction , L _d 1000 beyond; T16 @ 150 hoops × 3 layers + T16 @ 150 radials ; slab thickened 900 → 1200 over a 600 annulus
Stair void free edge	Thickened 900 → 1200 over 600; 6-T25 top + 6-T25 bottom, T12 closed @ 150
Stair void side bands	6-T25 each face, top and bottom, in a 900 band over W6 and W7
Stair void re-entrant corners (x2)	4-T25 each face at 45°, 2000 long each way , anchored L _d beyond
Band beneath headhouse wall HW3	4-T25 extra top and bottom in a 1200 band , lapped 1250
Construction joints (~6 m)	Reinforcement fully continuous ; two waterstops + welded Cu / galvanised EMP strap

12.2 Stairs

Element	Size	Cover	d	Main	Distribution	Top steel
Main stair flight waist	200	30	164	T12 @ 150 (754)	T10 @ 200 (393)	T12 @ 150 for 800 into the flight, L_d 480
Main stair landings L1/L2/arrival	200	30	164	T12 @ 125 bottom (905)	T10 @ 200	T12 @ 125 for 900 from each support
Entry stairwell flight waist	250	30	214	T16 @ 200 (1005)	T10 @ 200	T16 @ 200 for 1200 into the flight, L_d 640
Entry stairwell top landing	250	30	214	T12 @ 200 EF EW (565)	—	—
Entry stairwell platform	250	30	214	T12 @ 200 EF EW	—	on fill
Entry stairwell side walls	250	50	194	T12 @ 200 EF EW	—	minimum steel governs
Entry stairwell raking roof	250	30	204	T12 @ 200 EF EW	—	minimum steel governs
Entry stairwell headwall	250	30	214	T12 @ 200 EF EW	—	—
Entry door lintel (1000 clear)	250 × 350	30	306	3-T12 top + 3-T12 bottom	T8 @ 150	—
Stepped raft	300	50	244	T12 @ 200 EF EW	—	on compacted fill
Opening-corner U-bar (top of entry flight)	—	—	—	T16 @ 200 U-bar, bars CROSSED and anchored L_d 640 into the opposite face		SP 34 Cl. 5.5

12.3 Headhouse

Element	Size	Cover	d	Main	Links
Roof	500	75	415	T20 @ 150 EF EW (2094)	T12 4L @ 175 in the end 1200 each side; @ 250 elsewhere
Walls HW1–HW4	400	50/40	342	T16 @ 150 EF EW (1340)	T12 4L @ 250 throughout
Door jambs, HW2	—	—	—	2-T20 each jamb, each face, L_d 800 beyond	—
Door head edge band	400 × 800	40	742	4-T20 top + 4-T20 bottom	T12 4L @ 150, over the opening + 600 each side
Wall–roof haunch	300 × 300	—	—	diagonal T16 @ 150	—

12.4 Sentry post

Element	Size	Cover	d	Reinforcement
Slab S1 bottom	150	30	116/108	T8 @ 150 c/c BOTH WAYS (335 mm ² /m)
Slab S1 top, discontinuous edges	150	30	116	T8 @ 300, 400 mm (0.1 L) into the span, all four edges — Cl. D-1.6
Slab S1 corner torsion	—	—	—	T8 @ 200, FOUR LAYERS, 700 × 700 at all four corners — Cl. D-1.8
Beam B1 (3650)	250 × 450	30	404	4-T16 top at supports; 2-T16 bottom continuous
Beam B2 (4650)	250 × 450	30	404	3-T20 top at supports; 2-T20 bottom continuous
Beam hoops, both	—	—	—	T8 2-leg @ 100 over 2d = 810 from each face; @ 150 elsewhere; first hoop ≤ 50 from the face
Column C1	350 × 350	40	d' 56	8-T16 (1608 mm ² , p = 1.31 %)
Column confining hoops	—	—	—	T10 hoops + 1 cross-tie each way @ 85 c/c over 500 mm from every joint face, top and bottom, AND through the joint
Column general ties	—	—	—	T8 @ 150 c/c elsewhere
Plinth beam PB	250 × 400	30	354	3-T12 top + 3-T12 bottom
Footing F1	1500 × 1500 × 600	50	542	T12 @ 150 both ways bottom (754 mm ² /m)
Lintel L1 × 11	190 × 150	30	—	2-T10 bottom + 2-T8 top, T6 2-leg links @ 150, bearing 200 each end
Wall ties	—	—	—	6 mm MS at every 5th course (≈ 450), 200 into the bed joint, 10 mm dowel

12.5 The six that matter

ROOF	T25 @ 150 EF EW	.	T12 4L @ 250 end 1500 / 2L @ 300 mid	51 %
WALLS	T16 @ 150 EF EW	.	T12 closed @ 200	68 %

W6/W7	T20 @ 150 EF EW .	T12 4L @ 200	69 %
MAT	T16 @ 150 EF EW .	T12 @ 250 x 250 grid	84 %
HH ROOF	T20 @ 150 EF EW .	T12 4L @ 175 / 250	90 %
HH WALLS	T16 @ 150 EF EW .	T12 4L @ 250	59 %

12.6 Quantities — package SC1

Sentry post excluded — no sentry post bar, quantity or weight appears in the structural CAD schedules.

Bar	Unit mass kg/m	No. of bars	Total length m	Weight kg	% of total
T8	0.395	13	14.6	5.7	0.0 %
T10	0.617	85	102.6	63.3	0.1 %
T12	0.888	13 814	22 885.7	20 318.3	28.8 %
T16	1.578	3 698	15 020.2	23 707.1	33.6 %
T20	2.466	1 440	4 429.1	10 922.8	15.5 %
T25	3.853	654	4 006.7	15 439.4	21.9 %
		19 704	46 458.9	70 456.7	100 %

Element group	Weight kg	Weight t	%
FOUNDATIONS	14 665.2	14.665	20.8 %
WALLS	24 725.6	24.726	35.1 %
ROOF SLAB	20 490.5	20.490	29.1 %
BEAM-TYPE ELEMENTS	297.1	0.297	0.4 %
MAIN STAIRCASE	281.3	0.281	0.4 %
HEADHOUSE	8 364.2	8.364	11.9 %
ENTRANCE / ENTRY STAIRWELL	1 632.8	1.633	2.3 %
TOTAL	70 456.7	70.46	100 %

Concrete, in scope: mat 81.8 + pressure slab 112.0 (net of openings) + perimeter walls 103.7

- W6/W7 12.8 + W5 3.2 + headhouse 32.7 + main staircase 3.0 + entry stairwell 19.9 =

369.2 m³, excluding blinding and the sump pit.

Average steel density $\approx 191 \text{ kg/m}^3$. For a blast-hardened buried box whose bar spacing is set by the **150 mm EMP rule** rather than by strength, and whose sections are thick (900 roof, 600 walls and mat), that is a plausible figure. **It is a sanity check, not a design check.**

Quantity basis — declared, not assumed:

- **PBR-1** — cut length = Σ scheduled leg dimensions, **no bend deduction taken**; links add $2 \times 10 \phi$ for 135° hooks. Conservative by $\approx 2 \phi$ per 90° bend. Bend deductions to BS 8666 Table 3 shall be applied by the fabricator on the approved schedule. [A] — **no bar-bending standard is in the workspace; IS 2502 is NOT held and NOT cited.**
- **P-1** — stock bar 12 000 mm; longer runs split into equal pieces with 50ϕ laps added.
- **PBR-2** — a 4-legged link @ s is scheduled as **two** closed links per node on an s x s grid; a 2-legged link @ s as one.
- **PBR-3** — the stair void and the two escape openings are **deducted zone by zone**.
- Quantities are for **tender and estimating** and must be re-measured on the approved bar bending schedule.

Not quantified, and declared as such: the five blast valves with their sleeves and trimming (sizes are on the services sheet, not in the structural design); the service entry plate (size [N]); and the ventilation / CBRN duct penetrations (no penetration schedule exists). **All three are NOT DETERMINABLE from the information the project holds, and none is fabricated.**

PART 13 — DRAWINGS AND DRAWING QUALITY

13.1 Two distinct DXF sets — do not confuse them

Set	Origin	Revision	Purpose
Input set (10 files)	Supplied by the project owner	Rev F	Architectural / civil geometry — the primary geometry source . Directly editable
Output set (S-01 ... S-08)	Generated in this project	Phase 2 Rev A	Structural drawings with reinforcement. Generated — edit the script, never the DXF
Discipline sets	Generated in this project	per package	Structural CAD, drainage, HVAC, finishes, fire, concealment, EMP, electrical, site

13.2 The package as it stands: 80 DXF, 74 PASS

Discipline	Sheets
Architectural / general — Rev F	11 (A-101...A-301, plus the services sheet S-06)
Architectural — finishes	3 (A-601, A-611, A-612)
Structural — reinforcement	30 (R-001...R-805)
Drainage	11 (D-001...D-305) + 2 handout
HVAC	6 (M-001...M-203) + 2 handout
Fire and life safety	2 (F-101, F-102)
Site and concealment	1 (C-101)
EMP protection	6 (EM-001...EM-302)
Electrical	1 (E-001)
Site selection and geotechnical	5 (SG-001...SG-202)
TOTAL	80 DXF · 75 A1 · 4 A4 · 1 A0

The drawing index is generated from the DXF files themselves, so it cannot drift from the drawings. Filenames are deliberately unchanged — every document in the project cites the current filenames — so the drawing number lives in the title block and the index carries both.

13.3 Drawing QA/QC — QA1 and QA2

QA1 (9 September 2026) inspected and corrected 65 DXF in place, at the same filename and folder. Scope: drafting, annotation, sheet composition and title blocks only.

Measure	Before	After
DXF inspected / edited in place	—	65 / 65
Text-on-text overlaps, all 65 drawings	67	2
· of which, the 54 generated sheets	49	0
· of which, the 11 Rev F drawings	18	2
Annotation crossing hard line work (Rev F sheets)	50	10
A view drawn on top of its own notes panel	3 sheets	0
Entities outside the sheet border	0	0
Drawings with a border and title block	54 of 65	65 of 65
Panel / table body text below print size (1.4 mm on A1)	most of the package	0 — floor is now 2.0 mm
Generated packages rebuilding clean	54/54	54/54, 0 errors

No engineering value was changed anywhere by the QA pass. No dimension, level, bar mark, bar size, spacing, load, material grade, thickness or room size was touched, and the frozen staircase geometry was not touched.

QA2 (11 September 2026) re-scanned the whole package at **80 DXF, 74 PASS**, after **twelve sheets added by the EMP**, **electrical and site packages were found to be invisible to the index tool**.

That is a process failure with a process fix, and it is recorded as a working rule. A new drawing package must register itself with the QA tool — add its DXF/ prefix to DISCIPLINE in `qa_report_data.py` and its name to ORDER in `make_index.py`, then re-run both. Twelve sheets were invisible to the index for four revisions because this was not done.

13.4 Sheet standard

SHEET SIZE	A1, 841 x 594 mm, DRAWN IN PAPER MILLIMETRES, PLOT 1:1 (one A0 -- the front elevation, which is 880 mm at 1:50 and cannot fit A1; four A4 handout sheets)
DXF FORMAT	AutoCAD R12 ASCII -- opens in AutoCAD, DraftSight, BricsCAD, LibreCAD and QCAD without translation
BORDER	outer 0,0-841,594 ; inner 10,10-831,584
TITLE BLOCK	180 x 62 mm, bottom right, at x0 = 651, y0 = 10
VIEW SCALES	each view carries its own scale note; views are mapped from model millimetres to paper millimetres by $\text{paper} = \text{origin} + \text{model} / \text{scale}$
TEXT HEIGHTS	3.4 view titles 2.4 panel headings 2.0 scale notes 1.6-1.7 general 1.55-1.62 panel body 1.4-1.5 small
VALIDATION	SECTION/ENDSEC balanced, EOF present, no undeclared layer, all geometry within the sheet extents

22 layers, identical in every generated file — CONC, CONC-HIDDEN, REINF-MAIN (red), REINF-DIST (green), REINF-LINK (magenta), REINF-SEC (cyan), DIM, TEXT, TEXT-TITLE, HATCH, GRID, SOIL, CENTRELINE, TITLEBLOCK, WATERPROOF (orange), STEELWORK (yellow), LEVELS, NOTES, TABLE, SERVICES, BLAST (red), and 0. **Dashed lines are drawn as explicit segments, not by linetype**, for maximum compatibility.

13.5 Two rulings about the drawings that are worth reading

QA-1 — the front elevation cannot be plotted at 1:50 on A1, and A0 is the only answer. The drawing is 880 mm wide at 1:50 against an A1 printable width of 821 mm. **The scale is a measurement statement and must not be changed to fit the paper**, and splitting the sentry post onto its own sheet would break a continuous 44 m elevation, which is the whole point of the drawing. **A0 keeps both the scale and the drawing intact.**

QA-2 — six sheets do not fill their paper, and that is accepted as drawn. Six sheets share an empty strip. **Every one of them is correct, complete and legible; emptiness is not an error.** Re-scaling would make the stated scale a layout variable instead of a measurement statement — 1:20 is the right scale for a 400 mm wall section regardless of how much paper it leaves. Combining or renumbering would break cross-references, the index and the sheet numbering **for zero engineering benefit**. And reflowing the blocks **moves** the void, it does not remove it.

13.6 What cannot be regenerated, and is not invented

Seven of the eight S-series structural sheets are ABSENT and CANNOT be regenerated. The Part E.4 toolchain — `proj.py`, `dxflib.py` and the eight sheet generators — is **not in the workspace**. S-01 ... S-05, S-07 and S-08 therefore cannot be rebuilt, and **they are not fabricated**. S-06 is present and is already at M1.

Also not in this workspace and not invented: the Phase 1 Rev D report, `SK02_Underground_Plan.png`, and the nineteen STAAD screen captures.

And the sentry post beams, columns and footings are not yet drawn. A future sheet s-09 would carry B1 4-T16 / 2-T16, B2 3-T20 / 2-T20, C1 8-T16 with T10 confining hoops @ 85, and F1 1500² x 600 with T12 @ 150 B/W. **Everything it needs is in Part 11; the sheet does not exist.**

13.7 The two most recent drawing changes

DR-A1 and DR-A2 put the drainage items onto the Rev F architectural sheets — the sump pit on the side section, the septic tank and soak pit on the front elevation beyond a break, and then at DR-A2 the sump pit's **cover** at (-)6.100 and a **1:200 key plan** carrying the sump, the septic tank and the soak pit at true project X and Y. **No design value changed in either.**

And DR-A2 found something the drawing could not avoid asking. The mat is genuinely interrupted over the clean sump — the reinforcement register trims a **1 500 × 1 500 opening** with 4-T20 each face each side, the bar schedule calls it an opening, and the bill measures the mat gross by its 1.35 m³. **A 1 500 × 1 500 hole through the floor is what the project records, so that is what was drawn — and then Bay 5 is 1 560 mm clear and the pit is 1 500 of it.**

With the opening open there is no route past it to the two filter trains, the CO₂/O₂ plant or the dehumidifier; and the room finish schedule falls that room **1:80 direct to the sump**, so whatever closes it has to pass water. **A cover is not optional here — and the project contains no cover: no type, no depth, no duty, no fixing, no lifting arrangement, in any schedule, drawing or document.** The cover is drawn as one diagrammatic line, tagged [A], with its specification tagged [N], and DR-A2-V1 **was opened for it.** Nothing was converted.

PART 14 — SERVICES, CBRN, EMP AND LIFE SAFETY

14.1 HVAC and CBRN filtration

14.1.1 The basis, and the eleven figures that reproduce

The ventilation basis existed on the issued services sheet before the HVAC package started. **That package re-derived every figure from first principles rather than carrying them forward**, and all eleven reproduce:

Stated	Reproduced from	Result
Gas-tight envelope 67.8 m² / 217.0 m³	Bay widths 2900+1800+3500+1800+1560+2000 × 5.000 × 3.200	67.80 / 216.96
Clean zone 57.8 m²	Envelope less the 10.0 m ² decon airlock	57.80
Occupied volume, airlock shut 185.0 m³	57.80 × 3.200	184.96
FEMA 453 rate 264 m³/h	57.8 m ² = 622.2 ft ² × 0.25 cfm/ft ²	264.3 (0.10 %)
Leakage 32.5 m³/h	0.15 vol/h × 216.96	32.5 = 10.8 % of one train
Time to 1.0 % CO ₂ 9.9 h	(0.0096 × 184.96) / (9 × 0.02)	9.9
Airlock purge 13 min	5 × (2.0 × 2.0 × 3.2) = 64 m ³ at 300 m ³ /h	12.8 min
DN100 throat 10.6 m/s	300 / 3600 / (π × 0.05 ²)	10.61
DN350 throat 7.5 m/s	2600 / 3600 / (π × 0.175 ²)	7.51
Survival / working rates	5 × 9 = 45, 15 × 9 = 135 m ³ /h	reproduces
O ₂ store 15 m³	2 × 50 L at 150 bar	80 h at 4.5 m ³ /day

Every figure reproduces. The sheet is internally consistent.

14.1.2 Design basis

Parameter	Value	Class
Occupancy / endurance	9 persons / 96 h	[C]
Design flow	300 m³/h , 2 × 300 trains, true N+1	[C]
— against the three criteria	6.7 × survival · 2.2 × working · 1.14 × FEMA	[R]
— per person	33.3 m³/h	[R]
Air changes, whole envelope / clean zone	1.38 / 1.62	[R]
Leakage	≤ 0.15 vol/h at +300 Pa = 32.5 m³/h , 11 % of one train	[C]/[R]
Operating overpressure	+50 to +100 Pa , tested at +300 Pa	[C]
Cascade	0 → +10 → +20 → +35 → +50 Pa	[C]
Closed mode	CO ₂ 0.18 m ³ /h; 9.9 h unscrubbed, 48 h on soda lime, 80 h on O ₂	[C]/[R]
Airlock purge	64 m ³ , 12.8 min , 4–5 persons/hour	[R]
Ductwork + plenum loss	161 Pa	[R]
Total fan duty	VENDOR DATA REQUIRED — 5 of 8 components	[N]
Cooling load	See Part 2.11 and RC4-V1	—

Room-by-room distribution — developed in this project. The nine occupants are not in two places at once: they work in the ops room and sleep in the berthing bay. A single fixed split would starve whichever of the two is occupied, so the distribution is **balanced in two modes on two volume control dampers**:

Room	Use	Volume m ³	Day m ³ /h	Night m ³ /h	Extract
U-01	Emergency stores / ESC 1	46.40	30	30	—
U-02	Lavatory + medical	28.80	30	30	45
U-03	Ops room and hazard plotting	56.00	135	45	—
U-04	Berthing, 9 berths	28.80	45	135	—
U-05	CBRN plant + sump	24.96	60	60	—
U-06	Decon airlock	32.00	<i>300 transfer</i>	<i>300 transfer</i>	<i>to BV-3</i>
	TOTAL		300	300	

The occupied room of the pair gets $135 \text{ m}^3/\text{h} = 15.0 \text{ m}^3/\text{h}$ per person — exactly the working-shelter rate the sheet names as its second criterion. Holding that rate in *both* rooms at once would take 270 of the $300 \text{ m}^3/\text{h}$ and leave 30 for the stores, the lavatory and the plant room. If a reviewer requires that, the total must rise above $300 \text{ m}^3/\text{h}$ — which is a change to a confirmed value. Raised, not taken.

Nothing is supplied to Bay 6. The airlock is the exhaust path: the whole $300 \text{ m}^3/\text{h}$ transfers through the three decon stages and out at BV-3, which is what makes the confirmed cascade work. The cascade maps exactly onto the three airlock stages.

14.1.3 Findings

Ref	Finding
HV-F1	The soda lime, not the oxygen, limits closed mode — 9.9 h unscrubbed, 48 h on soda lime, 80 h on O_2 . The consumable to count, and the one to write on the drill card. Closed mode covers 48 of the 96 h; the other 48 h require filtration
HV-F2	The raw-air duct is a protective element, not a duct. The fresh-air blast valves are in the west wall of Bay 1 and the filter trains are in Bay 5. Between them the air is unfiltered, and the duct carries it 11.2 m through the clean zone. Over that length the duct wall is the only barrier. Positions are not changed — they are confirmed on an issued sheet — but the duct is specified as protective: fully welded, no push-fit or slip joint, tested to the +300 Pa envelope standard, run visible and never boxed in, labelled RAW AIR — UNFILTERED at $\leq 2 \text{ m}$ centres, re-tested after any work in bays 1–4. Should the trains move to Bay 1? — raised, not taken
HV-F3	Maintenance access is the tight dimension, not headroom. Each train is 1 450 wide in a 1 560 clear bay — 110 mm at the sides — so all access is along the bay, and a HEPA or carbon cassette must come in through Blast Door 1 (1200×2100) and along bays 6 and 5. Confirm the cassette dimensions against that route before the trains are ordered
HV-D1	No filter bypass for peacetime running. Without one, every peacetime hour spends carbon-bed life replaceable only in closed mode. A bypass is a deliberate leak path around the filters — a protective decision. Not added
HV-D2	BV-3 discharges into Bay 7, and the onward path to atmosphere — up the stair shaft, out through the headhouse — is not recorded anywhere . A path exists; its resistance is unknown and it is in series with the relief valve
HV-D4	No noise criterion exists anywhere in the project. Terminal selection cannot be closed, and a berthing space for nine is exactly where a noisy diffuser is felt
HV-D6	Fan static pressure — 5 of 8 loss components are vendor data. The derivable part is 161 Pa. Quoting a total without the vendor figures would be a fabricated number . The fan must be selected on the dirty figure, and the hand crank sized on the same duty

14.1.4 Operating modes

Mode	Condition
1 NORMAL	Peacetime, unfiltered or lightly filtered
2 FILTERED / PROTECTIVE	Full seven-stage train, overpressure held
3 CLOSED	Detonation to all-clear. All five blast valves shut . Soda lime and O_2 only. 48 h limit
4 PURGE	Airlock purge cycle
5 GENERATOR RUNNING	BV-4 and BV-5 open . Bay 8 only — does not touch the gas-tight envelope. Independent of modes 1–4

Modes 3 and 5 contradicted each other, and the contradiction was worth a factor of twelve. Mode 3 shuts all five valves; Mode 5 needs two of them open. **RULED — RC2: the valves shut at the shock and hold 1.3 s, then BV-4 and BV-5 are REOPENED for generator operation**. Bay 8 is outside the gas-tight envelope, so the clean zone is unaffected, which is exactly what the Mode 5 note always said. **The battery is therefore a 4 h cabinet (149 Ah) and not a 48 h room (1 783 Ah, 2.4 tonnes, 4.8 m^2 of floor) — and there is nowhere to put the latter.**

One consequence, recorded not resolved: reopening BV-4 and BV-5 leaves **two DN350 bores open through the post-attack period** — the two that fail both EMP criteria (Part 2.8.3). That **sharpens** the EMP boundary question without deciding it.

14.2 Drainage

Concept — a tanked box, not a drained one (Part 6.4). Collected flow:

structural seepage	200 L/day	[C]
condensate and washdown	200 L/day	[C]
TOTAL 400 L/day = 0.00463 L/s		

And the seepage figure's basis, reconstructed because the sheet did not state it:

submerged wall height	(-)2.000 -> (-)6.700	= 4.700 m
external perimeter	2 x (22.0 + 6.2)	= 56.40 m
submerged wall area	56.40 x 4.700	= 265.08 m^2
mat underside	22.0 x 6.2	= 136.40 m^2

0.5 L/m²/day × 401 m²TOTAL = 401.48 m²
= 200 L/day

It is the external envelope standing below the design water table — not the internal wetted area. Recording this matters: **if the water table is confirmed higher than (–)2.000, the wetted area, the seepage and the sump duty all move together, and the relationship is now explicit.**

Item	Adopted	Class
Floor falls	transverse 1:80 in wet areas, 1:100 elsewhere; spine 1:400 east to the sump along Y 3100	[A]
Fall formed in	floor screed — 25 mm minimum at the sump edge rising to 78 mm at the far corner; area-average 52 mm	[A]/[R]
Gravity drains	DN100 minimum at 1:100 minimum	[A]
Capacity, DN100 at 1:100	6.72 L/s at 0.85 m/s — self-cleansing	[R]
Capacity ratio against the design flow	1 450 : 1	[R]
Traps	75 mm deep seal throughout, primed	[A]
Rising main	DN50 , 0.76 m/s at 1.5 L/s; welded or flanged, no push-fit joint inside the envelope	[A]
Clean sump	3.375 m³ , invert (–)7.600, base slab (–)8.000; 2 × 1.5 L/s submersible + hand pump PU-03	[C]

Internal drainage is governed by minimum bore and self-cleansing velocity, not by flow — the capacity ratio is three orders of magnitude. **Stated so that no reviewer looks for a hydraulic sizing calculation the flows do not justify.**

Trap seals against shelter overpressure (Part 2.7.1):

Seal depth	Holds	vs +100 Pa operating	vs +300 Pa test
50 mm	491 Pa	4.9 : 1	1.6 : 1
75 mm (adopted)	736 Pa	7.4 : 1	2.5 : 1
100 mm	981 Pa	9.8 : 1	3.3 : 1

Sewage. In protective mode the shelter is sealed and uses **sealed-cassette chemical toilets** — **nothing is discharged**. The septic tank serves **peacetime use only**: design population 10, 45 lpcd = **450 L/day**, tank **1.5 × 0.75 × 1.0 m** liquid to IS 2470 (Pt 1) Table 1, two compartments, baffle at $\frac{2}{3}$ L, inlet and outlet tees, 50 mm cowled vent ≥ 2 m above grade. Every IS 2470 (Pt 1) check reproduces: **450 L detention + 600 L sludge = 1 050 L required against 1 125 L provided (+7.1 %)**; L/B = 2.0; B = 750; depth 1.00 m; freeboard 300 → overall 1.30 m. **PASS.**

Findings.

Ref	Finding
DR-C2, ruled and closed	The soak pit was 2.3 % short on its own stated requirement — $\pi \times 2.0 \times 3.5 = 21.99 \text{ m}^2$ printed as "22.0 OK" against "22.5 required". Arithmetic, not judgement. RULED: SK-01 is WIDENED, 2.0 → 2.200 m diameter, depth UNCHANGED at 3.500 m → 24.19 m², +7.5 %. Widened rather than deepened because deepening drives the pit further below the design water table, where it cannot soak at all. The mandatory percolation test still governs the final size
DR-C1, ruled and closed	A superseded catchment. The 0.10 L/s for stairwell surface water is the Rev E open-cut figure — 7.2 m ² of open pit at 50 mm/h. At Rev F the approach is covered, the door is at grade, and the sheet's own note states the catchment with the door shut is zero. RULED: Rev F governs, and 0.10 L/s is RETAINED as a declared conservatism — removing it changes no pump, pipe or pit (the stairwell pump is 20 × it) and deleting a superseded number would hide the history. Both figures are shown and labelled
DR-F1	The clean sump cycles once every 3.4 days , 15 min per start, 0.31 % duty ratio. Cycling is fine — but at that duty a failed standby would never be discovered by use. A witnessed monthly test of the standby path and the hand pump is required
DR-F2	A drained screed reduces the finished clear height from the structural 3 200 to between 3 122 (far corner) and 3 175 (sump edge). A client-facing number must say which one it is
DR-F4	A drained floor cannot be built inside the 1.0 kPa mat allowance. At 24 kN/m ³ that buys 42 mm of screed; the adopted grading averages 52 mm = 1.24 kPa , an excess of 0.24 kPa = 25 kN over the 104 m ² floor — 1.2 % of the mat's own weight, 0.4 % of the uplift, and acting in the FAVOURABLE direction for flotation. Referred to the structural engineer, not assumed
DR-F5	The protective boundary creates three hydraulic zones and two have no drainage destination. Zone 1 (bays 1–5) → clean sump, complete. Zone 2 (bay 6 decon) → its tank, which the sheet draws in bay 8, across the boundary — route undefined. Zone 3 (bays 7–8) → nothing recorded. A protective-design decision
BW-01	Builder's work, requested and not accepted. The one buried drain inside the envelope needs a 300 wide recess in the top of the mat, 150 deep rising to 216, leaving 384 of the 600 mat locally on the line of the peak transverse sagging moment. Structural engineer to accept or refuse; a fallback with no buried drain at all is drawn
DR-D1	Rainfall intensity 50 mm/h is the only intensity in the project. No return period, duration or IDF source

14.3 Electrical and power

A basic package, on purpose. It stops at board level: sources, a load schedule, three boards, the essential / battery system, one single-line diagram. **No circuit schedule, no cable sizing, no luminaire or socket layout, no protection study** — that is a **declared scope boundary**, not a gap.

Tag	Load	Board	kW	Class
L-01	General lighting, bays 1–8, LED	DB-E	0.520	[A]
L-02	Emergency lighting, maintained, DC	DB-E	0.100	[A]
P-01	Small power, socket outlets	DB-M	1.000	[A]
F-01	Filter train fan, 1 duty of 2	DB-E	0.379	[R]
D-01	Dehumidifier	DB-E	1.000	[A]
U-01	Clean sump pump, 1 duty of 2	DB-E	0.334	[R]
U-02	Stairwell pump, 1 duty of 2	DB-M	0.223	[R]
Z-01	EMP Zone 2 ops / comms	DB-Z2	1.500	[A]
S-01	Fire detection and alarm panel	DB-E	0.100	[A]
B-01	Battery charger / inverter	DB-M	0.800	[A]
G-01	Generator auxiliaries	DB-M	0.300	[A]
	CONNECTED LOAD		6.256 kW	
	at power factor 0.85		7.360 kVA	

Standby units are not counted twice — the standby fan and the two standby pumps never run simultaneously with their duty partners.

GEN-1 rating	= 15.0 kVA	[C]
Connected load	= 7.360 kVA	
UTILISATION	= 49.1 %	
Spare	= 7.64 kVA	
Largest motor, the filter fan 0.379 kW; DOL start	~ 2.7 kVA	

The confirmed 15 kVA is about twice the connected demand, and 49 % sits in the healthy loading band for a diesel set — high enough to avoid wet-stacking, low enough to carry growth. **There is no starting problem and the project's own figure needs no change.**

Three boards, one cable entry: DB-M main LV in Bay 8 (outside the gas-tight envelope, mains

- generator with changeover); DB-E essential in Bay 5 (inside the envelope, fed from DB-M and

from the battery inverter on loss of both sources); DB-Z2 inside the shielded enclosure in Bay 3, fed from DB-E **through the PCI on the Zone 2 boundary.**

The essential system — emergency lighting 0.100 + reduced general lighting 0.150 + fire detection 0.100 + EMP Zone 2 at 50 % 0.750 + clean sump pump intermittent 0.033 + CO₂ scrubber recirculation fan 0.100 + instruments 0.050 = **1.283 kW**, carried for **4 h** on **149 Ah at 48 V** (depth of discharge 0.80, inverter efficiency 0.90).

Groundwater does not stop because the shelter is sealed. The clean sump pump has to stay powered through the closed mode — and **hand pump PU-03 is what covers it if the battery fails.**

And two sources need no electricity at all, both already confirmed: a hand crank on both filter fans, and the hand pump. These are the real last line, the project already has them, and no electrical design should be allowed to obscure them.

The CO₂ scrubber, recovered from the project's own arithmetic (owner ruling 25, RC7):

$$Q = \text{production} / (\text{eta} \times C_{\text{target}})$$

target CO2	eta 0.50	eta 0.80	eta 0.95	
0.5 %	72.0	45.0	37.9	m3/h
1.0 %	36.0	22.5	18.9	m3/h

ADOPTED 75 m3/h recirculation loop -- the worst cell, because designing to it costs almost nothing.

CHECK a 75 m3/h fan at an assumed 250 Pa across a packed bed draws ~ 11.6 W against the 0.10 kW allowance -- roughly 8 x margin.
NO ELECTRICAL VALUE CHANGES; the allowance is now CHECKED

rather than assumed.

Not designed, not invented [N]: the absorber vessel, bed depth, face area, residence time; the **soda lime charge** (17.28 m³ of CO₂ is the duty — converting it to a mass needs vendor absorption capacity); the single-pass efficiency; and **where in Bay 5 the absorber stands**, the tightest bay at 1 560 clear where the filter trains already leave 110 mm at the sides.

The generator day tank (owner ruling 20, RC5):

```

600.6 kWh over 96 h x 0.35 L/kWh [A] = 210.2 L
ADOPTED 210 L usable . 250 L nominal . 275 L bund (110 %)
        BAY 8 -- outside the gas-tight envelope, INSIDE the EMP boundary
        welded steel, banded, contents gauge, low-level alarm to the panel

THE FILL AND THE VENT ARE THE REAL DESIGN QUESTION, NOT THE TANK.
Both cross the protective boundary, and bay 8 is inside the EMP boundary,
so two new bores would each need blast, gas AND EMP treatment -- in the
same wall where BV-4/BV-5 already FAIL the EMP criteria.

ADOPTED route both up the EXISTING SH-2 bore and add NO NEW PENETRATION
-- DN25 metallic fill with a lockable cap at the SH-2 head,
  DN25 metallic gooseneck vent, both bonded to the shaft earth.
    
```

Use what already crosses. That instinct — never make a new hole in a protective boundary if an existing one can be shared — appears twice in the rulings, here and on the heat rejection. **It still needs an HVAC and EMP fit check: two DN25 lines sharing a 600 × 600 bore with two DN350 blast valves is a fit that has not been checked** [A]. Fuel type unconfirmed, the 0.35 L/kWh rate still [A], no vendor set, no fuel polishing, no tank fire suppression.

14.4 EMP protection

14.4.1 The zone model — adopted by the project owner

Zone	What it is	Performance	Verification
EMP ZONE 0	Everything above grade — sentry post +7.000, headhouse +0.900 with no earth cover , covered stairwell +2.450 (expendable), shaft heads, burster slab	None credited	n/a
EMP ZONE 1	The buried box, all eight bays . The reinforcement cage	99.99 dB at 10 kHz → 0 dB at 999 MHz , pierced by an 8.85 m ² stair void and two 1 400 shafts. 80 dB only below 99.93 kHz	Cannot be surveyed — no accessible exterior under 2 m of cover
EMP ZONE 2	Welded steel enclosure, Bay 3	80 dB, 10 kHz – 1 GHz, STANDING ALONE	IEEE Std 299 full survey — HOLD POINT

The design rule, adopted with the model — the whole package in one line

EMP Zone 2 is designed to the full 80 dB standing alone. No attenuation from the concrete box is credited at any frequency.

The cage is then **margin, not design** — which is the only defensible way to use a shield you can never survey, buried under two metres of engineered cover.

14.4.2 What the design already does right — and none of it had been written down as EMP

1. **The 150 mm bar spacing is a deliberate EMP decision, and it was the right one.** Stricter than IS 456 Cl. 26.3.3 needs; recorded in the master, in the structural package, in eight bar schedules and on the drawings.
2. **The cast-in frames are already specified as EMP bonds** — every blast door frame *"cast in and welded to the cage"*, and the rule is applied even to the **non-blast-rated** headhouse door, because the shield must be electrically continuous.
3. **Every construction joint already carries a welded Cu / galvanised EMP strap.** A construction joint is where a cage loses continuity, and the project already knew it.
4. **There are no movement joints inside the protective envelope, and the reason is recorded:** *a movement joint is a guaranteed blast, gas and EMP discontinuity.*

14.4.3 EMP Zone 2 — the enclosure

Every dimension in this section is [A]. The project confirms the enclosure is **required** and contains no specification for it.

Host	Bay 3, internal 3 500 × 5 000 × 3 200
External	2 400 × 1 600 × 2 200, at X 5820–8220, Y 3700–5300

Shielded panel	50 mm
Internal clear	2 300 × 1 500 × 2 100 — 3.45 m ² / 7.245 m ³
Shielded envelope area	25.28 m ² — walls 17.60, roof 3.84, floor 3.84
Total seam length	24.80 m
Clearances	300 mm north, south and west; 800 east; 1 000 head — 850 clear above the 150 mm HVAC duct zone

A 300 mm gap is held on every free face so an IEEE Std 299 survey can physically reach every seam. A shielded room you cannot walk around cannot be tested, and **an untested shield is a claim, not a shield.** The enclosure is entirely clear of the Y 2500–3400 circulation route.

The seam length is the risk, not the area. Every one of those **24.8 metres** has to be continuously welded or continuously gasketed, and IEEE Std 299 will find the metre that is not.

The five ways in:

PoE	Treatment	Performance
PoE-1 ACCESS §5.4	RF-gasketed or knife-edge shielded door	≥ 80 dB · type and vendor [N]
PoE-2 VENTILATION §5.5	Honeycomb, 6 mm cell × 25 mm deep	cutoff 29.3 GHz · 133 dB (+53 dB margin)
PoE-3 POWER §5.7.2.1	PCI on every conductor, mounted on the boundary	[N] — no residual quoted, deliberately
PoE-4 SIGNAL §5.7.4.1	Fibre , no metallic strength member or armour	A dielectric is not a penetration
PoE-5 RF §5.7.6	—	Nothing to apply it to — EM-F6

PoE-3 quotes no residual figure on purpose. MIL-STD-188-125-1 specifies pulse-current injection performance **by pulse test**, and the project's register carries the section number only. **Inventing a number would be worse than** [N].

PoE-4 is the bargain of this package. Fibre with no metallic member is **not a penetration at all**. It is the one place this project can buy perfect performance for almost nothing, and it should take it.

14.4.4 Findings

Ref	Finding
EM-F1	The stair void is a 2 800 × 3 160 (8.85 m²) aperture that never reaches 80 dB anywhere in the band and is simply open above 47.4 MHz. It does not open into soil — it opens into the headhouse, which is at +0.900 with no earth cover , and thence to grade through a stairwell that is declared expendable. The entry route is an open electromagnetic path from grade to Bay 7 , and Blast Door 1 is the only thing across it — its RF performance is vendor data the project does not have. <i>This is not a criticism of the architecture: a shelter needs a staircase, and the void is where a staircase has to go. It is a statement that the blast boundary and the EMP boundary are not the same surface, and the project has only ever drawn one of them</i>
EM-F2	The cage meets 80 dB over one decade of the five required and gives 0.00 dB at 1 GHz. A genuine low-frequency measure; must not be described as a MIL-STD boundary
EM-F3	"EMP Zone 2" had been named since Rev F with no Zone 1 and no Zone 0 ever defined, and no specification for the enclosure itself. Four revisions, no shield
EM-F4	BV-4 and BV-5 (DN350) are the only penetrations failing both criteria — cutoff 502 MHz, 54.8 dB. RULED (ruling 3, RC4): bay 8 IS inside the EMP boundary, so the honeycomb panels move from conditional to REQUIRED — and because the generator may run in closed mode, the panels must work with the valves open
EM-F5	≤ 5 Ω is not achievable with rods in Deccan basalt (335 Ω per rod at the low resistivity bound) and is not an EMP requirement in any case. The structure is already the better electrode
EM-F6	There is no antenna, mast, feeder or communications design anywhere in this project. §5.7.6 is in the register with nothing to apply it to. An ops room that cannot transmit is an ops room in name only — and an antenna is by definition a deliberate conductor from outside to inside, the hardest EMP penetration there is

14.4.5 Verification, and what must not be claimed

EMP ZONE 2 — SURVEY IT. Full IEEE Std 299 survey, 10 kHz – 1 GHz, on the completed enclosure with every penetration made off and every panel closed. Acceptance **80 dB**. **This is a HOLD POINT: no equipment is installed before it passes**, because a failed survey means opening seams.

EMP ZONE 1 — DO NOT SURVEY IT, AND DO NOT CLAIM IT. A buried box under 2 m of engineered cover **cannot** be surveyed to IEEE 299 — there is no accessible exterior to put a transmitter on. **Any statement that the box gives 80 dB is unsupportable and should not be made.**

What can be checked on site, cheaply, at the only moment it can be fixed: continuity of the cage across every construction joint **before the pour**; continuity of every cast-in frame to the cage **before the pour**; **earth resistance, early**; and bond resistance at every strap after making off. **None of these prove shielding effectiveness. All of them catch the mistakes that destroy it.**

Every cage figure in this project is an upper bound. The classical $-10 \log_{10}(n)$ array correction is not applied, the crossings are **tied rather than welded** — and whether they are tied or welded is itself [N] — and no concrete absorption is credited. **A measured cage will be worse than the calculation.**

14.5 Fire and life safety

The governing fact is Part 2.10: this is a sealed shelter and it cannot be ventilated of smoke.

14.5.1 What the design already does well

1. **The largest fire load is already compartmented away from the occupants.** The generator is in **Bay 8**, behind **Blast Door 2** in the 400 mm wall W7, and Bay 8 is separated from the occupied bays by **Bay 7, the stair shaft. Two barriers and a buffer between the fuel and the people.**
2. **Bay 8 has its own air path that never touches the gas-tight envelope** — 2 600 m³/h through BV-4/BV-5. **A generator fire is therefore ventilated, and ventilated away from the occupied zone.** The one part of this shelter that can be cleared of smoke is the one part most likely to produce it.
3. **The two dead-end bays are exactly the two bays with escape shafts.** Bay 1 has ESC 1; Bay 8 has ESC 2. **No occupant is ever in a dead end without a way up out of it.**
4. **Travel distances are short** — the longest travel to a route is about **14.6 m**, and ESC 1 is 1.5 m away in that same bay.

14.5.2 Where the fire load is

Bay	Load	Comment
8	Generator, fuel system, exhaust	The largest by far. The set, its fuel system, exhaust and acoustic treatment are not specified and not priced
5	CBRN filter trains	ASZM-TEDA activated carbon. Carbon beds self-heat; a filter fire is a known hazard in protected ventilation
1	Stores	Combustible stores load not scheduled anywhere [N]
3	Ops room, EMP Zone 2 enclosure	Electronics and cabling
4	Berthing, 9 berths	Bedding and personal effects
all	Cabling	Cable type, route, containment and fire-stopping are not specified anywhere [N]

14.5.3 The evacuation plan

Route	Path	Emerges at
R1 PRIM ARY	Any bay → spine → Blast Door 1 → Bay 7 stair shaft → 24R @ 170.8333, 3 flights of 8, 1 200 wide	Headhouse (–)2.000, then 12R up the covered stairwell to the entry door at grade
R2	ESC 1 , Bay 1, 1 400 dia	Head +0.150 — a 6.250 m climb
R3	ESC 2 , Bay 8, 1 400 dia	Head +0.700 — a 6.800 m climb

R1 is the only route that does not require climbing a shaft, and the only one usable by an injured or unconscious person. R2 and R3 are escape shafts, not exits — treat them as the last resort they are.

Decision rule — where the fire is decides the route:

Fire in	Route	Why
Bay 8, generator	R1 , or R2 if the spine is smoke-logged	Blast Door 2 SHUT. Bay 8 is separately ventilated. R3 / ESC 2 is inside the fire compartment — do not use it
Bays 1–6, occupied zone	R1 if the route to Bay 7 is clear; otherwise R2 from the west end, R1 or R3 from the east	Bays 1–6 are one smoke compartment
Bay 7, stair shaft	R2 or R3	The primary route is the fire. Both blast doors shut
Headhouse or covered stairwell	R2 or R3	R1's surface end is blocked; both are outside the boundary and the stairwell is expendable

Immediate actions. Raise the alarm **by voice** — there is no alarm system, and in a 20.8 m shelter with 9 occupants and permanently open 900 mm partition gaps, voice carries. Attack it only while it is small and only if the route behind you is clear. **Shut the blast door between you and the fire — they are the only real barriers in the shelter.** Stop the generator if the fire is in Bay 8 or of unknown origin. Evacuate by the rule above; **do not pass the fire to reach a preferred route. Roll call: 9** — the occupancy is a confirmed figure, so the count is unambiguous. **Muster point — NOT DEFINED, because no site plan exists.**

There is no fire team. With 9 occupants, everyone is the fire team. No fire duty, warden role or drill frequency is defined anywhere in the project [N].

14.5.4 The findings a fire plan here is obliged to raise

Ref	Finding
FS-1	W5 was designated "fire and gas-tight" and no door existed in it anywhere in the project. A fire separation with an unfilled opening in it is not a fire separation — and the compartment it protects contains the activated-carbon filter trains , while escape route R1 has to cross it. RULED (ruling 9, RC4): design it now. FS-1 CLOSES
FS-2	Bays 1 to 6 are ONE smoke compartment, 20.8 m long. The four W8 partitions are 110 mm, non-structural, with permanent 900 mm gaps and no doors scheduled . Smoke starting in Bay 1 reaches Bay 6 unobstructed. <i>Not a criticism of the partitions — they were never fire barriers</i> — but it means the only real fire barriers in the shelter are Blast Doors 1 and 2, and W5 once it has its door
FS-3	ESC 2 is in the same bay as the generator. Both facts are confirmed and neither is wrong alone. The coupling had never been stated: the most likely fire in the shelter denies one of its three escape routes — which is exactly why the decision rule routes a Bay 8 fire away from ESC 2
FS-4	A fire during the closed mode cannot be ventilated at all , and clearing smoke means opening a blast valve, which breaks the protection the closed mode exists to provide. No rule exists anywhere in the project for which hazard takes precedence. That belongs to the client, and it should be made before it is needed rather than during
FS-5	Every active fire measure followed the missing electrical design. The electrical blocker is now gone — a detection and alarm panel and maintained emergency lighting exist on the essential board, on the battery. What remains is fire engineering — head layout, zoning, detector type, any suppression — and none of it is invented here
FS-6 / FS-V7	The six-metre unequipped climb out of each escape shaft. Raised by the drawings, not by the text : the written plan had all three numbers and never put them in the same picture; the escape-plan drawing's vertical profile put them side by side and the gap was immediate. RULED — a ladder is now designed (Part 10.6), and it still has no fall-arrest, no rest platform, and no answer on the injured person

What the plan cannot say, and does not invent: extinguisher type, number and siting; fire ratings of doors and finishes; cable type, containment and fire-stopping; generator fuel type and quantity; muster point and fire service access; drill frequency and warden duties. **Codes beyond NBC 2016 Part 4 and IS 13416 are not in the register, and their numbers are not invented.**

And one trade that must be a conscious decision, not an improvisation. The 1 000 L tank in Bay 1 is the **96-hour potable supply** — 27.8 L/person/day. **It is not firefighting water.** Using it against a fire trades the shelter's endurance for its fire response, and that is the commander's call.

PART 15 — SITE LAYOUT, FINISHES AND CONCEALMENT

15.1 External works — the reserve, and why everything is in it

An **external works reserve** of 18.0 × 10.5 m at X 33 000 – 51 000, Y 6 500 – 17 500 — 10.0 m clear of the main excavation's east face, **downgradient**, and entirely inside the owner's 50 m envelope (worst corner 42.5 m of 50 m).

Tag	What	Centre (X, Y)	Serves
ST-01	Septic tank 1.5 × 0.75	(36 750, 16 000)	10 users, 450 L/day
SK-01	Foul soak pit 2.2 dia	(44 000, 16 000)	Septic tank effluent — the furthest downgradient of everything
SK-02	Storm soakaway 2.2 dia	(35 000, 8 500)	Clean sump + entry channel. Moved 300 mm north by RC4 to give 2.10 m to the sentry post against IS 2470's 2.0 m
SK-03	Stairwell soakaway — footprint reserved	(41 400, 8 500)	Stairwell sump. Size recorded nowhere [N]
SK-04	Headhouse soakaway — footprint reserved	(47 800, 8 500)	Headhouse gully. Size recorded nowhere, and its pipe cannot be routed
IC-01 · IC-02	Inspection chambers 600 × 450	(7 500, 9 800) · (40 200, 16 000)	Change of direction; de-sludging

Everything in one reserve, on the downgradient side, for four stated reasons:

1. **Nothing recharges the ground upslope or alongside a box that is flotation-critical at FoS 0.33 in the mat-only stage** — and whose side backfill at 95 % MDD is **more permeable than the basalt around it**, so effluent released near it would run *into* the backfill and down the outside of the tanking.
2. **One percolation-test location, one keep-clear zone, one reserved fallback.**
3. **One trench** — three of the runs share a common services trench, and **where rockhead is 0.9–1.5 m the cost *is* the trench.**
4. **Concealment** — four cover slabs and a 2 m septic vent, grouped 11–22 m away, **mark the drainage field, not the shelter.**

The layout is anchored to confirmed geometry only. Nothing is dimensioned from the sentry post, so it survives the sentry position being resolved differently; and **every offset is relative**, so if the perimeter fence turns out to be closer than the reserve's east edge, **the whole reserve translates and not one offset changes.** That is why the fence distance — SG2-V2 — could be closed: **the layout is immune to the answer.**

The IS 2470 offsets — complete for the first time:

foul soak pit to the SEPTIC TANK	>= 5 m	5.40 m	DEMONSTRATED
soak pit to any BUILDING	>= 2 m	10.97 m	DEMONSTRATED, 5.5x over
foul soak pit to any WELL	>= 15 m	--	OWNER RULING (RC7)

The caveat, and it matters. The well clearance is an **owner ruling, not a survey result.** It is recorded as [C] owner ruling and **not** as [C] surveyed, and the distinction is deliberate: **if a well is later found within 15 m, the foul soak pit moves.** Foul effluent near a water source is the one clearance nobody should carry on an assumption, which is why it stayed open until the owner could answer it from knowledge of the plot.

28 of 28 clearance checks pass, and the five pipe runs are fixed except one:

Run	Length	Status
Septic tank → foul soak pit, DN100 at 1:100	5.40 m	FIXED
Service entry plate → storm soakaway, DN50 rising main	27.00 m	FIXED — its vertical leg waits on the entry plate's level, which is [N]
Entry channel → storm soakaway, DN100 at 1:100	32.35 m	FIXED — routed west first, because north would cross the stairwell excavation
Stairwell sump → stairwell soakaway, DN50 rising main	29.60 m	FIXED
Headhouse gully → headhouse soakaway, DN100 at 1:80	[U]	CANNOT BE ROUTED

15.1.1 SG2-F1 — the soak pit's problem is DEPTH, not arithmetic

The soak pit was widened rather than deepened when it was found 2.3 % short (Part 14.2). **The reason that choice was forced, rather than merely preferred, is the finding:**

21–43 % of the pit's required side area is above the design water table, and its only permeable horizon is 0.2–0.5 m thick. A pit driven deeper goes further below (–)2.000, where it cannot soak at all. So the fallback **dispersion field is reserved** rather than the pit being re-sized again — and **the mandatory percolation test governs the final size and form.**

15.1.2 Two more findings that the layout could not resolve

Ref	Finding
SG2–F 3	The septic tank is sized for a building that no pipe connects to it. It is sized for <i>"sentry-post shift crews + shelter maintenance"</i> , the drainage package excludes the sentry post from scope , and there is no pipe from the sentry post to the tank anywhere in the project. The layout positions the tank and leaves the connection to the drainage engineer; it does not prejudge it
SG2–F 4 → closed	The fresh-air intake had no plan position. The equipment schedule gave the generator shaft an X range and gave the intake only <i>"west of the box"</i> — no X, no Y, no coordinate. The fresh-air intake of a CBRN shelter is not a minor fitting. RULED (ruling 14, RC4): SH-1 centred (–2 400, 3 100), RECOVERED from the project's own recorded 12.3 m to the entry, to within 28 mm. SH-2's head level is +1.500 , the same gooseneck head

And the intake / exhaust separation was settled by geometry, not by meteorology. SH-1 and SH-2 are **25.30 m apart at opposite ends of a 22 m box** — robust whatever the wind does, which is exactly why the site package justified its orientation on access, fall, noise and end-to-end separation and **explicitly not on prevailing wind.**

But the plume question stays open, and it must. A CBRN shelter with no wind direction data **cannot state which way a release drifts**, and 25.30 m of separation says nothing about that. Closing both halves of that question on one argument is exactly the move the open-item register exists to prevent.

15.2 Finishes

No finish specification exists anywhere in this project. The master, the Rev F drawings, the services sheet and the structural package fix concrete grade, cover, waterproofing, crack control and bar spacing — **and nothing about what a surface is finished with.**

So the finishes package does **not** report finishes as project facts. **Every code is a performance requirement derived from something the project does confirm**, and the product that satisfies it is left open. **A schedule that filled them in would be inventing a specification.**

Requirement	Derived from
Decontaminable, coved, joint-free finishes in bays 1–6	The gas-tight CBRN envelope
Wet-area treatment in the lavatory, the plant bay and the airlock	The room uses
Non-slip decontaminable finish in the stair shaft	Those faces are designated a wet / dirty zone — it is why their cover is 30 and not 40
Non-slip external finish in the covered stairwell	Outside the boundary, declared expendable
Hose-down finish and 1:80 fall in the headhouse	Confirmed on the Rev F ground plan
EMP Zone 2 enclosure in the ops room	The finish is subordinate to the shielding
Cast-in fixings only	40 mm cover, 150 mm bar spacing as an EMP requirement, the membrane, the cage
No suspended ceilings or cavities anywhere in the envelope	Duct inspectability, drain inspectability, decontamination
Floor build-up limited	The 1.0 kPa mat allowance, and drainage finding DR-F4
Cast-in stair nosing, no tread build-up	The stair going is FROZEN. An applied nosing or a thick tread finish would change the effective going

Thirteen spaces are scheduled, each with its floor, skirting, wall, ceiling, door, waterproofing, wet-area status and falls. Two entries carry findings rather than finishes:

- **The decon airlock is the dirtiest surface in the shelter** — wet area, falls 1:80 to **segregated** gullies, 50 mm upstand at W5 and at Blast Door 1.
- **The generator bay falls to a gully that has no destination defined** — drainage finding DR-F5.

FN–U1 / D–05 — the finishes schedule recorded the missing W5 door as a data gap; **the fire plan made it a defect**, and it has since been ruled: *design it now.*

15.3 Concealment

The shelter is concealed. The installation is not.

What the design already achieves:

Measure	Value
Topsoil / turf over the whole buried roof	300 mm , 30.867 m ³ , the top layer of the cover — its function is named as <i>concealment</i> , <i>erosion</i> , <i>sheds rain</i>
Turf sourced from the site itself	92.82 m³ stripped and stockpiled for re-use — the re-laid surface is the surface that was there before
Finished grade	crowned, falling 1:50 away — a natural-looking shed, not a flat platform
Berm against the headhouse and covered stairwell	1.5:1 to +0.900
Surplus excavation	1 113.937 m³ re-used on site as engineered fill and crushed rubble
Roof penetrations for drainage	NONE. No roof outlet, no downpipe, no rainwater pipe anywhere on the buried roof
Burster slab	laid to a 1:50 crossfall so infiltration disperses at the berm toe without a pipe

Two of these matter more than the rest. Re-using the site's own stripped turf means the restored surface is **not a different green from its surroundings**, which is what usually gives a buried structure away. And **because the cover is drained without a single pipe, there is no manhole, no gully and no outfall anywhere on the roof to break the surface.**

The signature inventory — every element above ground:

Element	Above grade	Note
Sentry post	+7.000 parapet	The tallest thing on the site by 4.5 m
Covered entry stairwell	+2.450 head	Outside the boundary, declared expendable
Fresh-air shaft SH-1	+1.500 gooseneck	600 × 600, west of the box
Generator air shaft SH-2	+1.500 gooseneck	600 × 600, east of the box — level ruled by RC4; the signature is now complete for the first time
Headhouse	+0.900 roof top	4 800 × 5 800, no earth cover. The largest above-ground mass
Escape shaft ESC 2	+0.700 head	1 400 dia clear, 1 900 OD collar
Escape shaft ESC 1	+0.150 head	as ESC 2
Soak pit / soakaway covers	at grade	Positions fixed by the site layout
Sentry post spiral stair	external	1 000 R, 250 dia pole

The buried box under 2 m of graded, turfed cover is genuinely hard to see. But a 7 m sentry post 10 m away, a 4.8 × 5.8 m headhouse with no earth cover, a 2.45 m stairwell head and two 1.5 m goosenecks are not concealed by anything, and nothing in the project asks them to be.

This is a statement of fact about the design as recorded, not a criticism of it — the sentry post is a manned observation position and is *meant* to be seen, and the headhouse is the entrance. **But the distinction had never been written down, and a concealment policy that did not say it would be misleading.**

Policy. The turf is a **continuous surface, not a patch** — re-lay the stockpiled site turf; do not import a different species or a different soil colour. **No new spoil heap** — all 1 113.937 m³ of surplus is re-used on site, because **a spoil mound is a permanent signature that outlasts the works.** The berm should **read as ground, not as an engineered line.**

And what the policy cannot say. The concealment bill item — "*camouflage and concealment measures beyond the 300 topsoil / turf layer*" — is tagged [N], quantity *to be verified from final measurement*, note "**no concealment**". **No net, screen, paint scheme, thermal treatment or detection criterion exists anywhere in this project, and none is invented.** Nor can a concealment **layout** be drawn: there is no site plan, so nothing's position on the ground is known except relative to the box.

PART 16 — WORKS MANAGEMENT

16.1 What exists

Deliverable	Contents
Works management handout	44 pages, 25 sections , covering the whole project
Master programme, MSPDI XML	Microsoft Project's own published schema, plus CSV for any other tool
Programme PDF	7 A3 sheets — basis and milestone register, summary Gantt, 5 sheets of detailed Gantt with the critical path
WBS	279 activities, 18 milestones, 404 logic links , 3-level, 14 level-1 packages
Bill of quantities	90 items in 10 sections , whole project
Resource plan	33 resources, 426 assignments
Procurement plan	16 packages, all tied to programme activities; long-lead register
QA/QC plan	42 ITP items, 16 hold points that are activities in the programme
Safety and risk register	15 hazard classes, 25 risks
Codes and references	78 references
Quantity derivation	~800 lines of working with the source of every input
Consistency audit	65 executed checks, 65 pass

16.2 The programme

Start	Monday 2 November 2026 — carried from the owner's own R0 schedule [A]
Finish	Saturday 20 November 2027
Duration	326 working days (384 calendar days) against R0's 224
Calendar	Six-day week, Mon–Sat , Sunday non-working, plus five date-certain national holidays. Festival holidays are NOT fabricated — a 10-day contingency activity absorbs them
Critical path	75 activities at total float 0 : possession → site investigation and monsoon groundwater monitoring → excavation support design → rock excavation → formation → tanking → mat → walls → pressure slab (14-day cure + 14-day props, IS 456 Table 11) → roof membrane → the six-layer cover including the burster slab → external works → snagging → contingency → handover

The site investigation is on the critical path, and so is the groundwater monitoring. That is the correct answer to Part 4.4: the design's most important unknown is programmed as work, not as a hope. **And the monitoring window, 12 November to 4 December, is not in the monsoon** — which is what SG-V3 says and what makes it worth saying.

16.3 Cost

Three versions are published side by side, deliberately: the project's own estimate, the project owner's own estimate, and the owner's estimate with the rulings applied.

Cost head	Basis	Owner's package	With rulings applied
Total basic cost	subtotal of packages	₹ 24 336 022	₹ 24 220 254
Contingencies	3.0 %	730 081	726 608
Water and electricity provisions	1.0 %	243 360	242 203
Specialist CBRN / EMP and engineering consultant	6.0 %	1 460 161	1 453 215
Site supervision and quality assurance	2.0 %	486 720	484 405
Statutory clearances and liaison	1.0 %	243 360	242 203
Contractor's overhead and profit	10.0 %	2 433 602	2 422 025
FINAL PROJECT COST		₹ 30 033 306	₹ 29 790 913

The total is a LOWER BOUND, and it says so. Several items in the bill read *"to be verified from final measurement"* with no rate, because the project does not hold what they need: **the blast doors, the blast valves, the filter trains, the generator and its fuel system, the EMP Zone 2 enclosure, the escape-shaft hatches and ladders, and all concealment beyond the turf.** Every one of those is a specialist or vendor item, and none of them is priced by guessing.

16.4 The rules the works management package holds itself to

- **The project owner's own bill, estimate and schedule GOVERN.** The project's versions are published alongside, never in place of them.
- **Nothing is retyped.** The owner's documents are reproduced, not transcribed.
- **No quantity, rate, date or float is changed by any discipline package.** Every services and rulings revision in this project states that it changed none, and each states it separately so the claim can be checked.
- **One design change was instructed and is recorded as one** — the sentry post masonry (Part 5.9.1). One bill item was added by a ruling — the 8 m³ strip-and-replace under the stairwell raft (Part 4.5), **at a rate tagged [A] rather than invented.**

16.5 Risks that are design outputs, not contractor problems

Risk	Where it is answered
Flotation during construction	Part 9.3.5 — four mandatory mitigations, one of them formally departed from in writing
Rock excavation quantity exceeds the estimate	Part 4.7 — +118 m³, about 2 extra days , against a register entry that said "roughly one"
The excavation face across a monsoon	Part 4.6 — batter the soil cap; the slope-stability assessment is still outstanding
Site investigation findings	Part 4.2.2 — 5.3 m of unlogged ground under the whole structure
Specialist lead times	The long-lead register: blast doors, blast valves, filter trains, the EMP enclosure

PART 17 — WHAT THIS DESIGN DOES NOT DEMONSTRATE

This part must be reproduced in every future revision. It is the boundary of the design's validity.

17.1 What IS demonstrated

Demonstrated to IS 456, with IS 13920 for the sentry post frame: ultimate flexural capacity; diagonal-tension shear; minimum and maximum steel; bar spacing; cover; development and anchorage; deflection; crack-width provisions; biaxial column interaction; punching and one-way shear in the footing; and ductile detailing to IS 13920.

17.2 What is NOT demonstrated, and NOT claimed

1. **Support rotation and ductility ratio for the blast case.** $\theta \leq 2^\circ$ corresponds to $\delta \leq 87 \text{ mm} = \text{span}/57$ and requires a **non-linear SDOF check** (IS 4991 Fig. 6 / Biggs) or explicit non-linear FE with strain-rate-dependent materials. **Phase 3.**
2. **Shock-wave propagation down the entry shaft** and the resulting door loading — CFD or shock tube. **Specialist.**
3. **Transient soil-structure interaction** — participating mass, arching, interface pressure ratios.
4. **Transient soil-structure THERMAL behaviour** — the sibling of (3), and the missing piece of RC4-V1.
5. **Blast door, valve and hatch performance** — **vendor-tested, not a civil deliverable.**
6. **Global flotation cannot be shown by an elastic-mat model.** The springs take tension, so in the model the mat never lifts. It is a hand check on real dimensions (Part 9.3.5).
7. **The berm's actual pressure-transmission factor** — removed from the critical path by designing the headhouse walls to the upper bound.
8. **Any analysis result whatsoever.** STAAD.Pro has not been run in this project. No moment, displacement, shear or reaction exists for the underground box or for the three mesh models, and **the mesh convergence conclusion is outstanding.**
9. **The negative phase and rebound as a separate load case** — allowed for physically at the hatches, not analysed.
10. **EMP Zone 1 shielding effectiveness.** It is a calculation and it will stay a calculation; a buried box cannot be surveyed to IEEE 299.

A conventional static STAAD.Pro analysis gives the correct DEMAND. It is not, and must never be presented as, proof of blast resistance.

IS 4991:1968 Cl. 1.1 expressly EXCLUDES nuclear explosions from its scope. It is used here for its loading rules and dynamic material strengths only, as a documented conservative extrapolation. **Every structural capacity is computed to IS 456.**

PART 18 — ASSUMPTIONS, OPEN ITEMS AND RULINGS

18.1 How to read this part

Three registers, and the distinction between them is the point:

Register	What it holds
18.2 Assumptions	Values the design uses that have no test and no client confirmation . Seven are closed; the seven that remain are exactly the site investigation
18.3 Open items	Questions that need information the project does not contain . Eighteen are open . They are not inconsistencies — each is a single position the project holds, with nothing contradicting it
18.4 Rulings	Thirty-two items ruled by the project owner across RC4 to RC9 , item by item, plus the two earlier RC2 rulings and the conflicts closed on the evidence

18.2 Assumptions — seven closed, seven open

CLOSED (7) — every one by judgement, and every one the conservative or immaterial choice:

#	Assumption	Ruling
A6	$K_a = 1.0$ saturated used; dry berm ≈ 0.5 not relied on	ACCEPTED as the conservative choice — the full 383 kPa is taken rather than crediting unverifiable berm attenuation
A9	Sentry infill not separated $\rightarrow R = 3.0$	KEPT . $R = 5.0$ gives A_n 0.060 against 0.100, so 73.18 kN is $\approx 1.67 \times$ a separated SMRF's demand . Separating would need an SMRF this project does not claim, and would undo the tie detail
A10	Wind $k_1 = 1.08$, 100-year life	ACCEPTED — a 100-year life on a load path seismic beats 2.4 : 1
A11	$b_{eff} = 2.5$ m for the HW3 line load	ACCEPTED ON THE MARGIN — 41 % on the conservative one-way bound leaves more than half the capacity spare
A12	Trapezoid factor 0.7946 used for both BM and FEM	ACCEPTED ON THE MARGIN . The factor itself is exact; the simplification is worth ≈ 2 %, inside B2's 89 %
A13	Sentry frame has no member releases	CLOSED ON THE EVIDENCE, and it needed no ruling . The register asked for the .std; the file had been in the workspace all along and nobody re-read it . MEMBER RELEASE appears nowhere in it
A14	Poisson's ratio 0.20	ACCEPTED as the standard value for concrete

STILL OPEN (7) — and every one needs a physical test on this plot:

A1	rockhead 1.5–2.0 m	boreholes
A2	DESIGN GWT (–)2.000	piezometer, read through a FULL monsoon
A3	SBC 3240 kPa	plate load / core testing
A4	k_s 100 000 – 500 000 kN/m ³	plate load test -- BOTH bounds
A5	K_0 0.50, gamma 20/21	site investigation
A7	soak-pit absorption 20 L/m ² /day	PERCOLATION TEST, IS 2470 Pt 2 Cl. 4, MANDATORY
A8	structural seepage 0.5 L/m ² /day	packer permeability tests

That is the sentence worth carrying into a viva. The assumption register is no longer a mixed bag of judgement calls and missing data. **Everything a competent engineer could decide has been decided. What is left is precisely the site investigation the project has never had** — and the reason is Part 4.2.2: the only investigation available is **off-site** and **reached about 1.5 m against a formation at (–)6.800**.

A4 deserves one extra line. The upper-bound model **now exists** and is verified by diff against the reference model. **It is still [ASSUMED] at both bounds, and neither has been run.**

18.3 Open items — eighteen

18.3.1 The four a reviewer should look at first

Ref	Item
SG–V2	The investigation reached about 1.5 m against a formation at (–)6.800 . 5.3 m of unlogged ground under the whole structure — including the red-hole case that sizes the mat
EM–V6	Two 1 400 mm holes in the protective boundary whose bonding is still undesigned . The structural half is now designed (Part 10.5); the EMP half is not

Ref	Item
SG2-V4	A CBRN shelter that cannot state which way a release drifts. The intake/exhaust half is closed on geometry; the plume half stays open, and must
RC4-V1	A sealed box with a 4 kW heat surplus and an unmodelled rejection path (Part 2.11)

18.3.2 Open, narrowed by a ruling (9)

Ref	Where it stands now
U8	Parapet confirmed at 300 × 150 and exact; one dimension — the roof projection — is recorded nowhere. 4.162 kN/m continues to be used as given; no member force changes
WM-V9	Excavation face ruled — batter the soil cap, vertical in rock. The slope-stability assessment is still outstanding
FS-V7	Ladder designed (Part 10.6). No fall-arrest, no rest platform, and the injured-person question
EM-V6	Structural half designed, EMP half open
EL-V2	Day tank sized at 250 L nominal in Bay 8, fill and vent up the existing shaft. Fuel type, rate and vendor open ; the shared-bore fit is unchecked
EL-V7	Duty computed — 0.18 m ³ /h of CO ₂ , a 75 m ³ /h loop, 8 × margin on the fan allowance. Absorber and soda-lime charge open
SG-V6	Stairwell raft in black cotton soil — specified and measured (8 m ³). Rate [A]
SG2-V4	Intake/exhaust closed at 25.30 m separation; PLUME half open
RC4-V1	Narrowed from "no rejection path exists" to "the rejection path is the ground through the structure, and it has not been modelled"

18.3.3 Open, annotated but NOT narrowed (2) — and this pair is instructive

Ref	The problem
EM-V2	Every EMP Zone 2 dimension is [A], pending an equipment schedule
EL-V3	There is no equipment schedule for EMP Zone 2 — only a 1.50 kW allowance

These two are each holding the other's placeholder, and it is worth naming. The enclosure is **sized to fit a bay**; the 1.50 kW load is **chosen to give the enclosure a basis**. **Neither is evidence.** They corroborate each other and neither corroborates anything. **What breaks the loop is an operational equipment list — a client input, not a calculation.**

18.3.4 Open, untouched (6)

Ref	Item
WM-V6	Sentry seismic weight after the masonry change. The direction is certain and favourable — brick is lighter, so W falls, V _b falls, and every member is over-designed. Nothing is unsafe and nothing is inconsistent. What is missing is a STAAD re-run to put a number on the margin — confirmation, not risk
EM-V4	No communications design of any kind exists. No antenna, mast or feeder, so MIL-STD §5.7.6 has nothing to apply to
EM-V5	No pipe material is specified for any run in the project. Metallic → bond it 360° to the entry plate and its exterior becomes shield. Plastic → the bore is an aperture <i>and</i> the water column is a conductor, needing a metallic spool piece nobody has specified
SG-V1	The geotechnical data is off-site. No investigation has ever been made on the project plot
SG-V2	Depth of investigation — the governing item
SG-V7	The concealment turf may be an expansive clay. If the site's own topsoil is the CH horizon, it cracks in the dry season and pumps fines into the granular filter, which is the one thing the filter exists to stop. The load is unaffected

18.3.5 Open, new (1)

Ref	Item
DR-A2-V1	The clean sump has no cover, and Bay 5 cannot be crossed without one (Part 13.7). Needs the cover itself: type, depth, imposed load duty , frame and rebate, fixing, and how it is lifted to withdraw the pumps

18.3.6 A declared scope boundary — not a gap

Ref	What does not exist	Where it was declared
EL-V 5	No circuit, cable, luminaire or socket schedules. The electrical package stops at board level	The package's own opening paragraph. It closes when a detailed electrical design is commissioned, not by any work in this project

This row was moved out of the open-item register on purpose, so that a declared decision is never counted as an outstanding problem — which is what was happening.

18.4 Rulings and conflicts

18.4.1 The conflicts closed on the evidence

#	Conflict	Resolution
C1	Drawings Rev F, report Rev D — the report describes a dog-leg open ramp; Rev F is a covered entry stairwell	DXF governs. The precedent for everything after it
C2	The report designs the headhouse roof for 3.0 m span + 1.0 m cover; Rev F is 4.0 m + no cover	Recomputed. T20 @ 150 survives at 90 %; shear links new and mandatory
C3	The report calls the stair-shaft walls 600 thk; the Rev F plan shows 200	DXF governs → triggers Finding F1 → Modification M1
C4	Blast door leaf: plan 1200, section 900	1200 × 2100 adopted — agrees with the report's own 1 564 kN leaf force
C5	Plan dimension "7300 covered stairwell external" against 6 800 in the plotted geometry	6 800 adopted
C6	The report checks the mat in flexure only	One-way shear checked → the T12 @ 250 × 250 link grid
C7 / C9	The report uses W = 505 kN for the sentry post; the hand check gives 592.7 kN; STAAD gives 731.80 kN	The STAAD value governs , and Part 7.8.2 explains the gap to the kilonewton
C8	The report adopts a sentry post height of 6 850 mm; the elevation shows roof +6.700, parapet +7.000	Drawn levels adopted; h = 6.250 m from the base at +0.450
C10	The report designs the headhouse walls for 113 kPa drag only	Raised to the full 383 kPa envelope (Part 7.5)
C11–C15	Five defects in the entry stairwell and underground .std files — wrong concrete grade, missing roof imposed load, missing waterproofing load, wrong fill unit weight and berm level, and four mat corner joints with no vertical restraint	All five resolved , each by bringing the file to the master rather than the master to the file
C16	The roof/platform junction — one clause said the stairwell roof becomes the 500 headhouse roof over the platform	RULED AT 250. It was the only place in the project saying 500 (Part 5.8)
C17	Cover build-up: 40.65 stated, 39.15 summed	RULED. 40.65 held, and the table made self-consistent (Part 7.3)
C18	Sump pit base: the register says 400, a sheet note says 300	RULED AT 400. $(-)8.000 - (-)7.600 = 0.400$ is arithmetic
C19	Soak pit 21.99 m ² against its own stated 22.50 required	RULED. Widened to 2.200 m dia → 24.19 m², +7.5 %
C20	A superseded Rev E catchment on an issued sheet	RULED. Rev F governs; the figure is retained as a declared conservatism and both are labelled
C21	Filter duty: the master said 2 × 250 m ³ /h, the sheet said 300 in nine places	RULED AT 300 m³/h, and the master was corrected. 250 fails the sheet's own FEMA criterion, so it was not the cautious option but a demonstrably inadequate one
ERR-1	A yield-line coefficient computed with the simply-supported constant instead of the fixed-four-edges one — 324.4 against 162.2	Corrected, and recorded as an error rather than quietly fixed. No reinforcement changed, because the one-way value was the one adopted (Part 10.3)

18.4.2 The owner's rulings, revision by revision

Revision	Rulings	Net effect on the open register
RC2 (11 Sep)	2 — the three-zone EMP model adopted ; the generator may run in closed mode	Neither changed a number
RC4 (11 Sep)	16, item by item — 13 actioned, 3 deliberately left open	33 → 25
RC5 (12 Sep)	4 — heat rejection, the hatches, the day tank, and one left open — plus A1.3 closed on the evidence	25, three narrowed, none closed
RC6 (12 Sep)	The excavation face, and six assumptions closed	K.2 14 → 7
RC7 (12 Sep)	4 — the scrubber duty, the mains, four source defects, the well	25 → 19

Revision	Rulings	Net effect on the open register
RC8 (12 Sep)	3 — the fence, the EM/EL circularity named, one reclassified	19 → 17
RC9 (12 Sep)	None. It corrected a count	17, verifiable by reading
DR-A2 (12 Sep)	None. It drew two things and found one	17 → 18

Three of those deserve to be read as a set, because they are about the project's own discipline rather than about the design.

RC9 **corrected a count this project got wrong**. Revisions RC4 to RC8 ruled faithfully and recorded every ruling — **but thirteen rows of the register itself were never annotated**, so a reader counting the table got 34 while the narratives claimed 21, then 15, then 13. **All three counts were wrong**, and the first was wrong arithmetically on its own terms. The fix was **not to restate the number but to make the register countable**: every row is now classified, and the totals can be verified by reading rather than by trusting a sentence.

RC10 **found the same failure mode one level down**. RC4 to RC9 ruled on thirty-two items and **did not touch the discipline packages**, so six artefacts still say something the master has settled — the EMP penetration register still calls the honeycomb panels conditional, the HVAC schedule still locates the intake as "*west of the box*", the escape-route schedule still records two climbs with no means of climbing them. **Nothing was silently edited**: all six are build artefacts regenerated from their packages' scripts, and applying the register is a package-by-package job with a regeneration check that **has not been done**.

The lesson is one this project already wrote down: a change that stops at "reinforcement" and never reaches "drawings" is not finished. RC4 to RC9 stopped at the master. The failure mode appeared twice in two days — once inside the master, once between the master and its packages — **and both times the fix was to make the thing countable rather than to trust a narrative**.

PART 19 — VERIFICATION PERFORMED FOR THIS REPORT

19.1 What was checked, and how

This report reproduces several hundred numbers from the project record. **Every one of them was independently recomputed from its own inputs before it was printed here**, by a script that is part of this package:

```
Project Report/Scripts/report_verify.py
-> Project Report/Calculations/REPORT_VERIFICATION_OUTPUT.txt

python3 "Project Report/Scripts/report_verify.py"
```

Each check takes the **inputs** a value is derived from, does the arithmetic **independently** — the formulas are written out again in the script, not copied — and compares the result with the value the project prints.

What a PASS means, stated precisely. *The printed number follows from the printed inputs. It is not a design check and it is not an analysis.* It does not re-derive the engineering judgement behind a value, it does not run STAAD, and it cannot tell you whether an assumption is true. What it does catch is the class of defect this project has already been bitten by twice: a number that no longer agrees with its own derivation.

19.2 The result

CHECKS EXECUTED	498
PASS	494
KNOWN / RECORDED difference	4
UNEXPLAINED FAIL	0

Coverage, by section of the script:

§	Subject	What it recomputes
1	Blast loading and structural dynamics	DLF and its limit case, the design pressure, the reflection coefficient, 50 psi in kPa, E_c , the participating mass, the flexural stiffness, the roof natural frequency and period, and t_d/T at both ends of the duration band
2	The engineered cover and the roof total	All six layer loads, their sum, the declared allowance , the total thickness, the crossfall drop, COMB 103, the static ULS case and the blast : static ratio
3	Soil, water and flotation	γ' , the lateral gradient and the water share, the pressures at soffit and floor, the uplift intensity and total at both box lengths, all four flotation factors of safety , the COMB 102 check at both lengths, the net pressure at formation, every bearing utilisation, and the whole rock-excavation quantity table
4–8	Element design, box and headhouse	Every d , $M_{u,lim}$, A_{st} provided, M_u , utilisation, x_u/d , shear force, τ_v , p_t , V_{us} , A_{sv}/s_v and link spacing for W1–W4, W6/W7, the mat, the roof, the openings, the cantilever pad, the headhouse roof and walls, and HW3 — plus the shaft fill time that produced Finding F1
9	Stairs	Both pitch angles and their cosines, every load term, effective spans, moments, areas, capacities, utilisations, the total-rise closure, and both opening-corner angles
10	Escape shaft head hatches	The leaf force, the hole area, the plate moment and shear, the flat-plate thickness and its 505 kg mass , both climb heights and both rung pitches
11	Seismic and wind	Both A_h values, both base shears, the $R = 3.0$ conservatism factor , the model-versus-hand weight gap and its two explanations to the kilonewton , all three period estimates, the full wind chain, and the portal distribution
12	Sentry post members	Effective spans and the two-way ratio, slab moments and capacities, both beams' governing combinations, capacities, capacity-design shears and earthquake shares , the 5-T16 bar-fit check that forced 3-T20 , the column's slenderness, steel, P_{uz} , biaxial sum and confinement, the footing in service and at ULS, punching, anchorage, and the whole lintel design including the 60° arching bound
13	Development, laps and cover	L_d/ϕ for both grades in tension and compression, every tabulated L_d , and the lap policy margin
14	CBRN ventilation	All eleven figures the HVAC package reproduces , plus the air-change rates, the scrubber matrix at six duty points and the recirculation fan power
15	Thermal balance	The load that is dissipated inside the envelope, the occupant gain, the 96-hour energy, all three bounds — adiabatic, concrete and film — the 13.2 K rise, the cooling duty, and the ventilation rejection arithmetic
16	EMP	Cage shielding effectiveness at five frequencies, the mesh cutoff, the highest frequency meeting 80 dB , the wavelength coincidence at 1 GHz, four TE11 cutoffs, six waveguide attenuations, the stair void, the honeycomb, both bond-strap inductances and reactances , the earth rod, the structure electrode, and the Zone 2 enclosure's area, seam length and volume

§	Subject	What it recomputes
17	Drainage and external works	The wetted envelope and the seepage it produces, the sump storage, all three trap-seal depths , the soak pit before and after widening, the septic tank checks, and every catchment
18	Electrical	The connected load and its kVA, generator utilisation and spare, the essential load, both battery cases and the factor of twelve between them , and the fuel quantity
19	Quantities and cost	Every bar mass from its unit mass and length, both totals, the concrete take-off, the steel density, and both cost summaries head by head
20	Register arithmetic	The open-item counts, the assumption counts and the drawing counts add up

19.3 The four differences found — reported, never corrected

Master rule M.11 and the project operating guide both forbid editing a recorded value away. Every difference below is an observation offered to the master, not a change. **None of them moves a dimension, load, thickness, bar, quantity, rate, date or float.**

#	Where	Computed	Printed	Assessment
1	Underground box wall shear stress — $525 \times 10^3 / (600 \times 0.8 \times 21600)$	0.0506 N/mm²	0.063 N/mm ²	Not previously recorded. Both values are negligible by two orders of magnitude ; the conclusion "IS 13920 Cl. 10.4 boundary elements NOT triggered" is unaffected and no adopted value depends on it
2	Sentry footing F1, ULS eccentricity — $M_u/P_u = 43.9 / 276.8$	0.159 m	0.128 m	Not previously recorded. The footing is governed by IS 456 Cl. 26.5.2.1 minimum steel (720 against 136 mm ² /m required) and its depth by starter anchorage , so no adopted bar, spacing or dimension moves
3	Sentry footing F1, $q_{u,max}$ — the printed expression $276.8/2.25 \times (1 + 6 \times 0.128/1.5)$	186.0 kPa	190.0 kPa	Same item as 2. One-way shear at 0.012 and punching at 0.065 against $\tau_c = 1.369$ are unaffected on either figure
4	The project owner's cost summary — the seven heads against the printed final	₹ 2 99 33 306	₹ 3 00 33 306	ALREADY RECORDED — conflict R-14 , exactly ₹ 1 00 000 apart, " <i>reported, not corrected</i> ". The verification pass reproduced a known finding , which is the outcome you want from a check like this. The revised estimate reconciles to the rupee and has no such difference

Three observations about that result are worth making.

The two footing items are one item. An eccentricity and the bearing pressure computed from it are the same arithmetic seen twice, and both land on an element whose steel is set by a code minimum and whose depth is set by anchorage. **That is precisely why it survived four revisions without being caught: nothing downstream of it is sensitive to it.**

The wall shear item is the same shape. A figure quoted to show that something is negligible is a figure nobody re-checks, because the conclusion is right either way.

And the fourth was already in the register. Reproducing a recorded finding independently is the best evidence that the check is working — and reproducing it **from the owner's own published numbers** is what makes it safe to leave alone.

19.4 What this report could not verify, and did not attempt

Not verified	Why
Any analysis result	STAAD.Pro is not available. No result exists to check
The mesh convergence conclusion	Same reason. The three models are built and validated; none has been run
Every [ASSUMED] value	An assumption is not an arithmetic error. k_s, the design water table, the safe bearing capacity, K_o, the soak-pit absorption rate, the seepage rate and the rockhead band are all tests, not sums
Vendor performance	Blast doors, blast valves, filter trains, the shielded door, the PCI, the honeycomb, the escape hatches
Anything tagged [N]	There is nothing to check. The projection dimension, the ambient design temperature, the opening heights, the pipe materials, the entry plate's level and size, the sump cover, the fan's five vendor loss components
The drawings themselves	Checked by the drawing QA tool, not by this report — Part 13.3

19.5 Consistency checks made against the project record

Beyond the arithmetic, the following were checked by reading:

1. **Every reinforcement arrangement in Part 12 matches the calculation in Parts 9 to 11.** No bar size, spacing or count appears in the register that does not appear in a calculation.
2. **Every evidence tag is carried unchanged.** No [ASSUMED], [UNRESOLVED] or [NOT AVAILABLE] item is converted, downgraded or deleted anywhere in this report.
3. **The open-item and assumption counts in Part 18 agree with their own tables** — $9 + 2 + 6 + 1 = 18$ open, $18 + 16 + 1 = 35$ rows, $7 + 7 = 14$ assumptions.
4. **The drawing counts in Part 13 sum to 80**, and the sheet sizes $75\text{ A1} + 4\text{ A4} + 1\text{ A0}$ sum to 80.
5. **The frozen staircase geometry is reproduced exactly** — 24 risers, 170.8333 mm, 280 mm tread, 3 flights $\times$ 8, total rise 4 100 mm — **and nothing in this report changes any of it.**
6. **Every code clause cited here appears in the project's own register.** No clause was added, and codes held by title only are named as such.

APPENDIX A — NOTATION

Symbol	Meaning
p_{so}	Peak incident static overpressure of the blast wave
p_r	Reflected overpressure on a surface normal to the wave
q	Dynamic (kinetic) pressure of the air flow behind the front
t_d	Positive phase duration
T, f_1	Fundamental natural period and frequency
μ	Ductility ratio — maximum displacement / yield displacement
DLF	Dynamic load factor, $\mu/(\mu - 0.5)$
K_a	Soil transmission factor for a buried element
K_0	Coefficient of earth pressure at rest, $1 - \sin \phi$
$\gamma, \gamma_{sat}, \gamma', \gamma_w$	Bulk, saturated, submerged and water unit weights
k_s	Modulus of subgrade reaction (kN/m^3)
f_{ck}, f_y	Characteristic concrete and steel strengths
$f_{ck,dyn}, f_{y,dyn}$	The same, increased 25 % for the blast case only
E_c	Modulus of elasticity of concrete, $5000\sqrt{f_{ck}}$
d, d'	Effective depth; cover to the centroid of compression steel
$A_{st}, A_{sv}/s_v$	Tension steel area; shear link area per unit length
$M_p, M_u, M_{u,lim}$	Plastic mechanism moment; design capacity; limiting singly-reinforced capacity
$x_u, x_{u,max}$	Neutral axis depth; its limiting value, 0.46 d for Fe500
$\tau_v, \tau_c, \tau_{c,max}$	Nominal shear stress; permissible concrete shear stress; its maximum
p_t	Tension steel percentage
V_{us}	Shear to be carried by links
L_d, τ_{bd}	Development length; design bond stress
$A_h, Z, I, R, S_a/g$	Seismic design coefficient and its four inputs
V_b, W	Seismic base shear; seismic weight
P_{uz}, α_n	Column pure-axial capacity; biaxial interaction exponent
A_{sh}, l_o	Special confining hoop area; confining length
SE	Shielding effectiveness, dB
WBC	Waveguide below cutoff
T16 @ 150 EF EW	16 mm bars at 150 mm centres, Each Face, Each Way
4L	Four-legged links

Evidence classes — [C] confirmed · [R] reconstructed · [A] assumed · [U] unresolved · [N] not available.

APPENDIX B — WHERE EACH CODE CLAUSE IS USED

Clause	Subject	Used in
IS 456 Cl. 6.2.3.1	$E_c = 5000\sqrt{f_{ck}}$	Part 6.1
IS 456 Cl. 13.4	Construction joints	Part 6.4
IS 456 Cl. 22.2(a)	Effective span	Parts 10.1, 11.2, 11.7
IS 456 Cl. 23.2.1 + Fig. 4	Deflection, modification factor	Parts 10.1, 10.2, 11.2
IS 456 Cl. 24.4, 24.5 + Fig. 7	Two-way slabs; load to beams	Parts 7.7, 10.4, 11.2
IS 456 Cl. 25.1.2	Short column	Part 11.5
IS 456 Cl. 25.4	Minimum eccentricity	Part 11.5
IS 456 Cl. 26.2.1, 26.2.1.1, 26.2.5.1	Development length, laps	Part 6.3
IS 456 Cl. 26.3.3	Bar spacing	Parts 6.2, 11.2 — the project's 150 mm cap is stricter
IS 456 Cl. 26.4.2, .1, .2 + Table 16	Cover	Part 6.2
IS 456 Cl. 26.5.1.1, .2, .5, .6	Beam steel and links	Parts 9.1–9.4, 10.3, 11.3, 11.7
IS 456 Cl. 26.5.2.1, .2	Slab steel, maximum bar diameter	Parts 9.4, 10.1–10.3, 11.2, 11.6
IS 456 Cl. 26.5.3.1, .2	Column steel and ties	Part 11.5
IS 456 Cl. 31.6, .1, .3.1	Punching shear	Parts 9.3 (declared N/A), 11.6
IS 456 Cl. 32.2, 32.5(a)(b)(c)	Walls — minimum steel, two curtains	Parts 9.1, 10.2, 10.4
IS 456 Cl. 33.1(b), 33.2	Stairs — effective span, step load	Parts 10.1, 10.2
IS 456 Cl. 34.2.4.1	Footing one-way shear	Part 11.6
IS 456 Cl. 36.4.2	Partial safety factors for materials	Part 6.1
IS 456 Cl. 38.1, (c), (f)	Limit state, stress block, $x_{u,max}$	Parts 9–11 throughout
IS 456 Cl. 39.1, 39.3, 39.6	Columns, biaxial bending	Part 11.5
IS 456 Cl. 40.1, 40.2.1.1, 40.2.3, 40.4(a)	Shear	Parts 9.1–9.4, 10.1–10.4
IS 456 Tables 3, 5	Exposure, mix requirements	Part 6.1
IS 456 Tables 19, 20	$\tau_c, \tau_{c,max}$	Parts 9–11 throughout
IS 456 Tables 26, 27	Two-way slab coefficients	Parts 5.3, 10.3, 11.2
IS 456 Table 28	Effective length of columns	Part 11.5
IS 456 Annex D, D-1.1 to D-1.8	Two-way slab detailing, corner torsion	Part 11.2
IS 456 Annex G-1.1(b), (c)	$M_u, M_{u,lim}$	Parts 9–11 throughout
IS 875 (Pt 1) Table 1	Unit weights	Parts 6.1, 7.2
IS 875 (Pt 2)	Imposed loads	Parts 7.2, 7.7, 10.1, 10.2
IS 875 (Pt 3) Cl. 7.2 etc.	Wind	Part 7.8.3
IS 1893 (Pt 1) Cl. 6.3.2.2	Directional combination	Part 11.5
IS 1893 (Pt 1) Cl. 6.4.2	A_h	Part 7.8
IS 1893 (Pt 1) Cl. 7.6.2, (c)	Fundamental period	Part 7.8.2
IS 1893 (Pt 1) Cl. 7.6.3	Base shear and its distribution	Parts 7.8, 11.1
IS 1893 (Pt 1) Cl. 7.11.1	Storey drift	Part 11.5
IS 13920 Cl. 6.1	Beam section limits	Part 11.3
IS 13920 Cl. 6.2.1(b), .2, .3, .4	Beam steel limits	Part 11.3
IS 13920 Cl. 6.3.3, 6.3.4 , 6.3.5.1, .2	Capacity-design shear; $\tau_c = 0$	Parts 11.3, 11.4
IS 13920 Cl. 7.2.1	Strong column – weak beam	Part 11.5
IS 13920 Cl. 7.4, 8.1, 8.2	Column hoops, special confining reinforcement	Part 11.5
IS 13920 Cl. 10.4	Boundary elements — checked, not triggered	Part 7.8.1
IS 3370 (Pts 1, 2)	Crack width 0.2 mm; 0.35 % surface-zone steel	Parts 6.1, 9.1
IS 1904 Table 1	Presumptive bearing capacity	Parts 4.3, 11.6
IS 12070 Cl. 6, Table 2	Settlement on rock; broken bedrock	Parts 4.3, 9.3

Clause	Subject	Used in
IS 1498, IS 2720 Pts IV, VIII, X, XIII, XL, 28	Soil classification and testing	Parts 4.2, 4.5
IS 2470 (Pt 1) Cl. 6.2–6.9, Table 1	Septic tank	Part 14.2
IS 2470 (Pt 2) Cl. 4, 5	Percolation test — mandatory ; dispersion	Parts 14.2, 15.1, 18.2
IS 4991 Cl. 1.1	EXCLUDES NUCLEAR — quoted in every deliverable	Parts 3.1, 17.2
IS 4991 Cl. 6.2.1	Reflected pressure, recessing of blast valves	Parts 2.7.3, 14.1.4
IS 4991 Cl. 7.2 + Table 3	Buried roof and walls, K_a	Parts 2.1.4, 7.1, 7.5
IS 4991 Cl. 7.4	Drag on bermed faces	Part 7.5 (superseded basis)
IS 4991 Cl. 10.3.1	Dynamic f_{ck} and f_y	Parts 2.2.5, 7.1
IS 4991 Cl. 10.3.1.1	No dynamic increase on shear; +25 % bond, refused	Parts 2.2.5, 6.3, 9.3
IS 4991 Cl. 10.3.3	$DLF = \mu/(\mu - 0.5)$	Parts 2.2.3, 7.1
IS 4991 Cl. 11.1, 11.2	No wind/EQ with blast; no live load on the roof	Parts 7.4, 7.9
IS 4991 Fig. 6	SDOF support rotation — Phase 3	Parts 2.2.3, 17.2
SP 34 Cl. 5.5	Opening-corner detailing	Part 10.2.1
NBC 2016 Part 4	Stair geometry, 1 100 guarding, means of escape	Parts 9.5.2, 10.1, 14.5
UFC 3-340-02 §4-27, §4-30	Opening trimmers; direct shear	Part 9.4.4
MIL-STD-188-125-1 §5.4–§5.7.6	EMP — access, WBC, PCI, fibre, RF	Parts 2.8, 14.4
IEEE Std 299	Shielding effectiveness survey — HOLD POINT	Part 14.4.5
IEEE 142 / IS 3043	Earthing, $\leq 5 \Omega$	Parts 2.8.5, 14.3
FEMA 453	0.25 cfm/ft ² ventilation rate	Part 14.1
EN 1822	HEPA H14 at MPPS	Part 2.7.2
IS 1077	Sentry post brick masonry	Parts 5.9.1, 11.7
IS 732, IS 694, IS 1554 (Pt 1)	Electrical installation and cables	Part 14.3
IS 13416	Construction-phase fire safety	Part 14.5
BS 8666	Bar shape codes	Part 12.6

APPENDIX C — REPRODUCING THIS REPORT AND THE PROJECT

C.1 This report

```
Project Report/
Documentation/MASTER_PROJECT_REPORT.md      the report SOURCE -- edit this
Scripts/report_render.py                    Markdown -> PDF typesetter
Scripts/report_verify.py                    independent recomputation
Calculations/REPORT_VERIFICATION_OUTPUT.txt its output
MASTER_PROJECT_REPORT.pdf                  the deliverable

python3 "Project Report/Scripts/report_verify.py"    run the checks
python3 "Project Report/Scripts/report_render.py"    rebuild the PDF
```

Never edit the PDF. Edit the Markdown and re-run the renderer. The renderer adds no content of its own beyond the cover, the running head and foot, and the automatically paginated contents list.

C.2 The project

To rebuild	Do this
A discipline drawing package	<code>cd "<Package>/Scripts" && python3 <prefix>_build_all.py</code> . Edit the generator, never the DXF
The drawing index and QA report	Re-run <code>qa_report_data.py</code> then <code>make_index.py</code> . A new package must first register its DXF/ prefix in DISCIPLINE and its name in ORDER
The works management package	<code>wm_build_all.py</code> regenerates everything from <code>wm_data.py</code> + <code>wm_content.py</code>
The S-series structural sheets	Not possible. The toolchain is not in the workspace (Part 13.6) and must not be fabricated
Any STAAD result	Not possible. STAAD.Pro is not available in this environment

The ten Rev F architectural DXFs are source, not build artefacts, and may be edited directly. Everything else generated in this project is a build artefact: **edit the generator and re-run.**

C.3 The rule that makes regeneration safe

Every discipline package in this project has verified that it **regenerates byte-identically**. That property is worth not breaking casually — it is what makes it possible to apply a change to a generator and be certain that nothing else moved.

APPENDIX D — REVISION HISTORY IN BRIEF

The full record, with the reasoning for every change, is in the master's Part H. This is an index to it.

Rev	Date	What it did
A–C	—	Architectural layout development. Rev C lengthened the box to give ESC 2 its 750 mm clearance
—	—	Cover reduced 4.0 m → 2.0 m. Saved 2 m of rock excavation, 2 m of shaft, one stair flight and 2 m of headroom
D	—	Design report issued — basis of design, loads, materials, geotechnics. Current for loads and materials; superseded for the entrance and the headhouse roof
E–F	—	The approach was replaced: a dog-leg open ramp became a covered entry stairwell entered at grade. The open cut had no gravity outfall, a 2 m pit collects dense CBRN agents, and lidding at grade gave zero headroom
F	—	The ten architectural drawings issued. Current for all geometry
M1	3 Sep 2026	W6/W7 200 → 400; box 21 600 → 22 000. Approved and implemented. Finding F1
MS1	3 Sep 2026	Three mesh-density models built and validated
BIM-P1 / P1B	3 Sep 2026	Revit scripts authored — not executed; Revit is not available
SC1	4 Sep 2026	30 reinforcement drawings , 8 bar bending schedules, 70.46 t of steel
DR1 · HV1 · FN1	5 Sep 2026	Drainage, HVAC and the schedule of finishes
WM1	7 Sep 2026	Works management; and SP-B1, the one instructed design change
QA1	9 Sep 2026	65 DXF inspected and corrected in place
BS1 · SP-B2	10 Sep 2026	Burster slab laid to falls; sentry post lintels and wall ties
WM2 · WM3	10 Sep 2026	The owner's own bill, estimate and schedule adopted , then revised alongside
RC1	10 Sep 2026	Every inconsistency that could be ruled on the evidence, ruled
CAM1/2 · FS1/2	10 Sep 2026	Concealment policy and fire plan — neither existed before — then relocated and drawn
EM1	10 Sep 2026	The project's first EMP design
EL1	11 Sep 2026	The project's first electrical design , deliberately basic
RC2	11 Sep 2026	Two owner rulings. Neither changed a number
SG1 · SG2	11 Sep 2026	The project's first site and geotechnical section , and its first site layout
QA2 · MS2 · RC3	11 Sep 2026	Drawing re-scan at 80 sheets; the .std input-file corrections ; project-wide reconciliation
DR-A1	11 Sep 2026	Drainage items drawn on the Rev F architectural sheets
RC4	11 Sep 2026	Sixteen owner rulings, item by item

Rev	Date	What it did
RC5 · RC6 · RC7 · RC8	12 Sep 2026	Sixteen more rulings; K.2 falls from fourteen assumptions to seven
RC9	12 Sep 2026	A count this project got wrong, corrected — and made countable
RC10	12 Sep 2026	What the rulings left stale in the packages — six artefacts, register raised, not yet applied
DR-A2	12 Sep 2026	The sump pit head and a key plan; DR-A2-V1 opened
PR1	13 Sep 2026	This report. No design value changed; 498 checks executed; four differences reported and none corrected

Master project report, revision PR1 · 13 September 2026 · FOR REVIEW — NOT FOR CONSTRUCTION. Authority: master/MASTER_PROJECT_STATE.md. Where this report and the master disagree, the master governs. No design value, evidence tag, quantity, rate, date or float is changed by this report, no .std file was touched, STAAD.Pro was not run, and the main staircase is unchanged — 24 risers, 170.8333 mm riser, 280 mm tread, 3 flights × 8, total rise 4 100 mm.